

Land and Water Resources Engineering

- ✓ Hydrology
- ✓ Irrigation and Drainage Engineering
- ✓ Soil and Water Conservation Engineering



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PROFESSIONAL REGULATION COMMISSION
PROFESSIONAL REGULATORY BOARD OF AGRICULTURAL AND BIOSYSTEMS ENGINEERING
TABLES OF SPECIFICATION



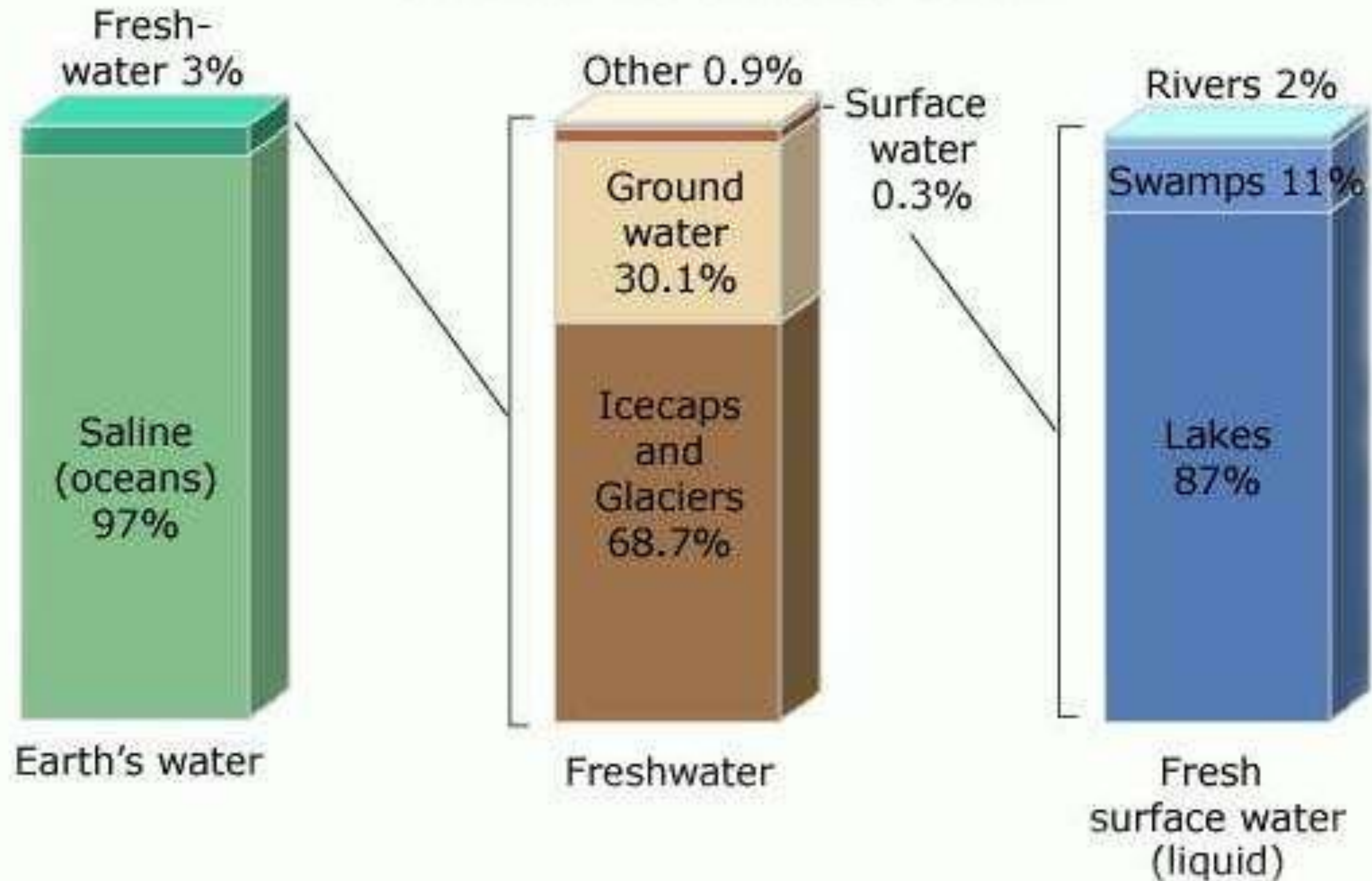
SUBJECT B. Land and Water Resources Engineering and Allied Subjects (32%)

I. Hydrology The registered and licensed Agricultural and Biosystems Engineer can perform the competencies under the following sub-topics:	Percentage Weight and Distribution	Number of Items and Distribution	Easy (30%)		Moderate (50%)	Difficult (20%)		
	32%	100	Remembering Recall information	Understanding Explain, Interpret, summarize, examine paraphrase, classify	Applying Using the information in another familiar situation	Analyzing Understanding, Comparing, organizing, deconstructing, interrogating, finding	Evaluating hypothesizing, critiquing, experimenting, judging	Creating Generating Designing, constructing, planning, producing, inventing
	2	8	Recognizing, listing, describing, retrieving, naming, finding		Implementing carrying out, using, executing			
1. Describe the hydrologic cycle; hydrology in engineering.	0.5	2	2					
2. Describe the principles of hydrometeorology; weather and hydrology; precipitation, evaporation and evapotranspiration; weather instruments.	0.5	2	1	1				
3. Describe the principles of watershed/surface hydrology; runoff, infiltration, seepage and percolation, stream flow, soil erosion and sediment transport.	0.5	2	1	1				
4. Describe the basic theories of aquifer systems and lithography; water quality analysis.	0.5	2	2					

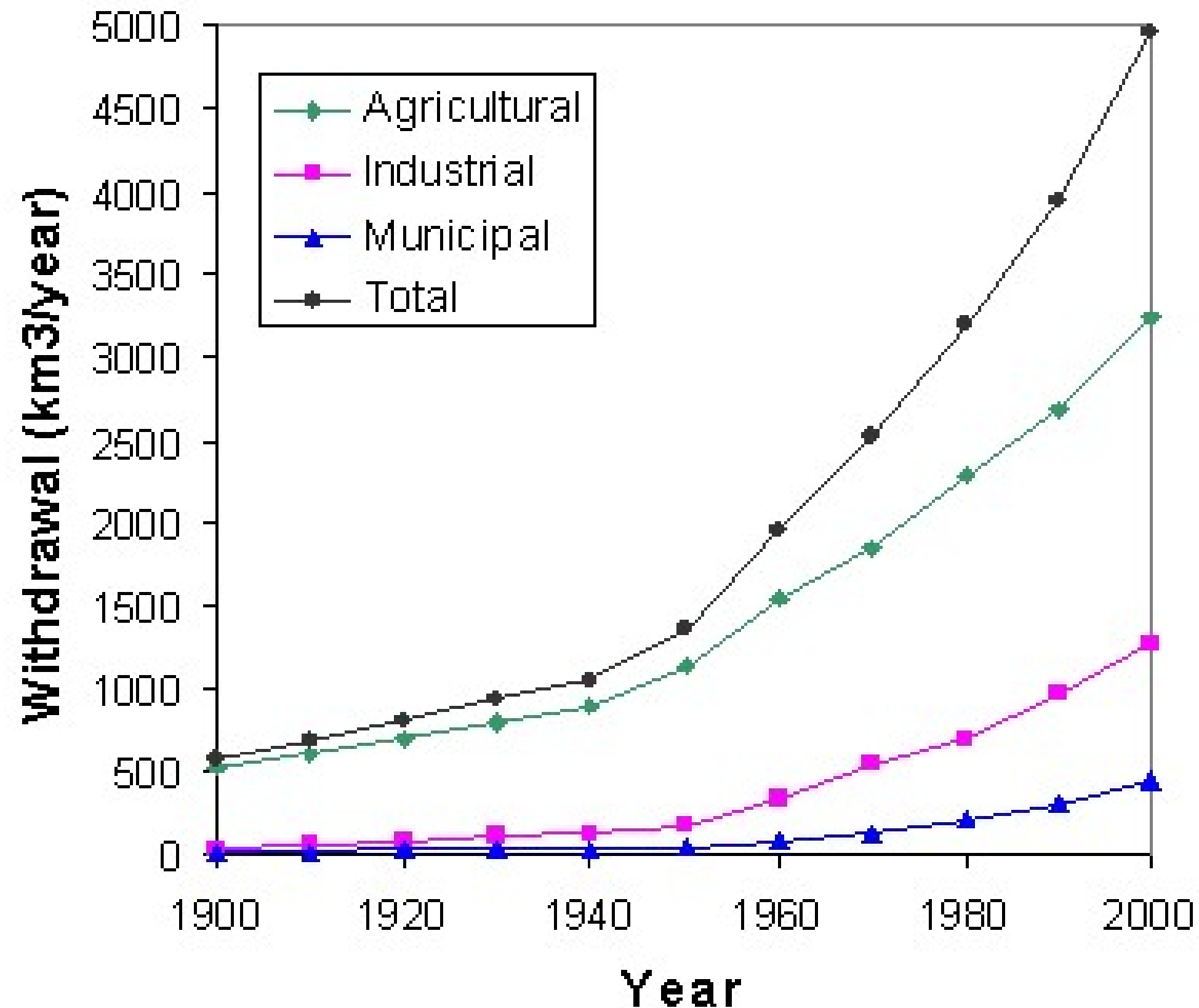
II. Irrigation and Drainage Engineering The registered and licensed Agricultural and Biosystems Engineer can perform the competencies under the following sub-topics:	Percentage Weight and Distribution	Number of Items and Distribution	Easy (30%)		Moderate (50%)	Difficult (20%)		
	32%	100	Remembering Recall information Recognizing, listing, describing, retrieving, naming, finding	Understanding Explain, Interpret, summarize, examine paraphrase, classify	Applying Using the information in another familiar situation Implementing carrying out, using, executing	Analyzing Understanding, Comparing, organizing, deconstructing, interrogating, finding	Evaluating hypothesizing, critiquing, experimenting, judging	Creating Generating Designing, constructing, planning, producing, inventing
	8	25						
1. Apply theories and practices in leveling, mapping, triangulation, topographic, route, and construction surveys, and global positioning system (GPS).	0.64	2	1		1			
2. Apply the theories and principles of fluid mechanics in the design, construction and management of agricultural and biosystems engineering production systems and facilities.	1.28	4	1	1	2			
3. Apply the basic theories of groundwater hydrology in planning water resource projects.	0.64	2	1		1			
4. Analyze the design, construction, operation and maintenance of hydraulics machinery and facilities (i.e. pumps, and turbines); pump size determination and specifications.	1.28	4				2	2	
5. Plan, design, construct, operate and manage irrigation and drainage systems and facilities; design small canal structures for the conveyance of irrigation or drainage water.	2.88	9		2	1	2	2	2
6. Apply the theories and principles in designing pressurized irrigation systems and facilities.	1.28	4			4			

III. Soil and Water Conservation Engineering The registered and licensed Agricultural and Biosystems Engineer can perform the competencies under the following sub-topics:	Percentage Weight and Distribution	Number of Items and Distribution	Easy (30%)		Moderate (50%)	Difficult (20%)		
	32%	100	Remembering Recall information Recognizing, listing, describing, retrieving, naming, finding	Understanding Explain, Interpret, summarize, examine paraphrase, classify	Applying Using the information in another familiar situation Implementing carrying out, using, executing	Analyzing Understanding, Comparing, organizing, deconstructing, interrogating, finding	Evaluating hypothesizing, critiquing, experimenting, judging	Creating Generating Designing, constructing, planning, producing, inventing
	6	20						
1. Explain and apply the basic soil-water- plant relations in crop production.	0.6	2		1	1			
2. Apply the principles of hydrometeorology in the planning and design and management of water resources facilities.	0.30	1			1			
3. Apply the principles of watershed hydrology in the planning and design of agricultural and biosystems engineering facilities.	0.30	1			1			
4. Design, construct, manage farm ponds, reservoirs, including earth dams, and spillways.	2.4	8			4	2	1	1
5. Design of soil and water control structures and facilities for stream banks and sloping cultivated agricultural areas.	1.2	4			2		1	1
6. Design of vegetative and lined water conveyance channels.	0.9	3			1		1	1
7. Maintain water conservation works, drainage and erosion control systems.	0.3	1			1			

Distribution of Earth's Water



Global Water Withdrawal by Sector, 1900-2000

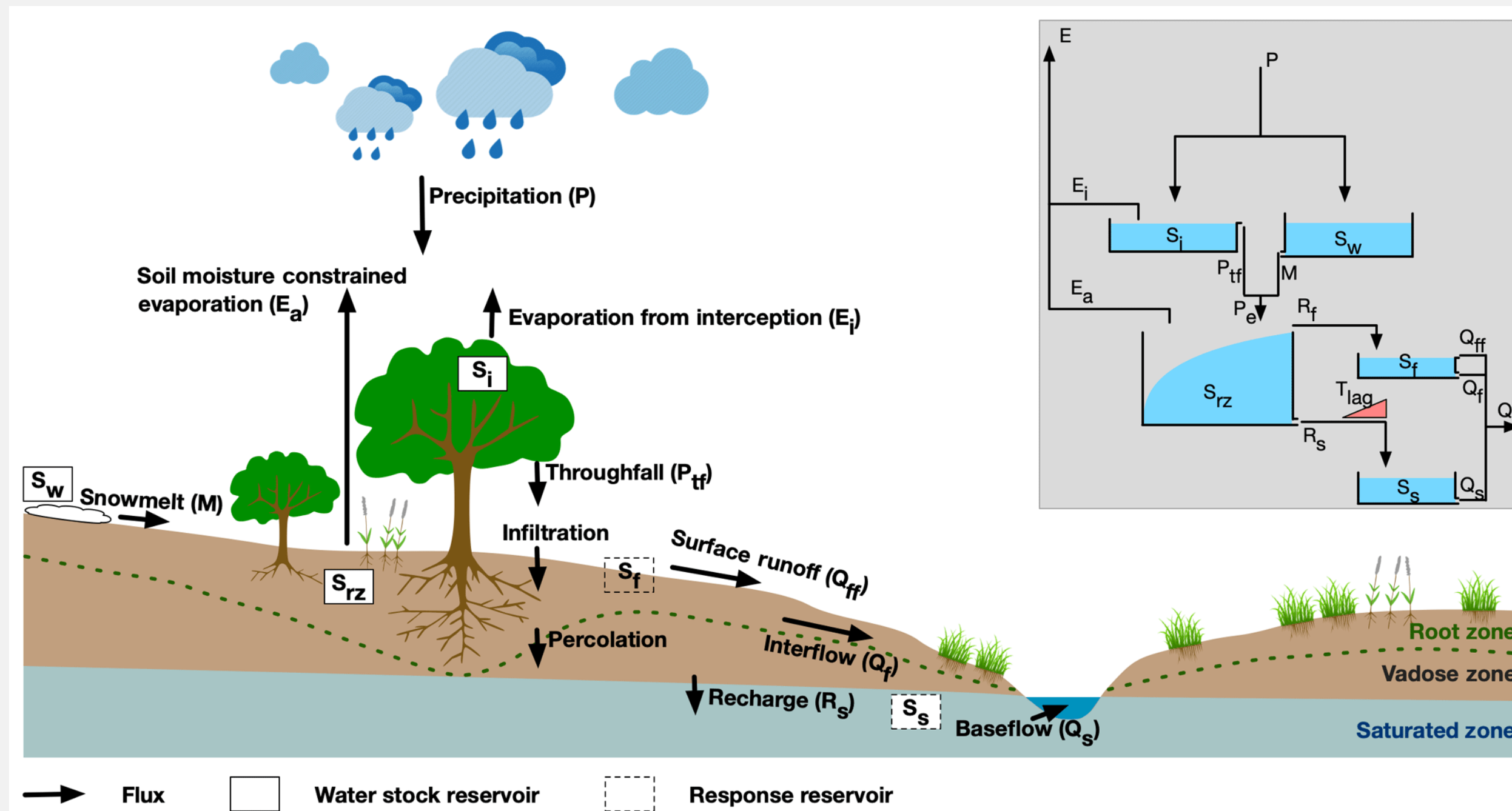


Compiled by
the Worldwatch
Institute from
P.H. Gleick (1993)
Water in crisis,
Oxford University
Press.

Hydrology

Hydrology

It is the science of water that is concerned with the origin, circulation, distribution and properties of water of the earth.

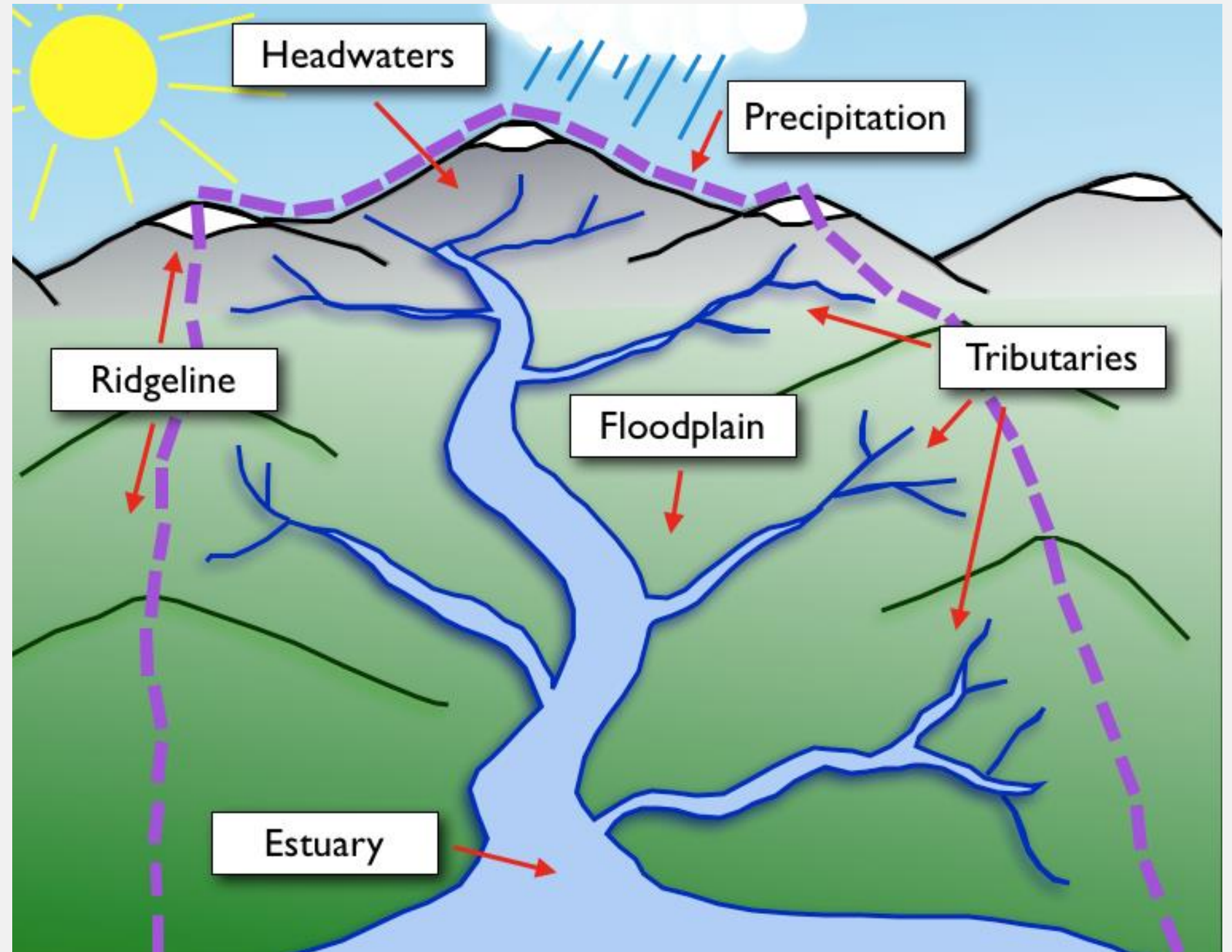


Watershed

- basic unit of hydrology
- area upstream of a point along a river where all water coming off mainly from precipitation drains towards that point.

Other Terms

1. Catchment Area
2. Drainage Area
3. Catch Basin
4. River Basin



Sample Question:

It is the science which deals with the occurrence, distribution and disposal of water on the planet earth.

- a. Hydrology
- b. Hydrometeorology
- c. Geology
- d. Hydrogeology
- e. All of the above

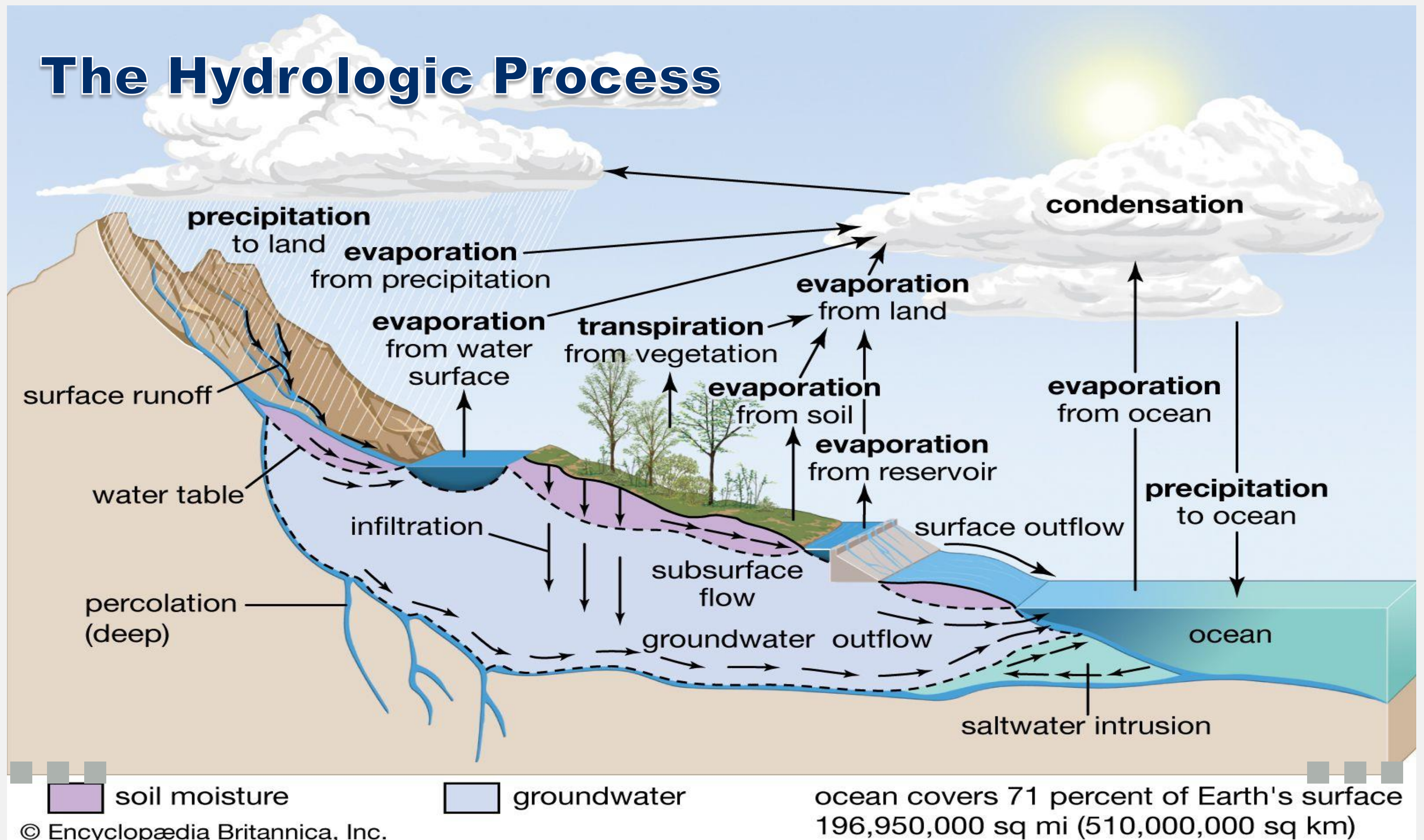
Sample Question:

It is the vertical and horizontal movement of water as either vapor, liquid, or solid between the earth's surface, subsurface, atmosphere, and oceans.

- a. Water Cycle
- b. Water Transfer Cycle
- c. Hydrologic Cycle
- d. All of the above

D

The Hydrologic Process





WEATHER

**SHORT-TERM STATE OF
THE ATMOSPHERE**



Rain



**Can change within
minutes or hours**

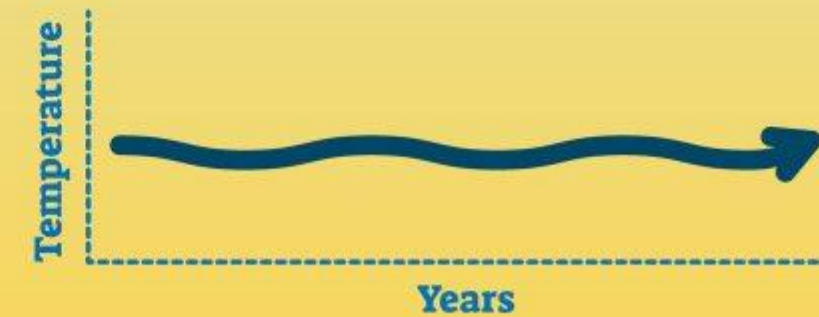


CLIMATE

**LONG-TERM PATTERN
OF WEATHER**



Tropical Climate



**Average weather over
many years in one
specific place**

VS

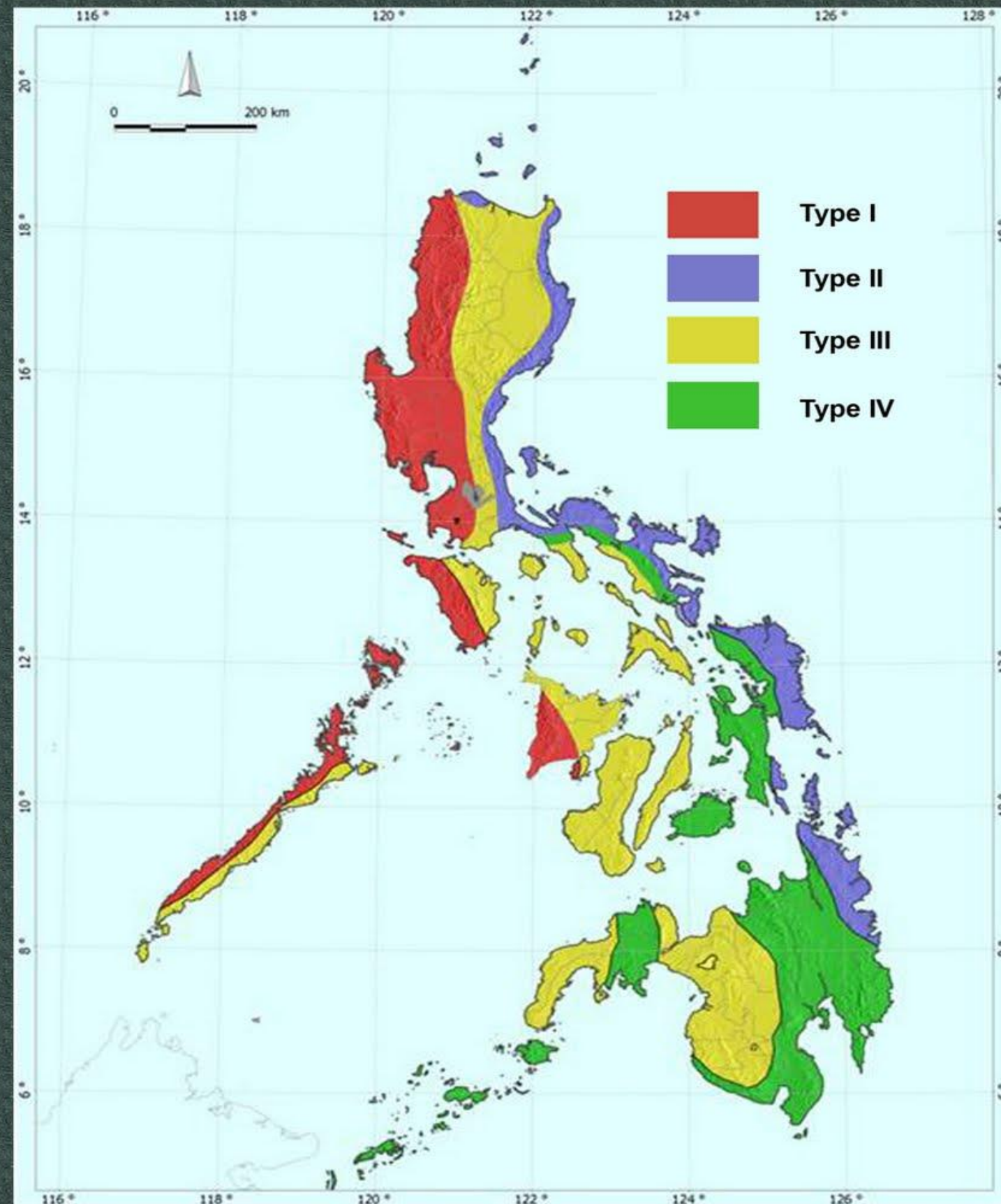


Climatic Controls for Philippine Climate

1. Topography and location
2. Trade winds
3. Fronts
4. Intertropical convergence zone (ITCZ)
5. Monsoon winds
6. Tropical cyclones
7. Easterly waves



Climate Map of the Philippines based on the Modified Coronas Classification



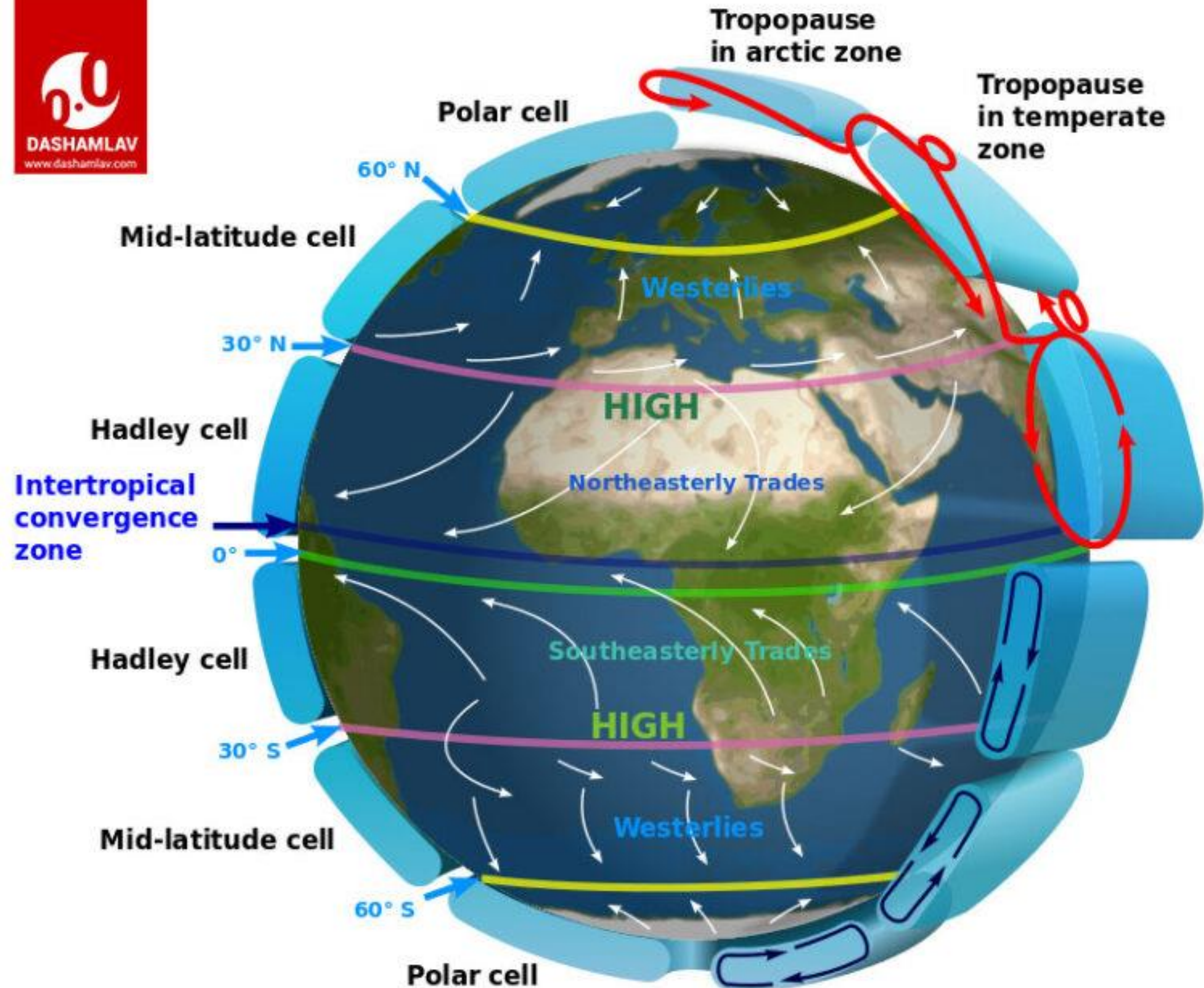
Description

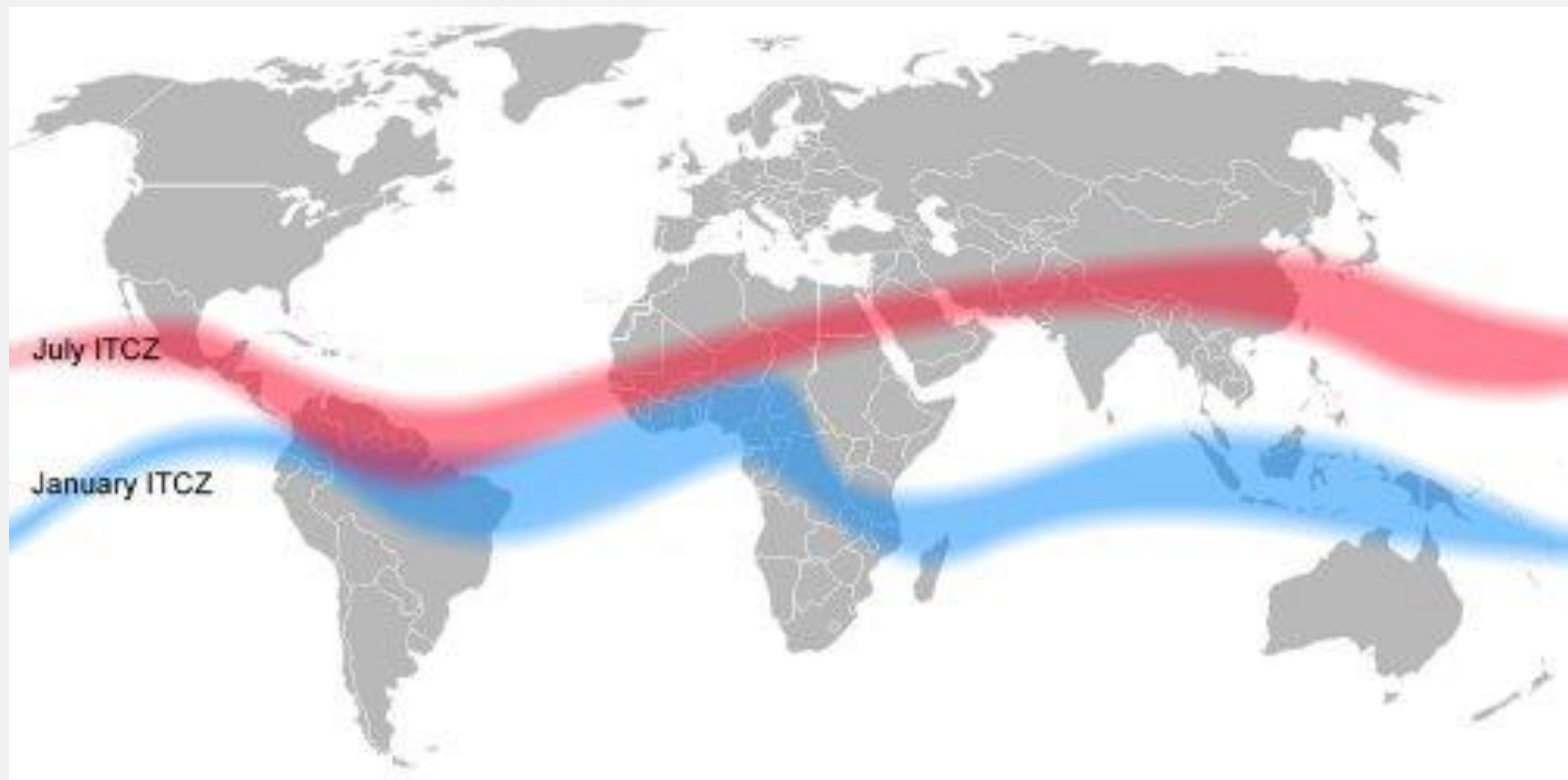
Type I - two pronounced season, dry from November to April and wet during the rest of the year. Maximum rain period is from June to September.

Type II- no dry season with a very pronounced maximum rain period from December to February. There is not a single dry month. Minimum monthly rainfall occurs during the period from March to May.

Type III- no very pronounced maximum rain period with a dry season lasting only from one to three months, either during the period from December to February or from March to May. This type resembles types I since it has a short dry season.

Type IV- rainfall is more or less evenly distributed throughout the year. This type resembles type 2 since it has no dry season.





MAP SYMBOL:



COLD FRONT

COLD AIR MASS



A **cold front** is the boundary along the leading edge of a **cold air mass that is pushing out a warm air mass**. As the cold front nears your region, the barometer falls. The cold air behind the front wedges under the warm air and lifts it sharply off the ground. **Large cumulonimbus clouds** appear. These clouds often **bring thunderstorms and rain showers**. As the cold front passes, the wind changes direction. The weather becomes clear and colder and the barometer rises again.

Further differentiation of cold fronts **on the basis of vertical velocities** is shown in two types:

Type I cold fronts are those observed **outside** the zone of cyclonic activity and are **generally slow moving** which may even be quasi-stationary in some cases. It is defined by a general upglide of warm air over the entire frontal surface so that the cloud system resembles that of a warm front.

Type II cold fronts are observed **within** the zone of cyclonic activity and as a rule move **very rapidly**. Warm air is lifted only along the leading edge of the intruding wedge of cold air. At higher levels, the warm air is moving faster than the cold air and therefore no ascending motions are observed in the warm air above the frontal surface.

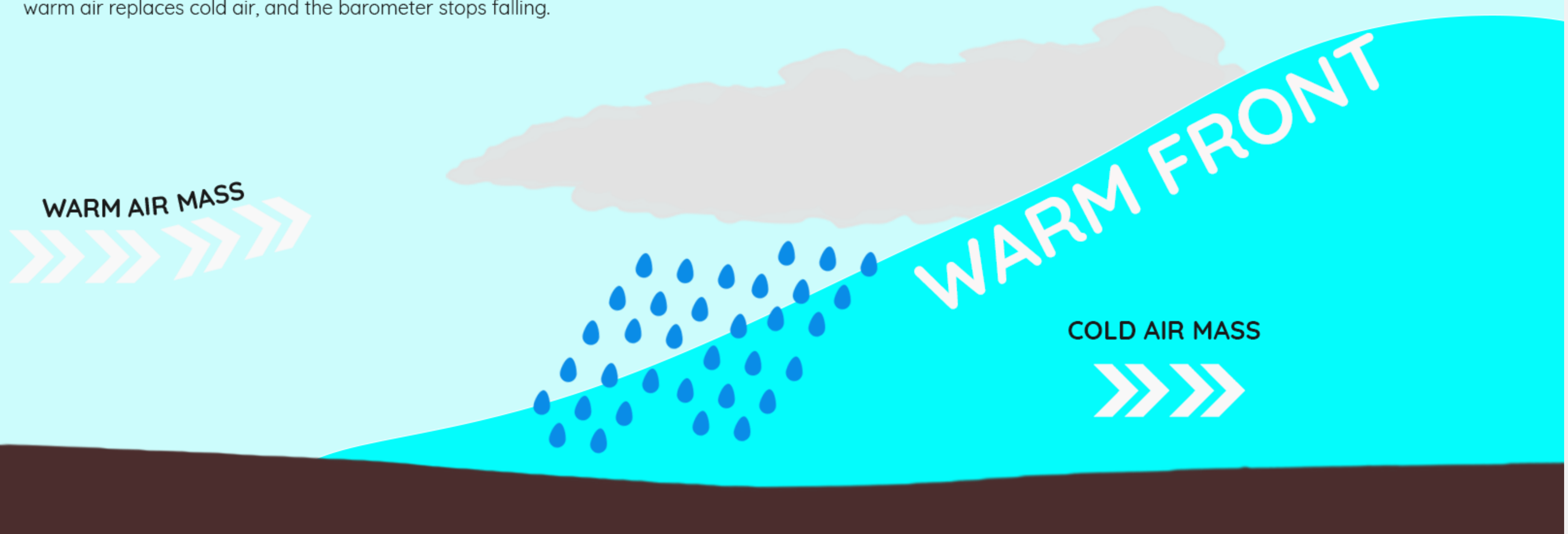
WARM AIR MASS



MAP SYMBOL:



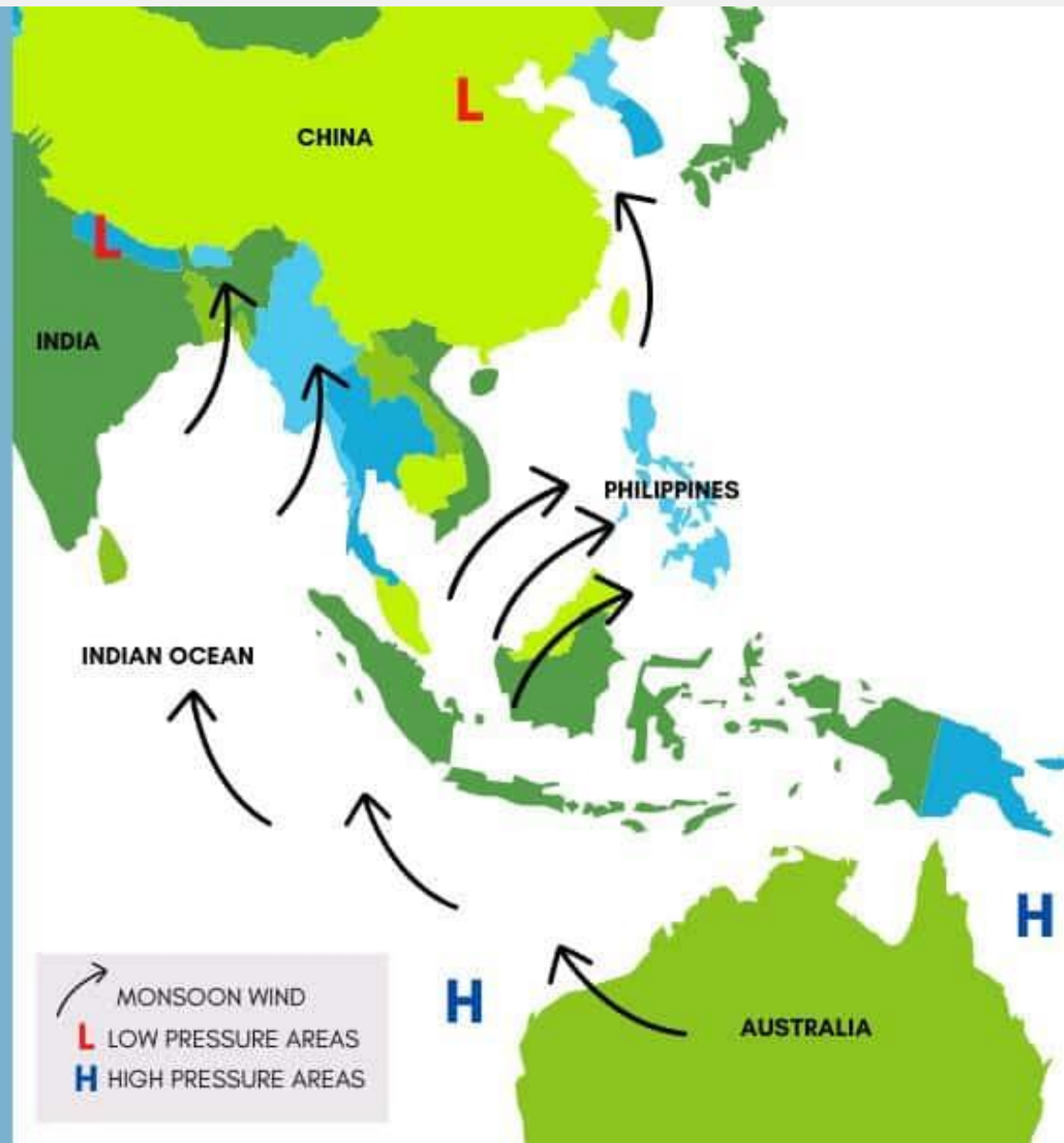
A **warm front** is the boundary along **a warm air mass that is pushing out a cold air mass**. The warm air behind the front glides up and over the cold air. As a warm front approaches, high cirrus clouds appear. These are followed by **stratus and nimbostratus clouds**. The barometer falls and a long, steady rain begins. Gradually, the front passes and the sky clears. Temperature rises as warm air replaces cold air, and the barometer stops falling.



HABAGAT

SOUTHWEST MONSOON

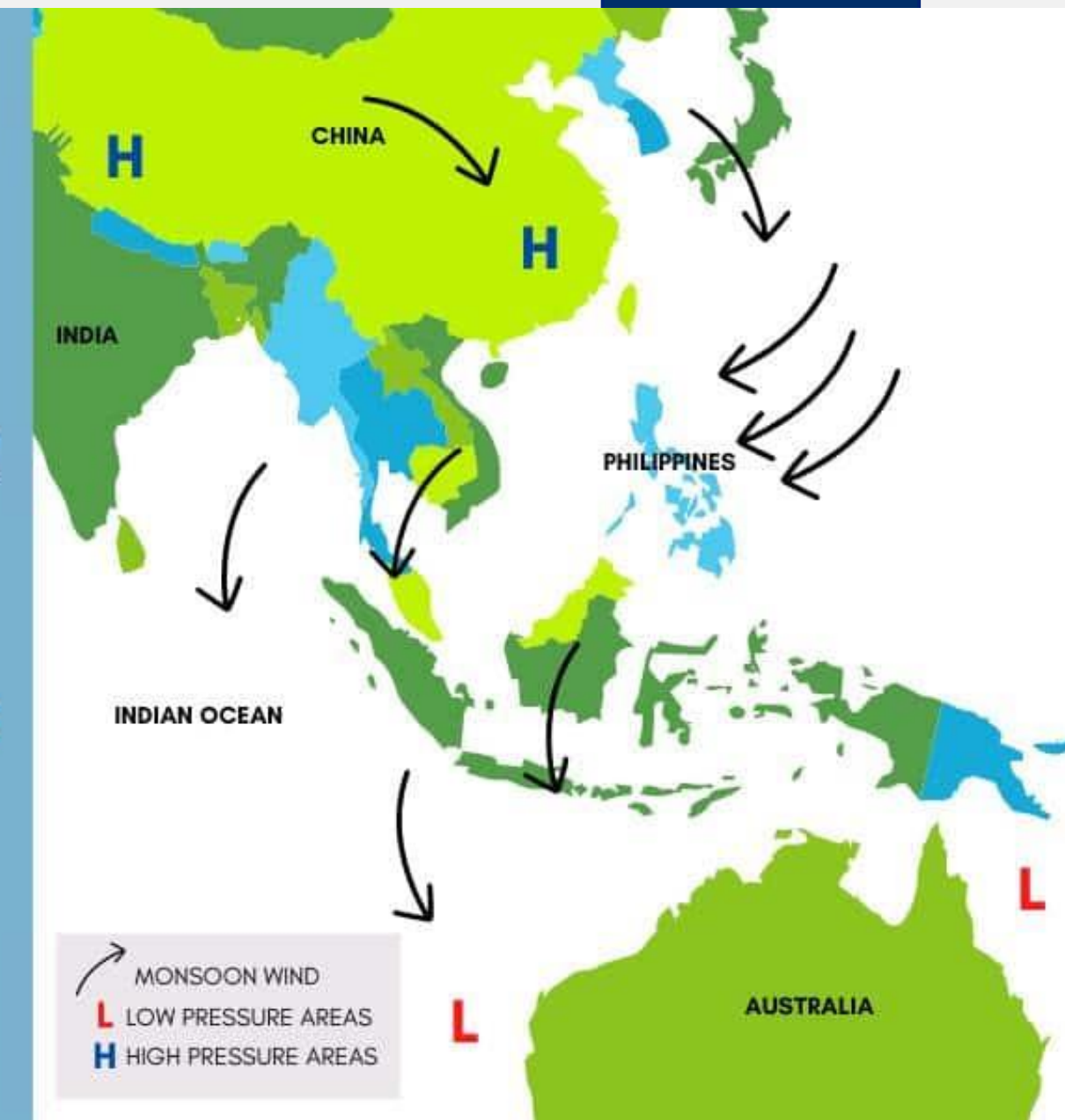
WINDS COMING FROM THE SOUTHWEST
WARM AND MOIST AIR
MODERATE TO HEAVY RAINFALL
JUNE TO OCTOBER



AMIHAN

NORTHEAST MONSOON

WINDS COMING FROM THE NORTHEAST
COOL AND DRY AIR
SLIGHT TO MODERATE RAINFALL
NOVEMBER TO MAY

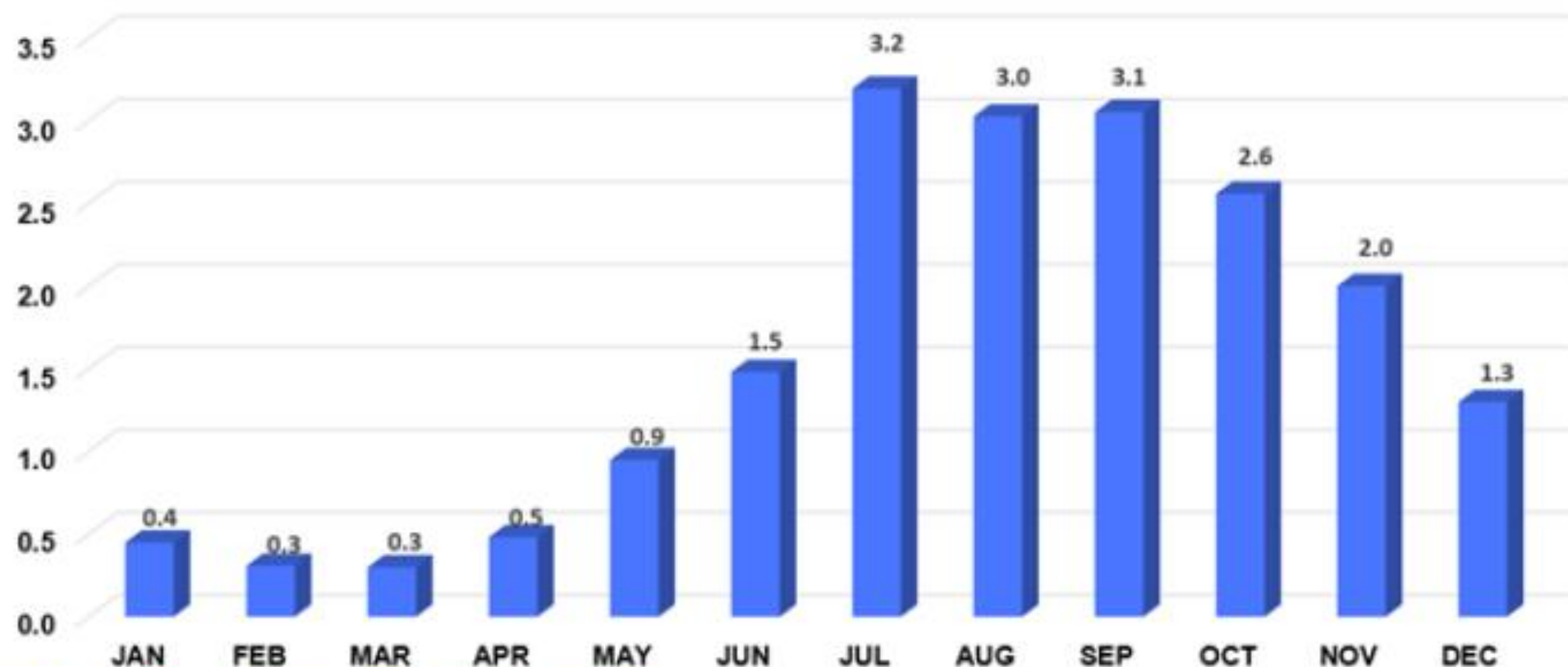




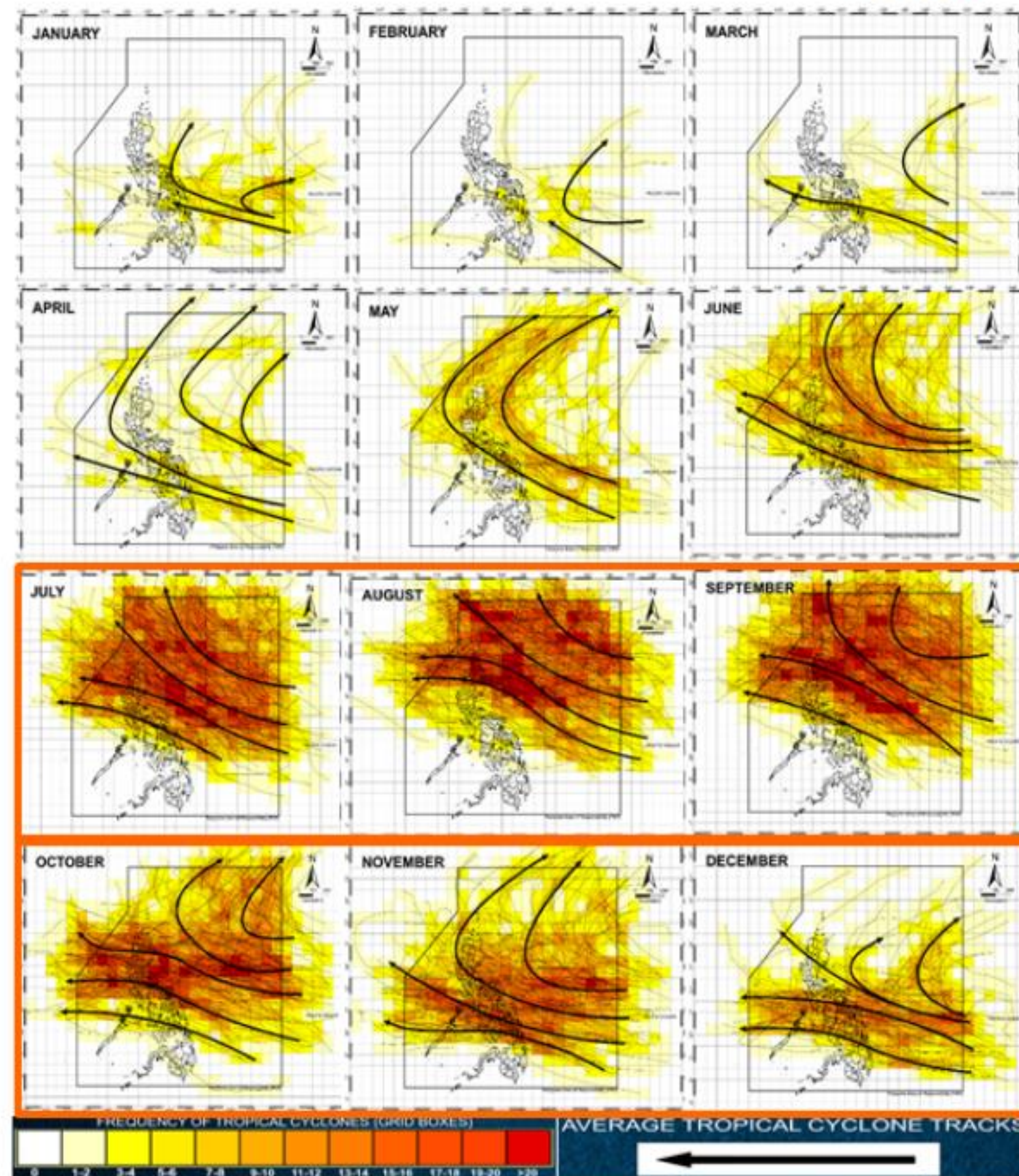
HOW MANY TROPICAL CYCLONES SHOULD YOU EXPECT?

JULY 2025	2 or 3	OCTOBER 2025	2 to 4
AUGUST 2025	2 or 3	NOVEMBER 2025	2 or 3
SEPTEMBER 2025	2 to 4	DECEMBER 2025	1 or 2
11-19 TCs expected			

Average Tropical Cyclone Frequency (1948-2024)



Tropical Cyclone Tracks Climatology with actual tracks from 1948-2015

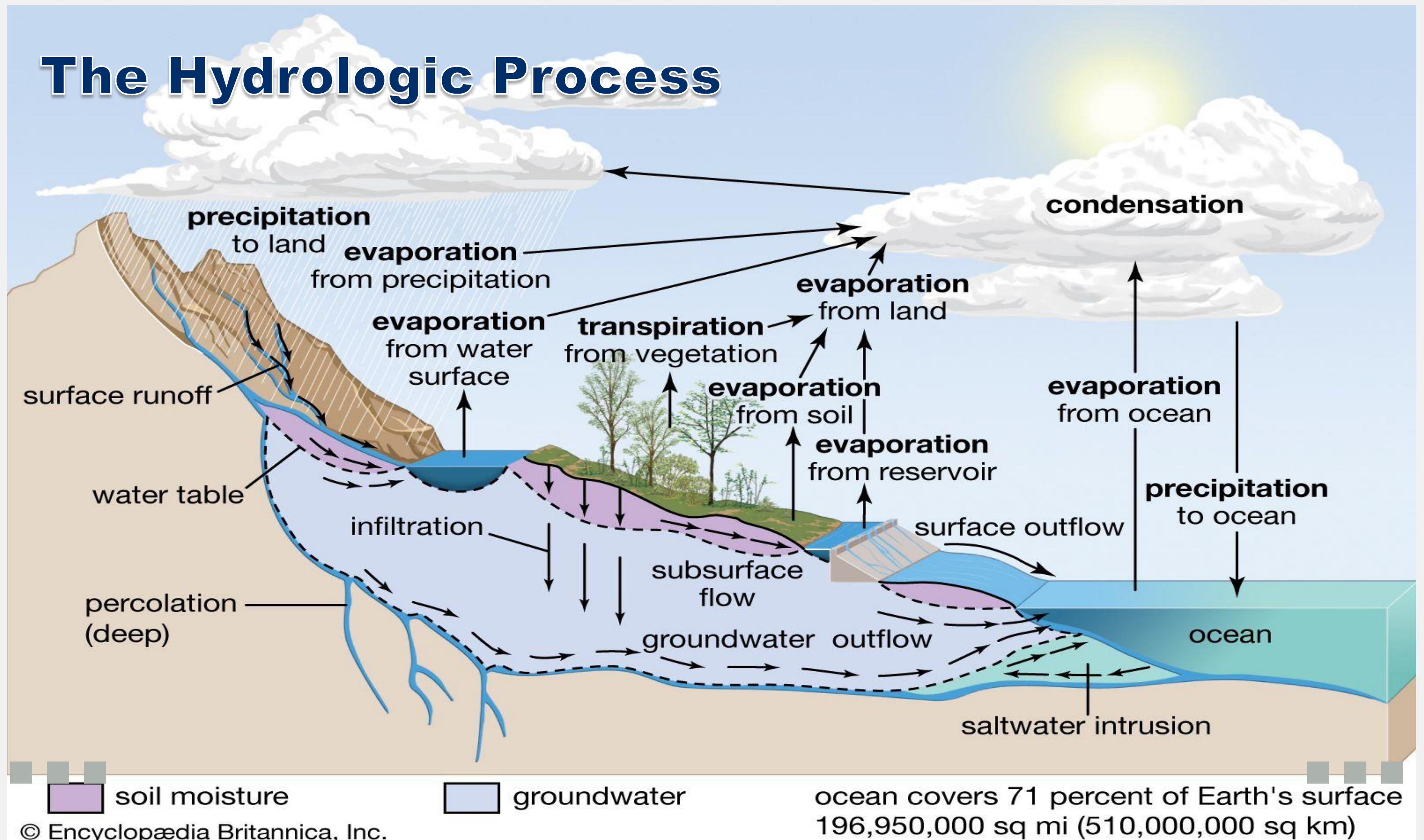




	Maximum wind speed
Tropical depression	30 – 64 kph
Tropical storm	65 – 117 kph
Typhoon	More than 117 kph



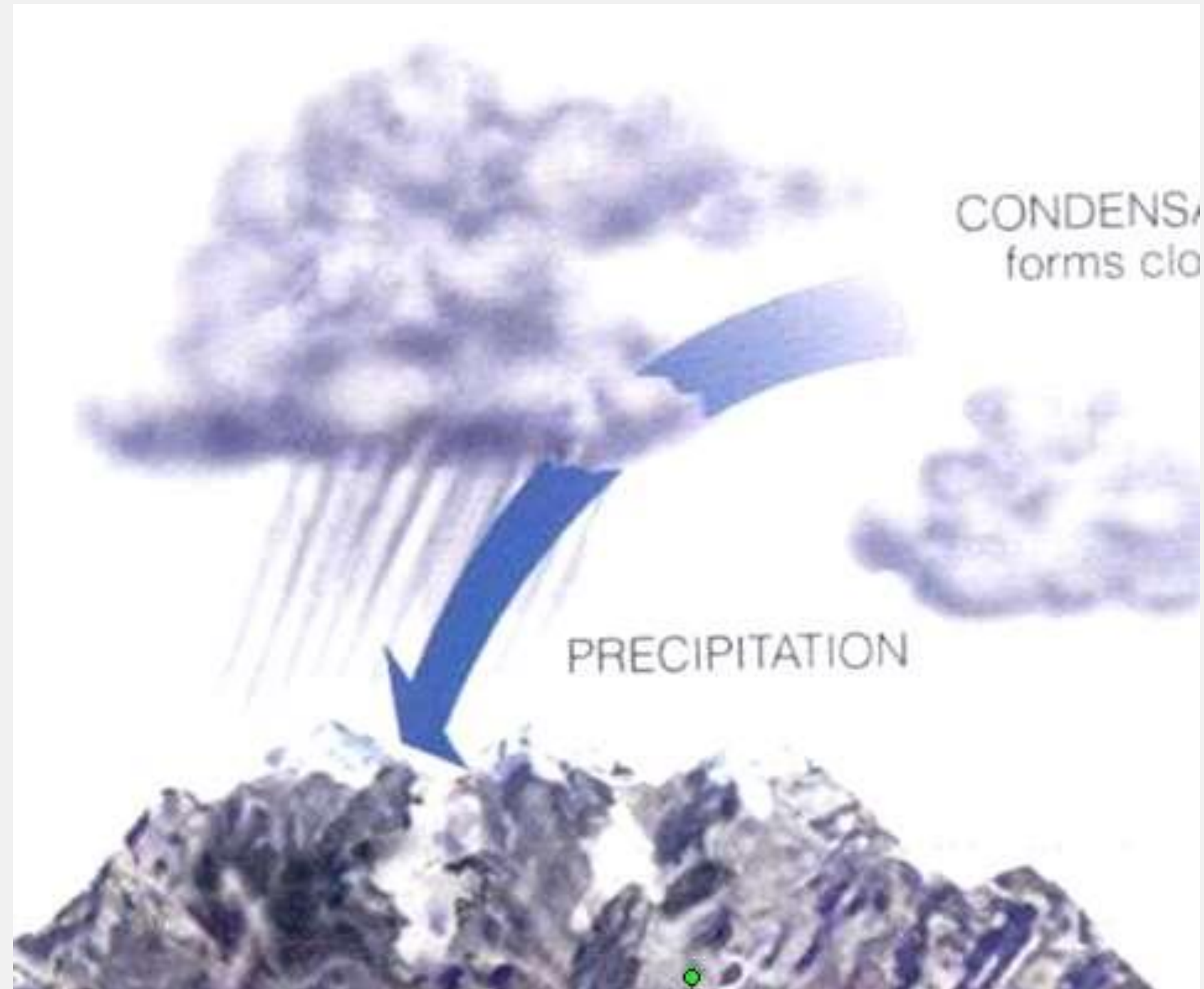
The Hydrologic Process



PRECIPITATION

Precipitation

that part of atmospheric moisture that falls on the earth surface.



Sample Question:

Condensed water vapor that falls to the earth surface. Most occurs as rain, but also includes snow, hail, fog drip, sleet, etc.

- a. Precipitation
- b. Run-off
- c. Water-falls
- d. Irrigation

Formation of Precipitation

PROCESS: condensation or sublimation of atmospheric moisture.

This transformation may be brought about by one or more of the following causes:

- (1) adiabatic cooling,
- (2) mixing of air masses of varying temperature
- (3) radiation cooling, and
- (4) contact cooling.



Types of Precipitation

1) Convective Precipitation – caused by the rising of warmer, lighter air in colder, denser surroundings.

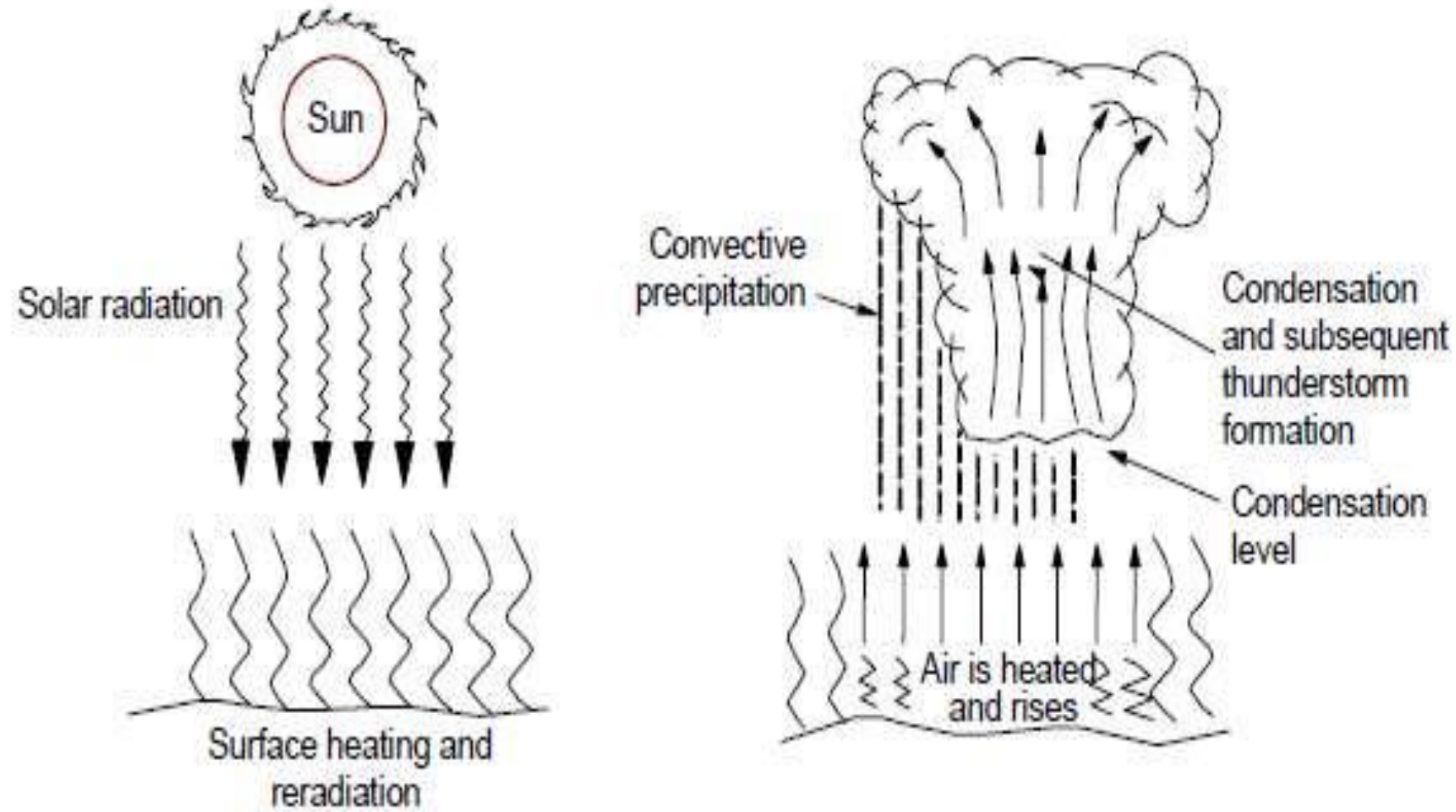
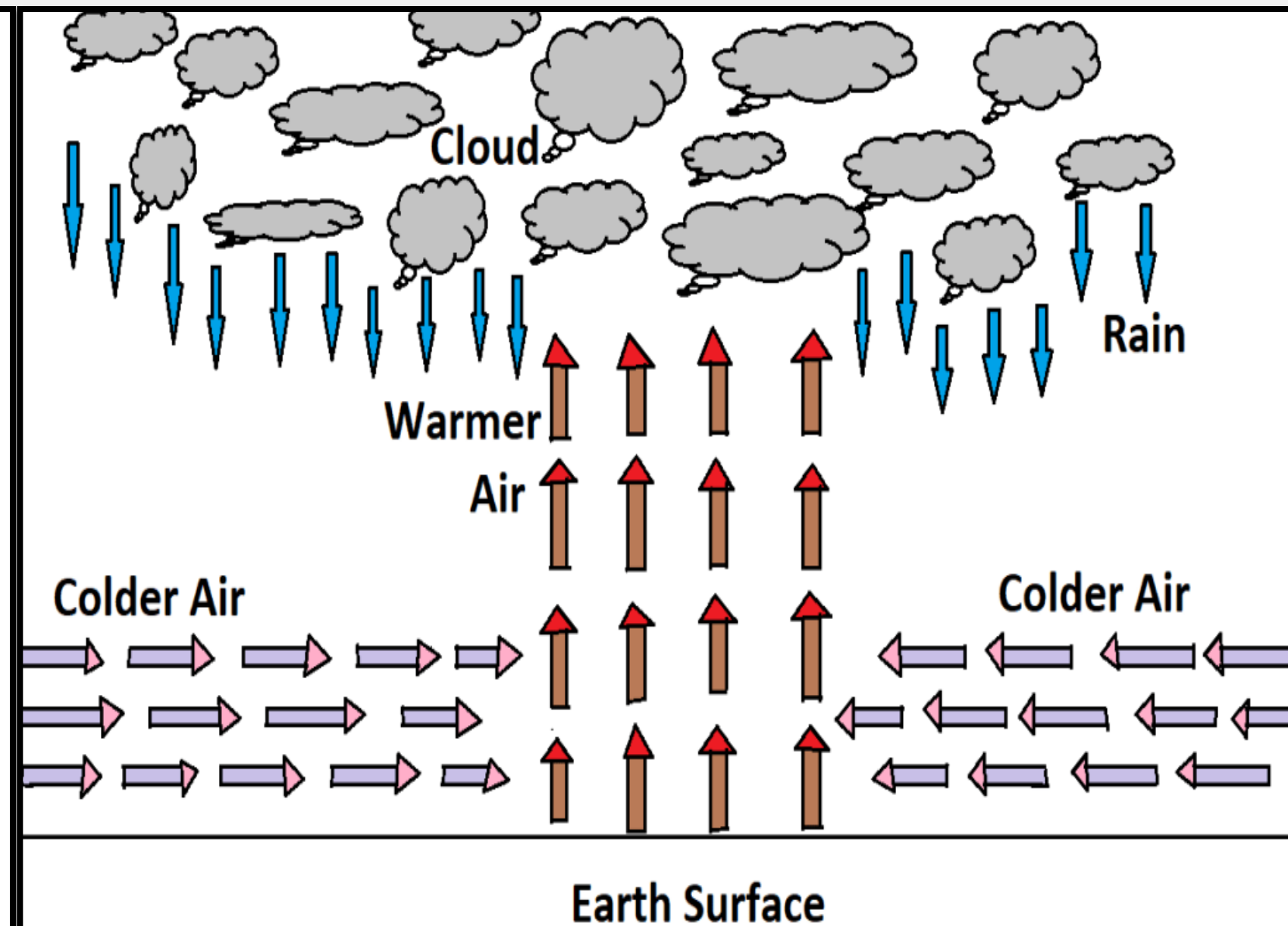
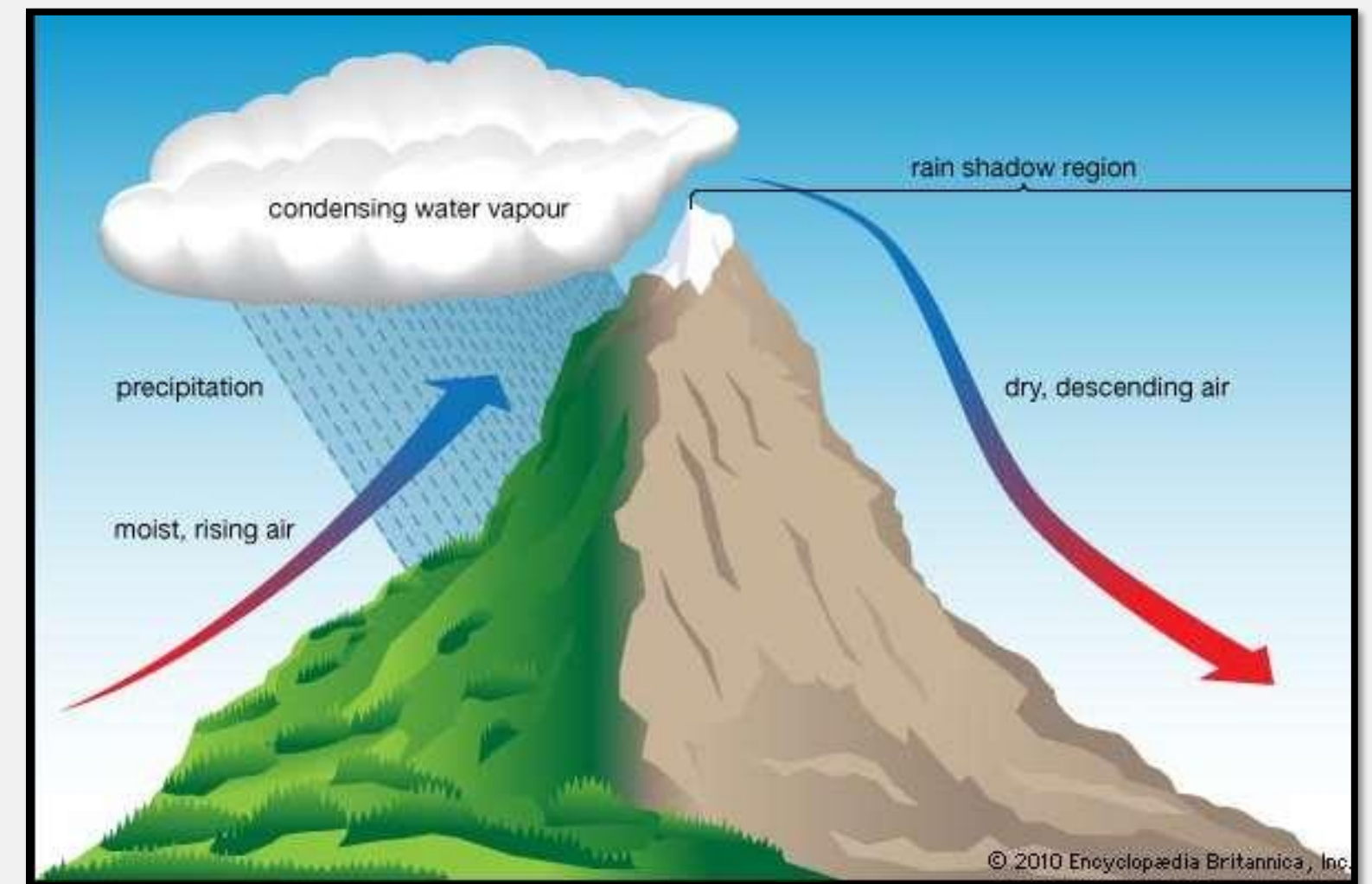
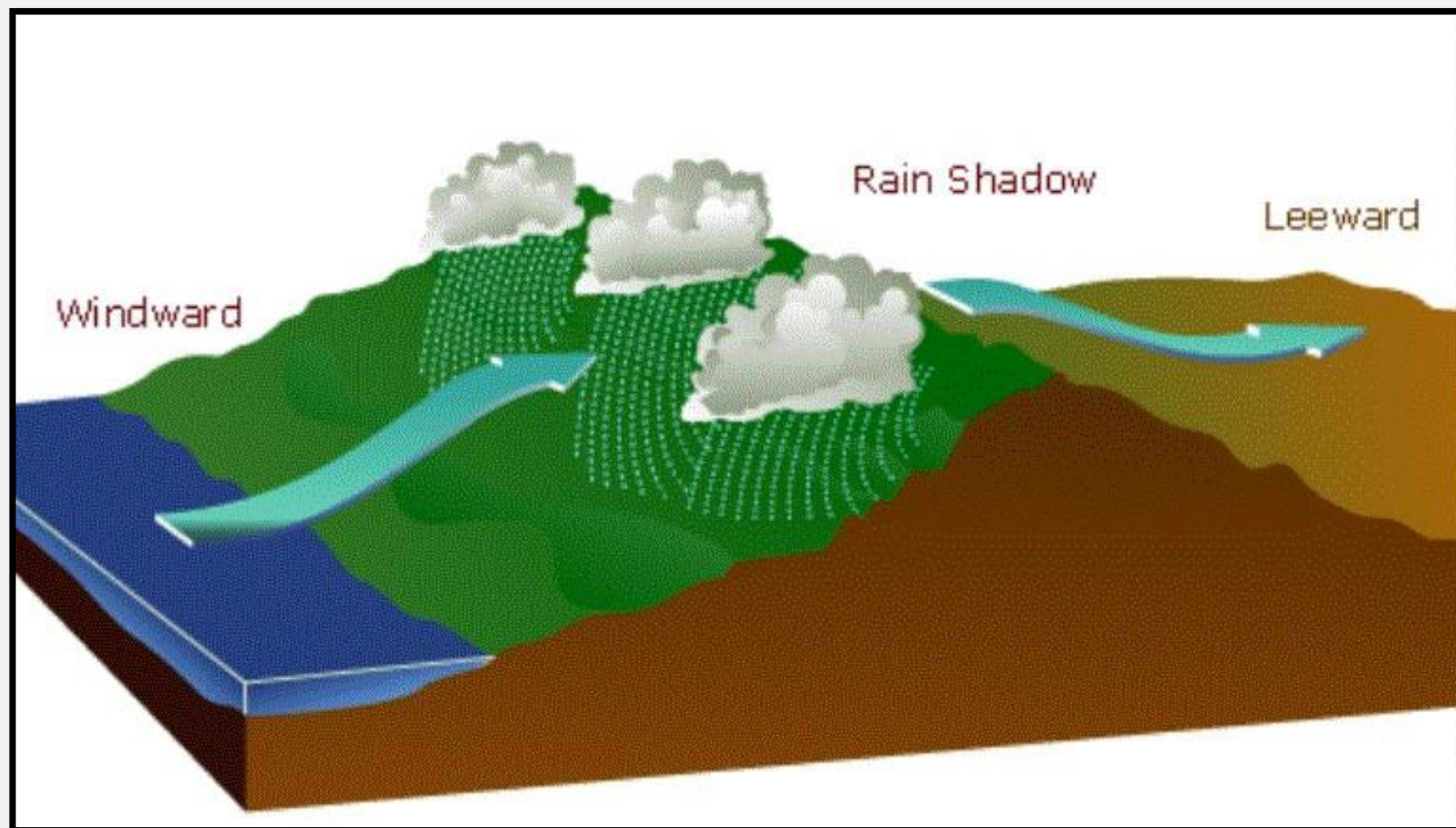


Figure 2.1. Convective storm



Types of Precipitation

2) Orographic Precipitation – results from mechanical lifting of the air mass over mountain barriers.



Types of Precipitation

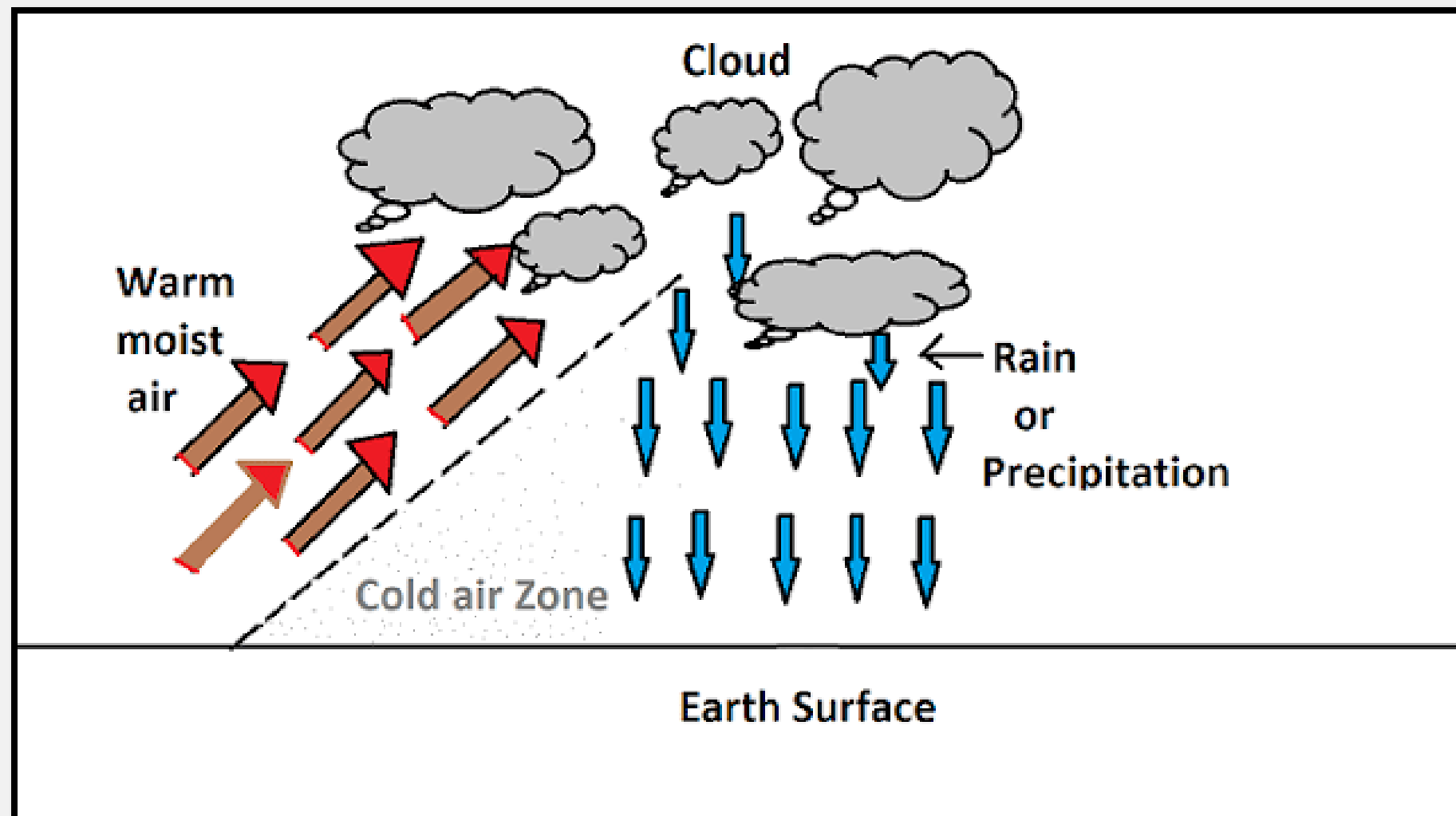
- 3) **Cyclonic Precipitation** – associated with the movement of air masses from high pressure to low-pressure regions. It occurs when air masses converge into a low pressure area resulting to their eventual ascension
 - a) **Frontal** – lifting of warm air over cold air at the contact zone between air masses having different characteristics
 - a) **Non-frontal** – air is lifted through horizontal convergence of the inflow into a low pressure area



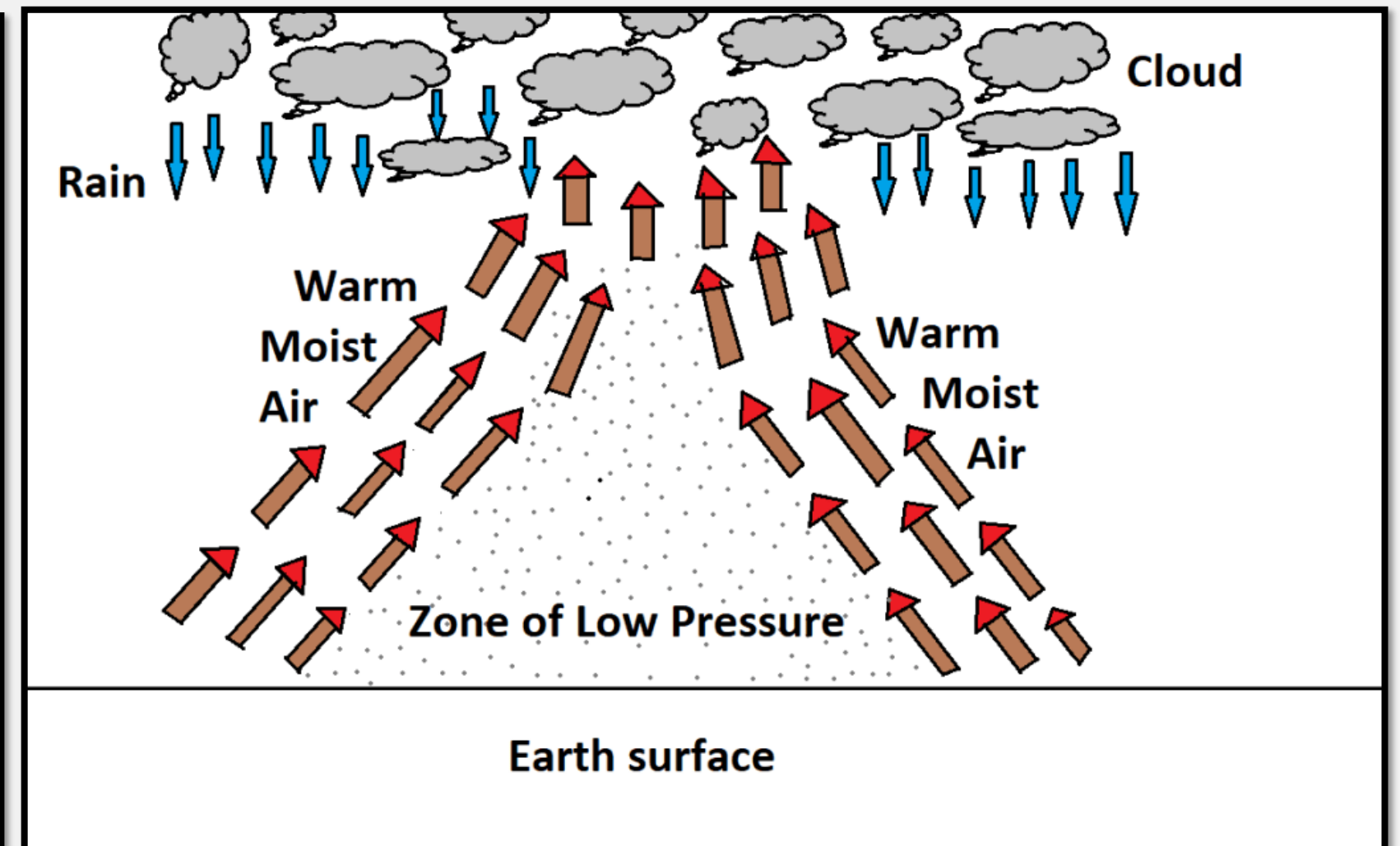
Types of Precipitation

3) Cyclonic Precipitation

a) Frontal



b) Non-frontal



Sample Question:

Type of precipitation which results from mechanical lifting of the air mass over mountain barriers.

- a. Orographic precipitation
- b. Cyclonic precipitation
- c. Convective precipitation
- d. Frontal precipitation

Sample Question:

Type of precipitation associated with the movement of air masses from high pressure regions to low-pressure regions

- a. Orographic precipitation
- b. Convective precipitation
- c. Cyclonic precipitation
- d. Artificial precipitation

C

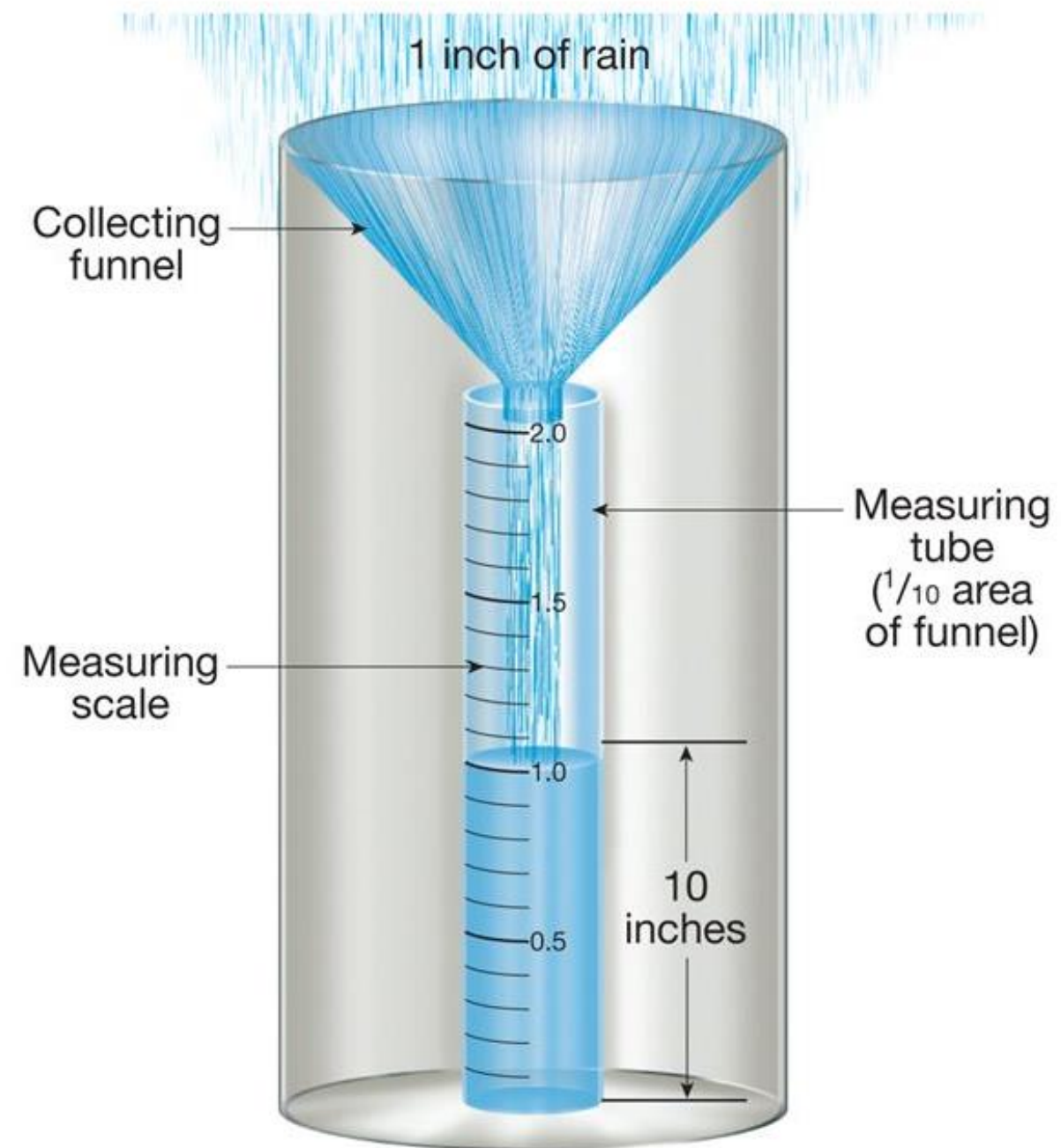
Sample Question:

The type of precipitation that occurs when air masses converge into a low-pressure area is called

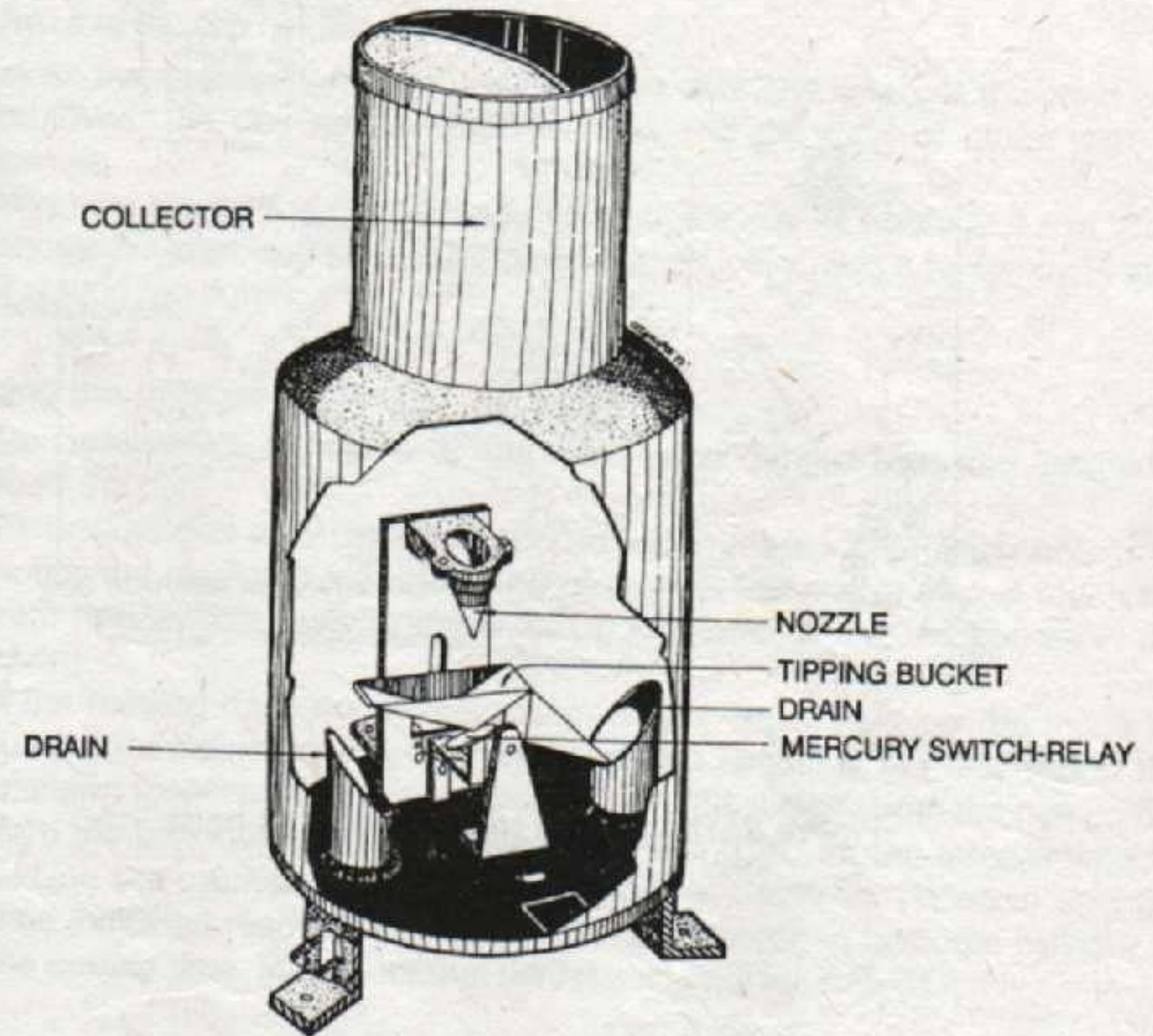
- a. Convective
- b. Orographic
- c. Cyclonic
- d. Frontal

Measurement of Precipitation

The standard rain gauge



Tipping Bucket Rain Gage



Estimation of Average Areal Precipitation

Estimation of Areal Precipitation

- 1. Arithmetic Mean**
- 2. Thiessen Polygon**
- 3. Isohyetal Method**

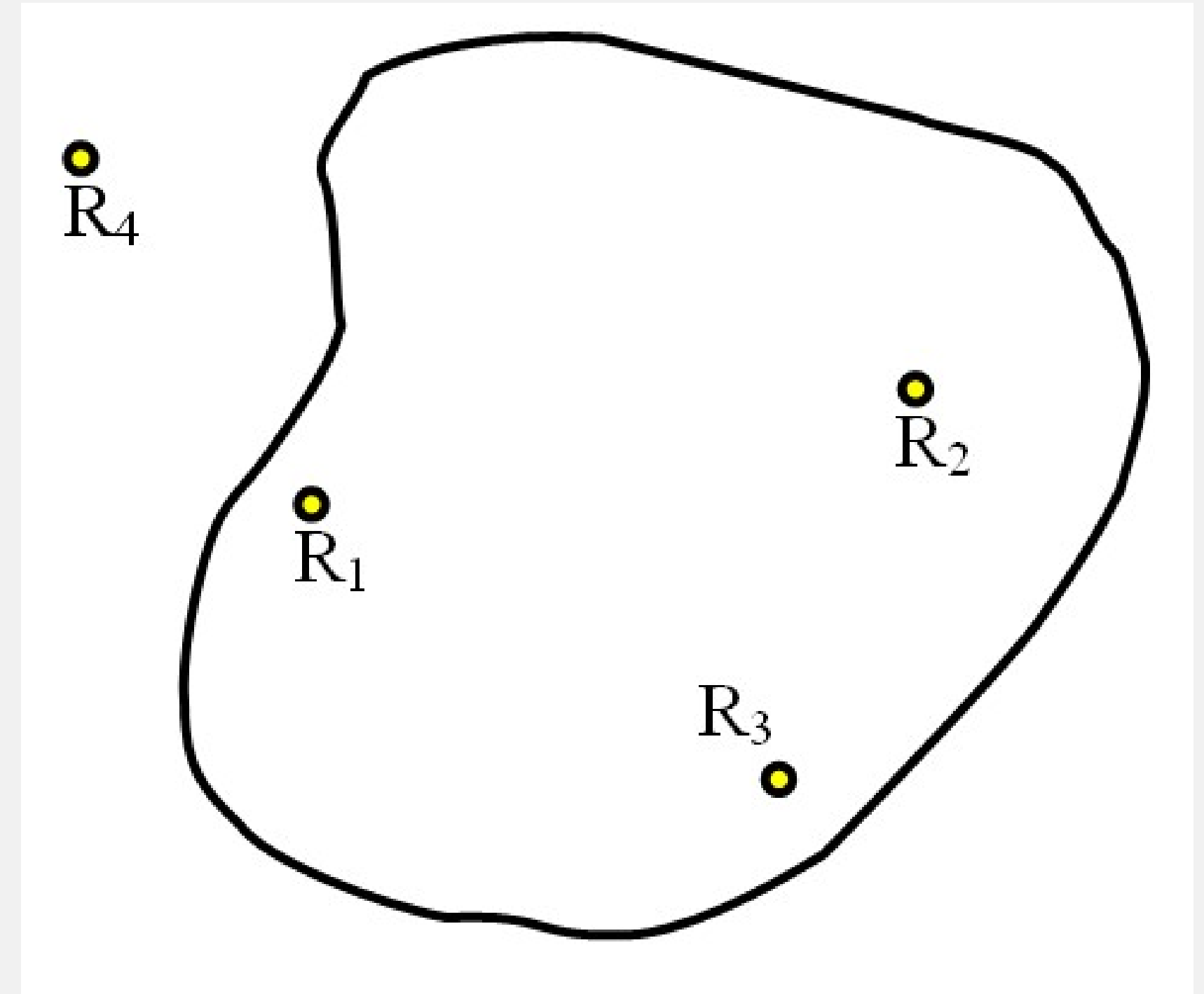


Estimation of Areal Precipitation

1. Arithmetic Mean

this involves the averaging arithmetically all the rainfall depths measured by the rain gages within the area.

$$\bar{R} = \frac{R_1 + R_2 + R_3}{3}$$



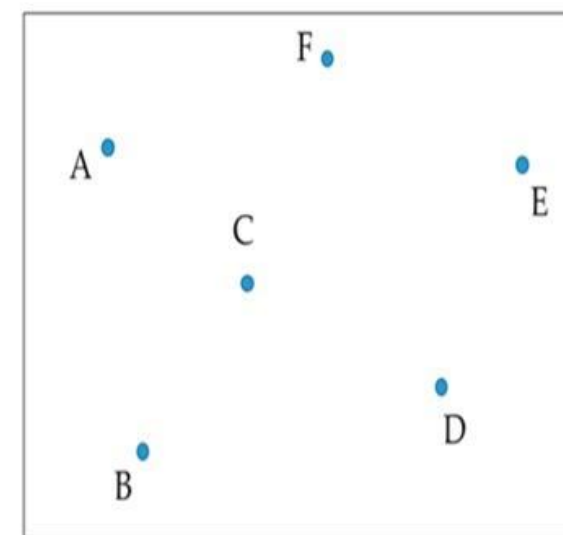
Estimation of Areal Precipitation

2. Thiessen Polygon

$$\bar{R} = \frac{\sum_{i=1}^n a_i R_i}{\sum_{i=1}^n a_i}$$

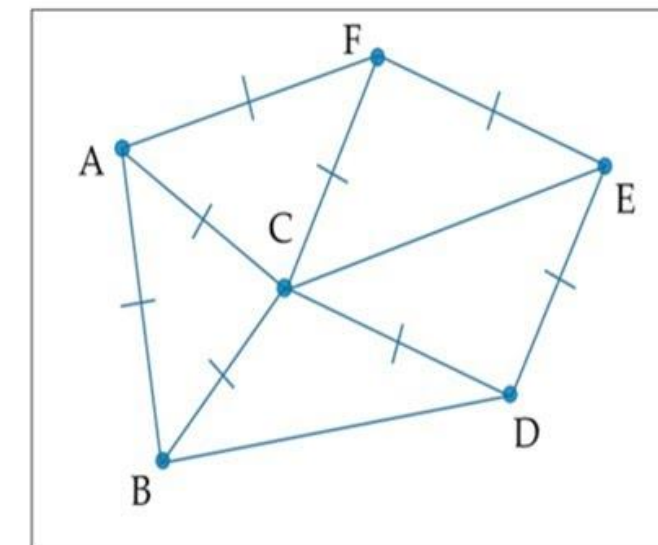
How To Create Thiessen Polygons

Step 1



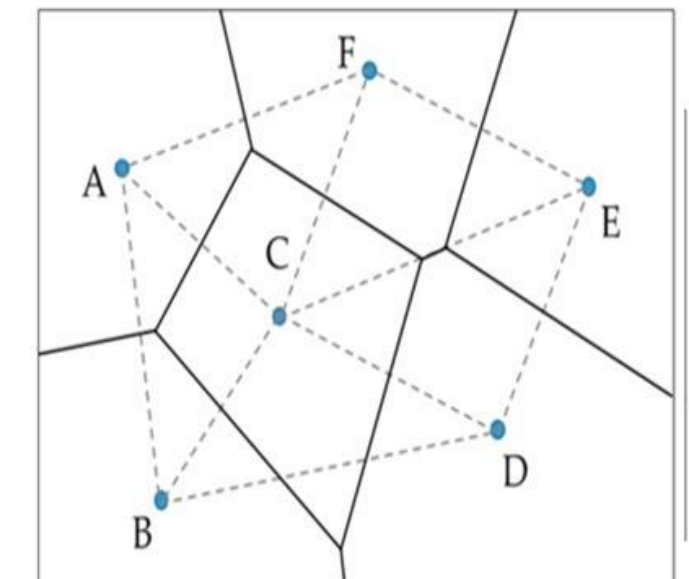
1. Sample POIs

Step 2



2. Breaking points

Step 3



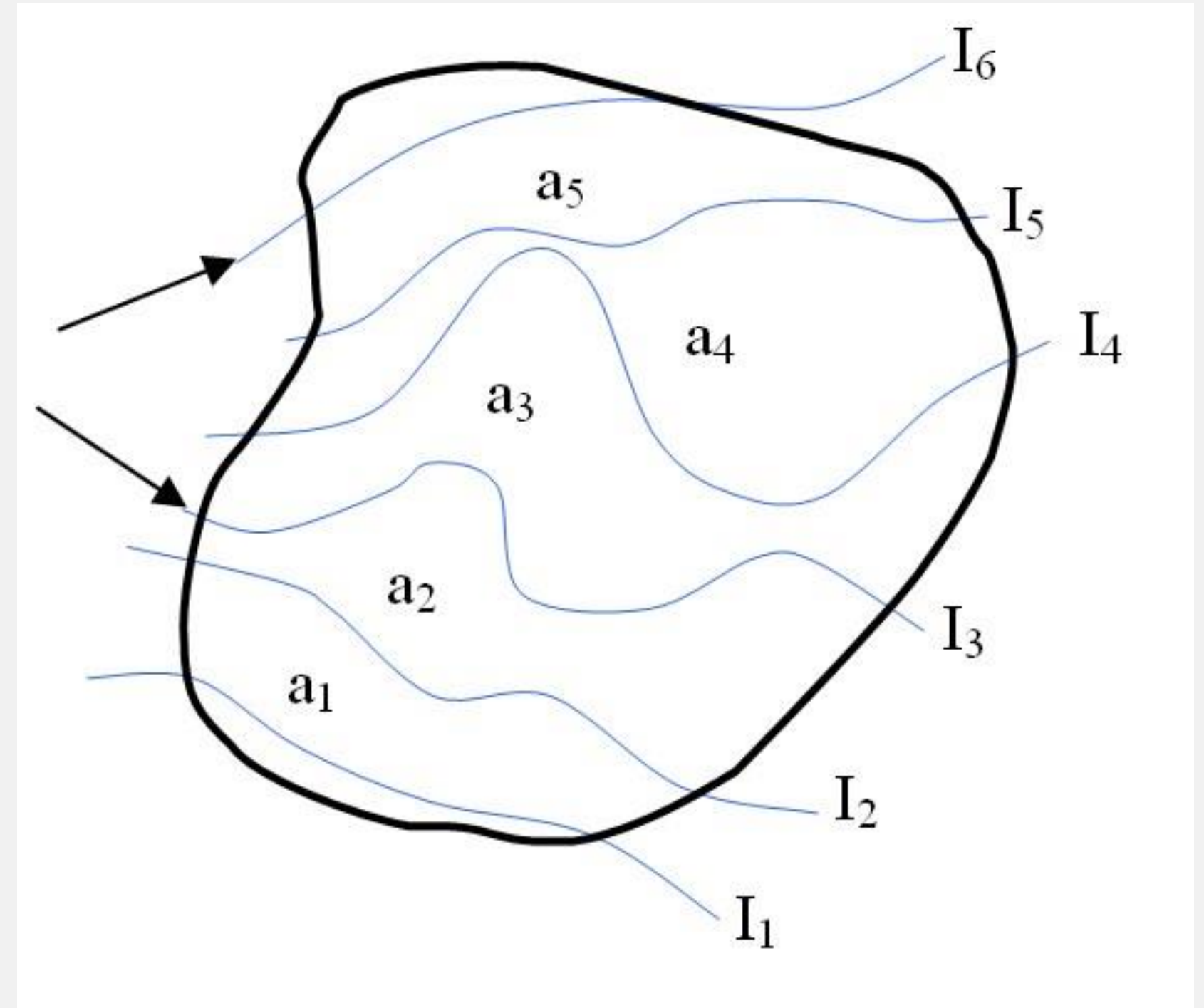
3. Service area generation



Estimation of Areal Precipitation

3. Isohyetal Method

$$\bar{R} = \frac{\sum_{i=1}^n a_i R_i}{\sum_{i=1}^n a_i}$$



Sample Question:

Determine the average monthly rainfall in a 500 ha watershed based on the rainfall data for September of 5 rainfall gaging stations. The areas of the corresponding Thiessen polygons are also indicated.

- a. 1747 mm
- b. 1796 mm
- c. 1832 mm
- d. 1914 mm

Station	Rainfall Total, September (mm)	Thiessen Polygon Area, ha
A	2190	22
B	1731	106
C	2040	75
D	1560	203
E	1974	300
F	1235	45

B

Sample Question:

A watershed with an area of 24 mi² was subdivided by Thiessen polygons with area in mi² and mean rainfall in inches provided: (5.9, 0.6), (13, 1.7), and (4.6, 5.5). Determine the average areal precipitation.

- a. 1.85 in
- b. 2.22 in
- c. 4.10 in
- d. 5.10 in

B

Sample Question:

The Thiessen weights of the 4 raingauges A, B, C and D covering a river basin are 0.15, 0.25, 0.30 and 0.30 respectively. If the average depth for the basin is 5 cm, and the rainfalls recorded at B, C and D are 5 cm, 4 cm, and 5 cm respectively, what is the rainfall (cm) at A?

7 cm

Sample Question:

For determination of average annual precipitation in a catchment basin, the best method is

- a. Arithmetical method
- b. Thiessen's mean method
- c. Isohyetal method
- d. None of these

C

Sample Question:

The isohyets drawn for a storm which occurred over a drainage basin of area 950 km^2 yielded the following information in the form of (isohyet interval in mm, area between isohytes in km^2): (85-75, 125), (75-65, 236), (65-55, 264), (55-45, 175) and (45-35, 150). Determine the average depth (mm) of rainfall over the basin.

60.12 mm

Rainfall Adjustments

Rainfall Adjustments



a) Estimation of Missing Rainfall Data

- Arithmetic Mean Method
- Normal Ratio Method

b) Double Mass Analysis



Estimation of Missing Rainfall Data

Arithmetic Mean Method

$$P_x = \frac{1}{m} (P_1 + P_2 + \cdots + P_m)$$

P_x = the missing precipitation at station X

$P_1, P_2, \dots, P_m$ = rainfalls at the m surrounding gages

Assumption: The normal annual precipitations at adjacent stations are **within 10%** of the normal rainfall of the station in question.

Estimation of Missing Rainfall Data

Normal Ratio Method

$$P_x = \frac{N_x}{m} \left(\frac{P_1}{N_1} + \frac{P_2}{N_2} + \dots + \frac{P_m}{N_m} \right)$$

P_x = the missing precipitation at station X

N_x = normal annual rainfall at station X

$P_1, P_2, \dots, P_m$ = rainfalls at the m surrounding gages

$N_1, N_2, \dots, N_m$ = normal annual rainfalls at the m surrounding gages

Assumption: The normal annual precipitations at the surrounding gages differ from the normal rainfall of the station in question by **more than 10%**.

Sample Question:

Precipitation station "X" was inoperative for part of a month during which a storm occurred. The storm totals at three surrounding stations A, B and C were respectively 10.7, 8.9 and 12.2 cm. The normal annual precipitation amounts at stations X, A, B and C are respectively 97.8, 112, 93.5 and 119.9 cm. Estimate the storm precipitation for station 'X'.

9.5 cm

Sample Question:

Find out the missing storm precipitation of station 'C' given in the following table:

Station	A	B	C	D	E
Storm precipitation (cm)	9.7	8.3	----	11.7	8.0
Normal Annual precipitation (cm)	100.3	109.5	93.5	125.7	117.5

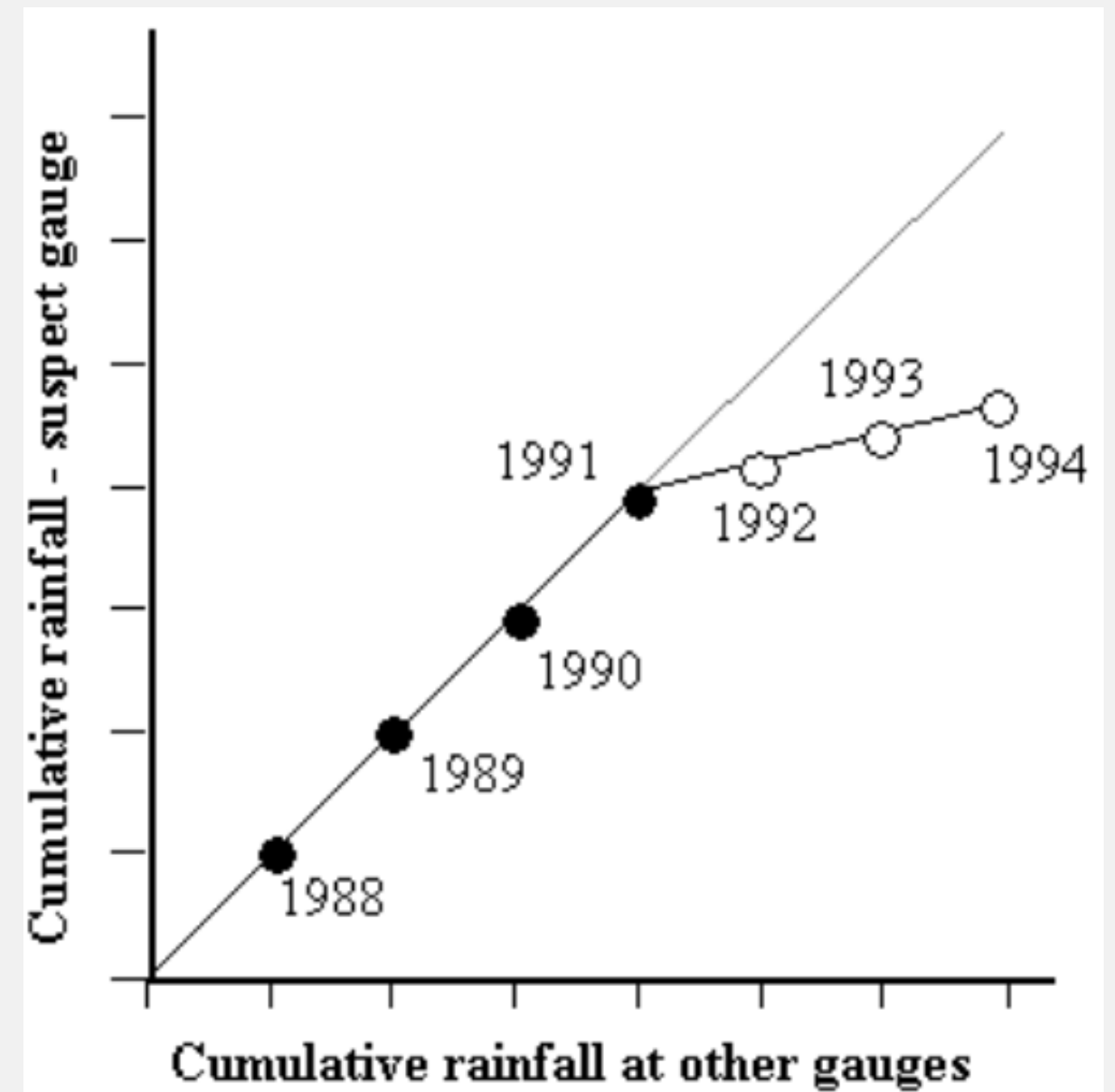
7.8 cm

Double Mass Analysis

double-mass curve is used to check the consistency of a rain gauge record:

- compute cumulative rainfall amounts for suspect gauge and check gauges
- plot cumulative rainfall amounts against each other (divergence from a straight line indicates error)
- multiplying erroneous data after change by a correction factor k where

$$k = \frac{\text{gradient of line before change}}{\text{gradient of line after change}}$$



HYDROLOGIC FREQUENCY ANALYSIS

September 23, 2:40 AM (Today)



25°C

RealFeel® 30°C

Partly cloudy

Precipitation: **25%**

Amount: **0 mm**

Wind: **S 1.7 km/h**

Humidity: **95%**

Sunrise: **5:27 AM**

Sunset: **5:34 PM**

Next 7 days

Next 24 hours

See more on AccuWeather >

Today



37° 23°

 25%

Mon 23



33° 23°

 89%

Tue 24



33° 23°

 45%

Wed 25



34° 23°

 84%

Thu 26



33° 24°

 59%

Fri 27



33° 23°

 41%



Characteristics of a Rainstorm

- **intensity** (mm/hr)
- **duration** (min, hours, days)
- **areal extent** (m^2 , ha)
area over which it is distributed
- **frequency**
once in 5, 10, 15, 20, 25, 50 or 100 years



Basic Terms in Statistical Hydrology

- time period of interest (N)
- probability of occurrence (P)
- recurrence interval or return period (T)
the average number of years during which a storm of given magnitude (maximum depth or intensity) may be expected to occur once, *i.e.*, may be equaled or exceeded.

$$P = \frac{1}{T}$$

$$P = 1 - \frac{1}{T}$$

**Probability that a T-year storm may
NOT occur in any series of N years**

$$P_{non-exceedance} = \left(1 - \frac{1}{T}\right)^N$$



**Probability that a T-year storm may
occur in any series of N years**

$$P_{\text{exceedance}} = 1 - \left(1 - \frac{1}{T}\right)^N$$



Sample Question:

A road culvert is designed for a 10-year flood with a magnitude of $1.5 \text{ m}^3/\text{s}$. Solve for the ff:

- a) Probability of exceedance in any given year
- b) Probability of non-exceedance in any given year
- c) Probability of exceedance in 10 years
- d) Probability of non-exceedance in 10 years

10%, 90%, 65%, 35%

Sample Question:

A flood way has been designed for a discharge with a return period of 75 years. The risk that the structure will experience the design flood one or more times in 20 years is

- a. 1.5%
- b. 23.5 %
- c. 98.5%
- d. 76.5%

B

Sample Question:

A flood flow of $560 \text{ m}^3/\text{s}$ was determined to have a 5% probability of occurrence. What is its return period?

- a. 2 years
- b. 20 years
- c. 5 years
- d. 50 years

B

Sample Question:

A flood flow of $560 \text{ m}^3/\text{s}$ was determined to have a 20% probability of occurrence. What is its return period?

- a. 2 years
- b. 20 years
- c. 5 years
- d. 50 years

Hydrologic Frequency Analysis

It is the probabilistic analysis of hydrologic processes.



Hydrologic Frequency Analysis

A hydrologic process is any hydrologic phenomenon which undergoes continuous changes with time.

Deterministic Process

one in which a definite law of certainty exists

Probabilistic Process

governed by chance; time series INDEPENDENT

Stochastic Process

governed by chance; time DEPENDENT



Sample Question:

In frequency analysis, most hydrologic processes, more or less, follow this process. This process is governed by chance and time dependent. What type of process is this?

- a. Deterministic Process
- b. Probabilistic Process
- c. Stochastic Process
- d. Hydrolistic Process



Common Probability Distributions Used/ Probability Density Curve

1. **Normal Distribution** – data is more or less normally distributed
2. **Log-Normal Distribution** – used when there is skewness in the data distribution
3. **Pearson Type III Distribution** - considers further the skewness of the logarithmically transformed hydrologic data; preferred for flood flow frequency; log-normal distribution is a special case of this method
4. **Gamma Density Function** – so far gives the best fit for most stations for 1-week, 2-week, 3-week and monthly rainfall totals of the country



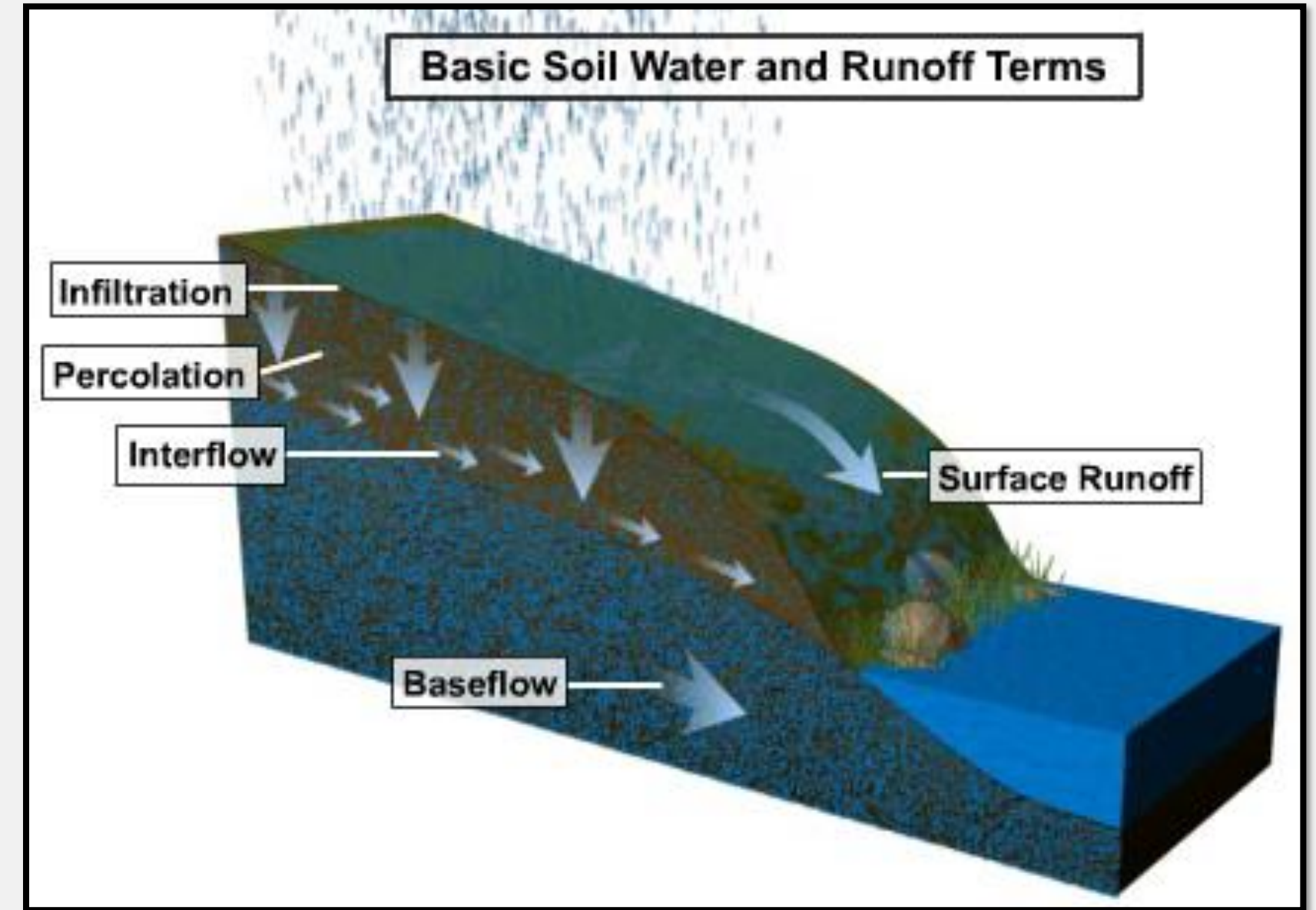
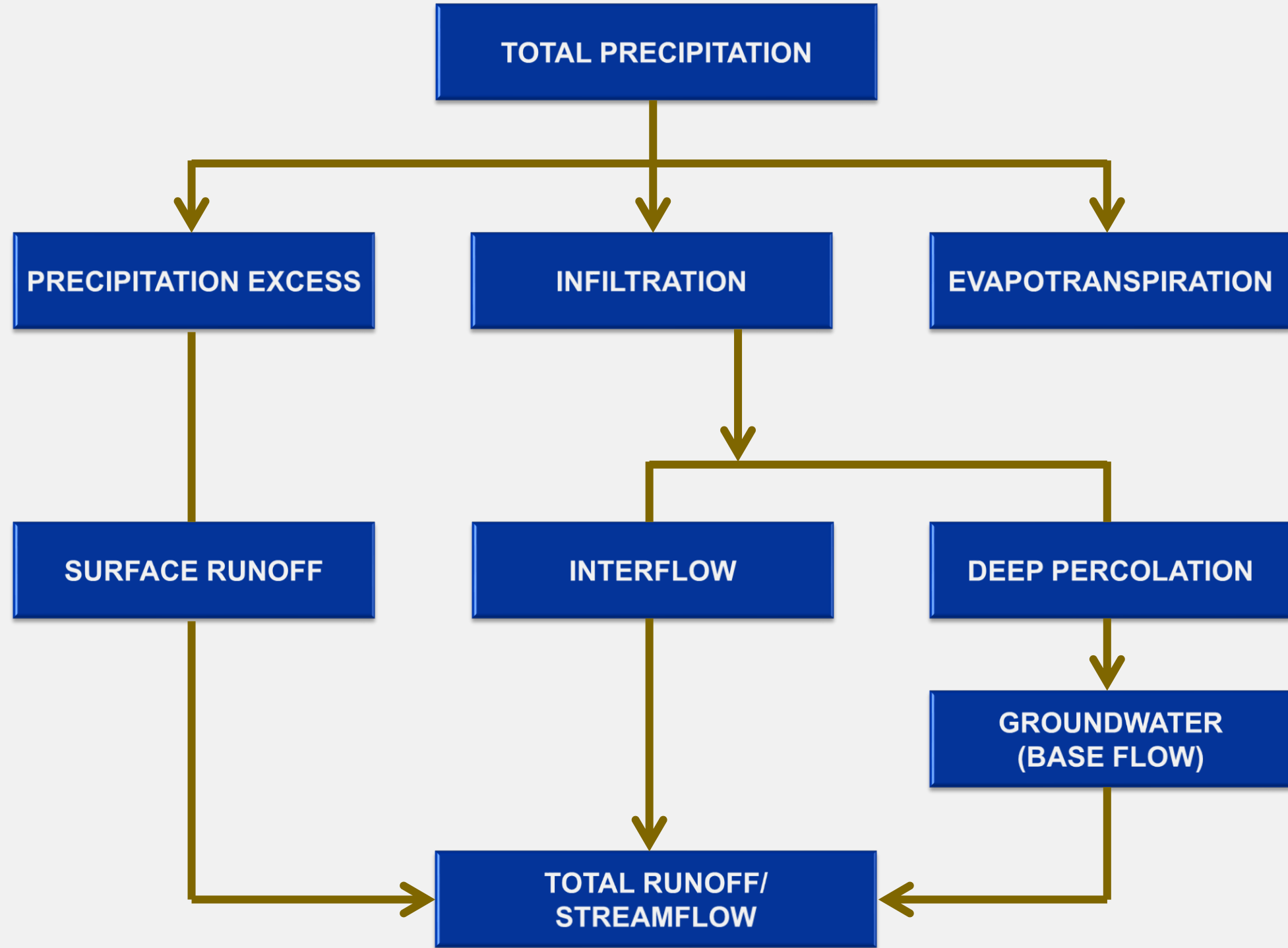
RUNOFF AND STREAMFLOW

RUNOFF

Runoff Components

- Surface Runoff (Overland Flow)
- Subsurface Runoff (Interflow)
- Groundwater Runoff (Base Flow)



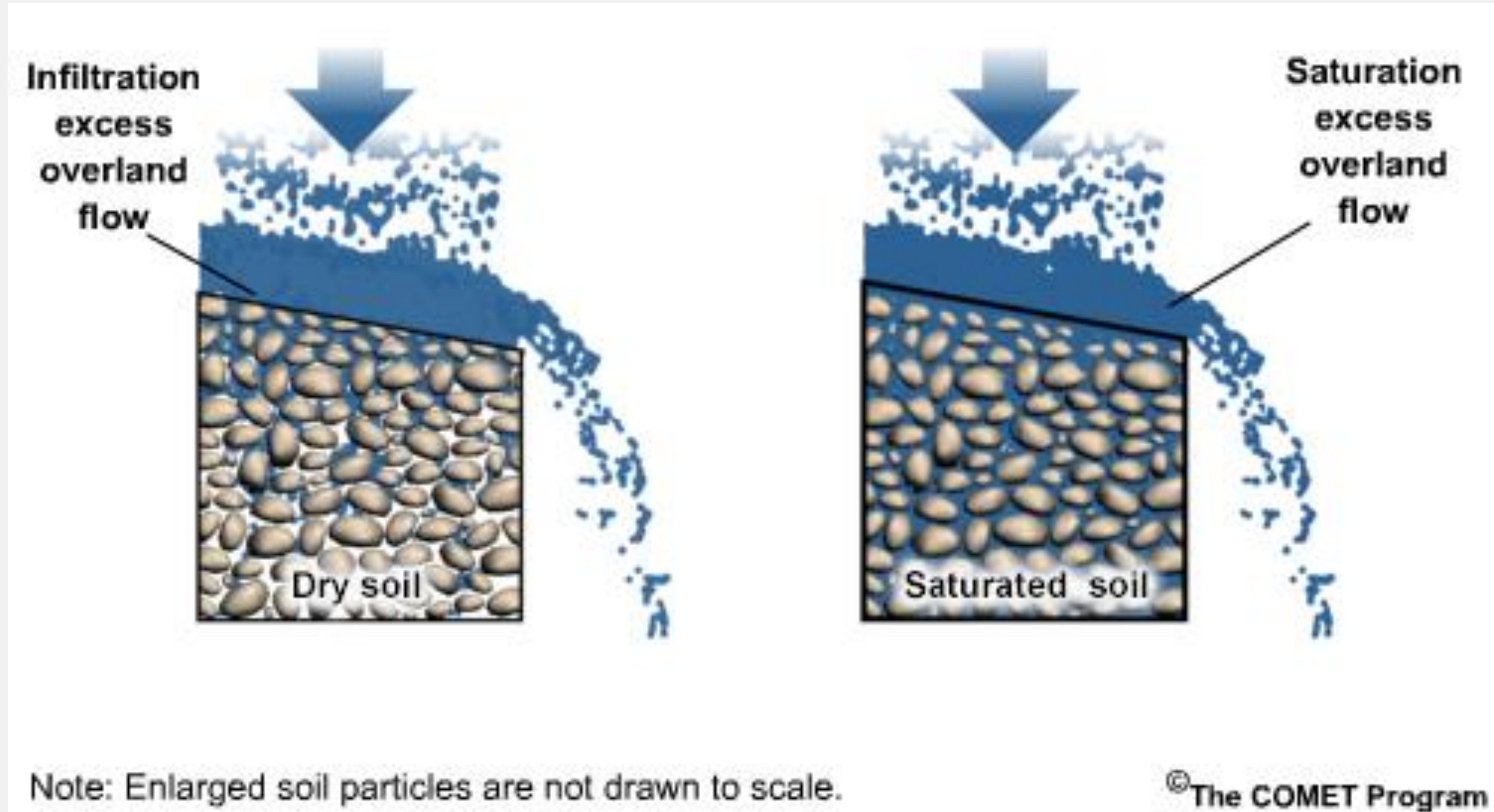


Runoff Components

1. **Surface Runoff** – that which travels over the ground surface and through the channels to reach the basin outlet.



Types of Surface Runoff



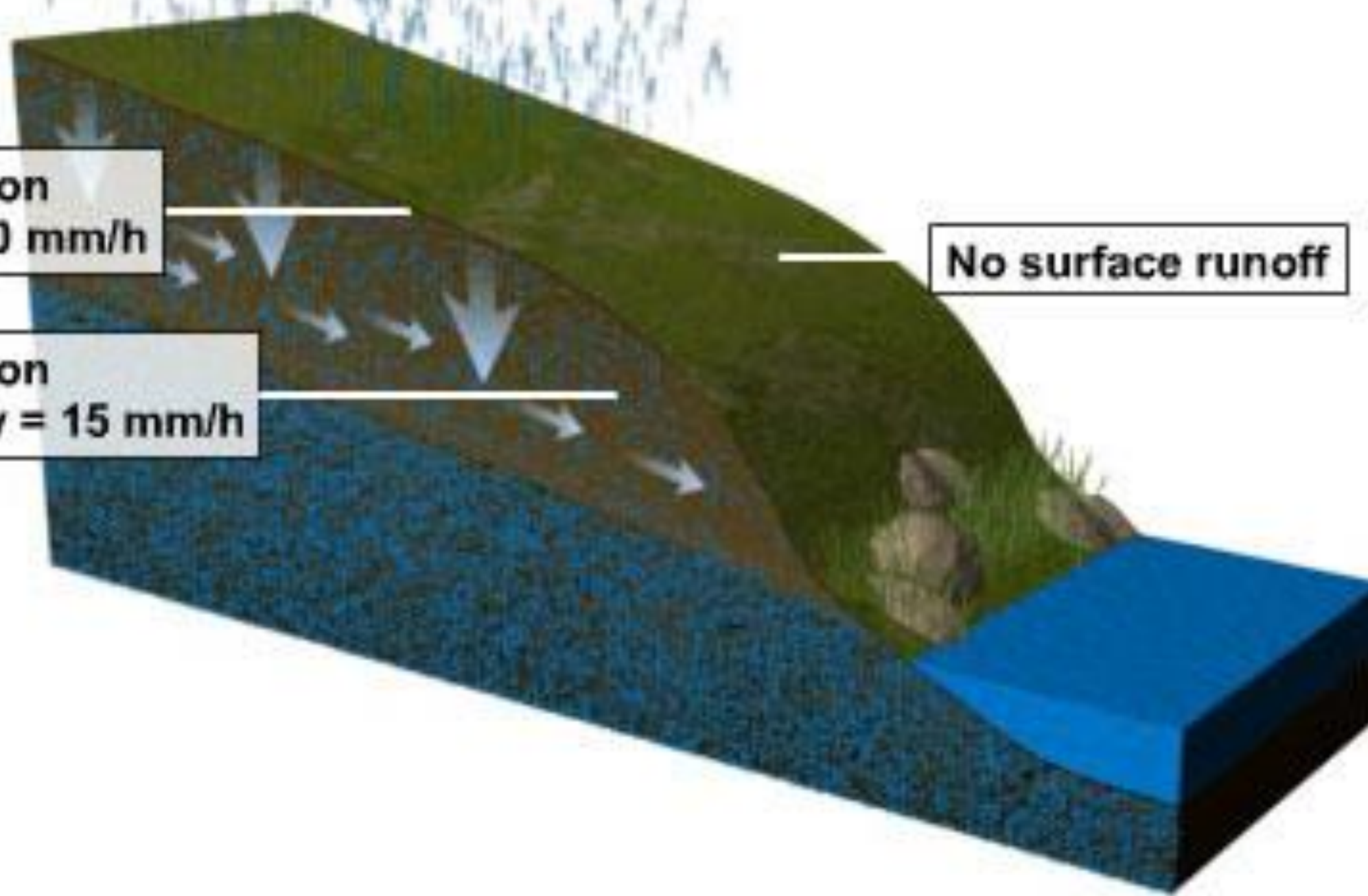
Relationship between Rainfall, Infiltration, and Runoff (Rainfall Rate $\leq$ Infiltration Capacity)

Rainfall rate = 10 mm/h

Infiltration
rate = 10 mm/h

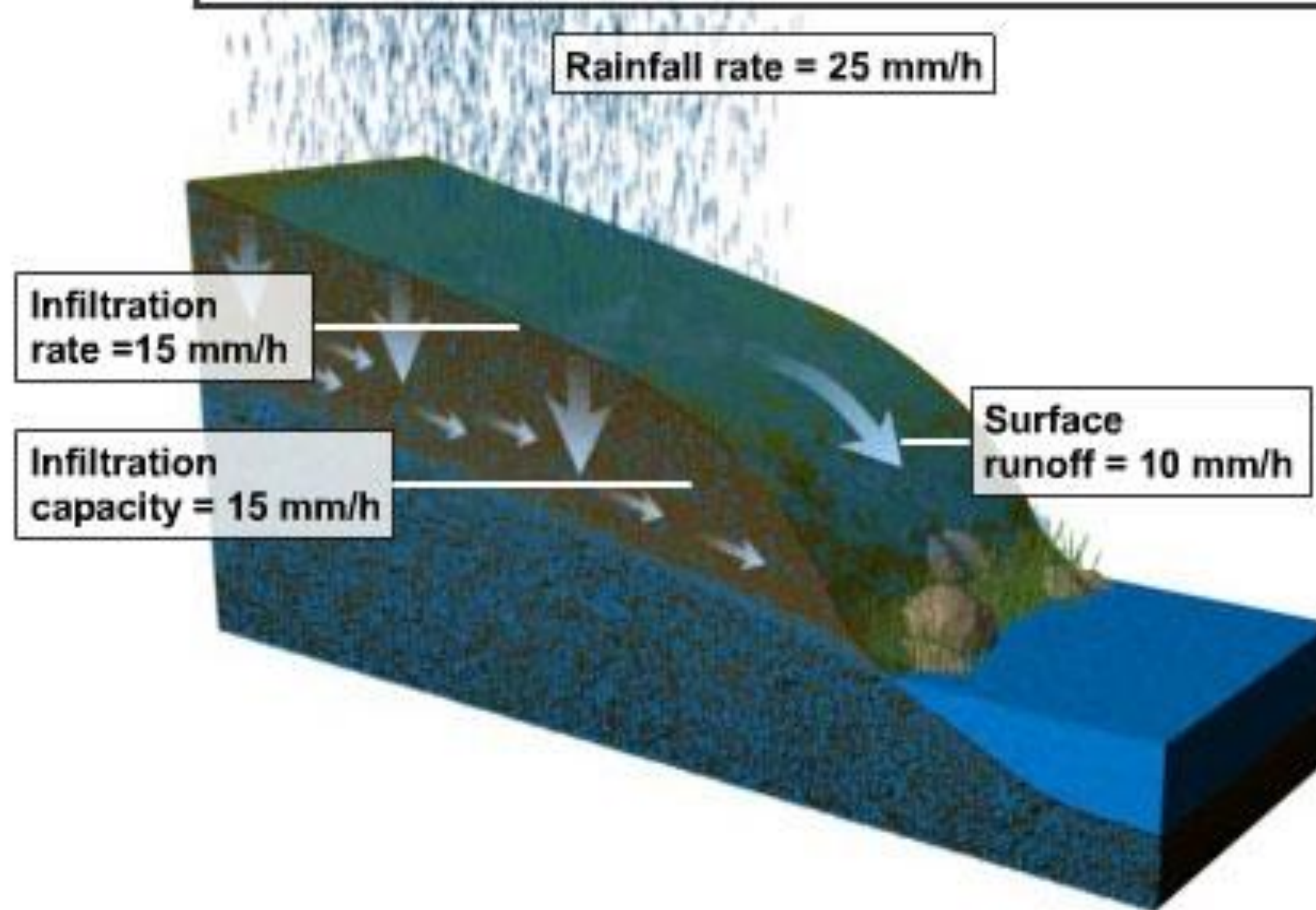
Infiltration
capacity = 15 mm/h

No surface runoff



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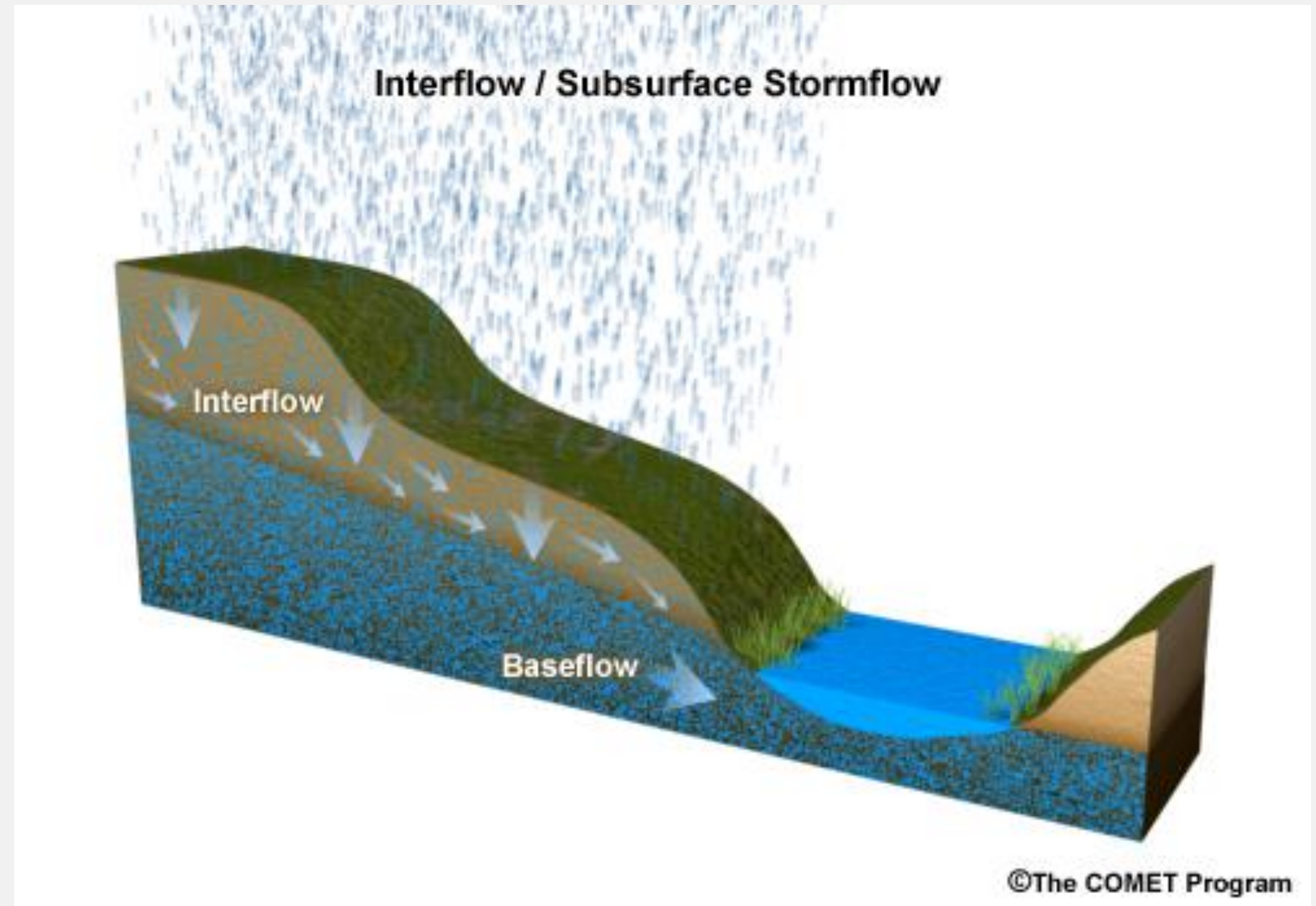
Relationship between Rainfall, Infiltration, and Runoff (Rainfall Rate > Infiltration Capacity)



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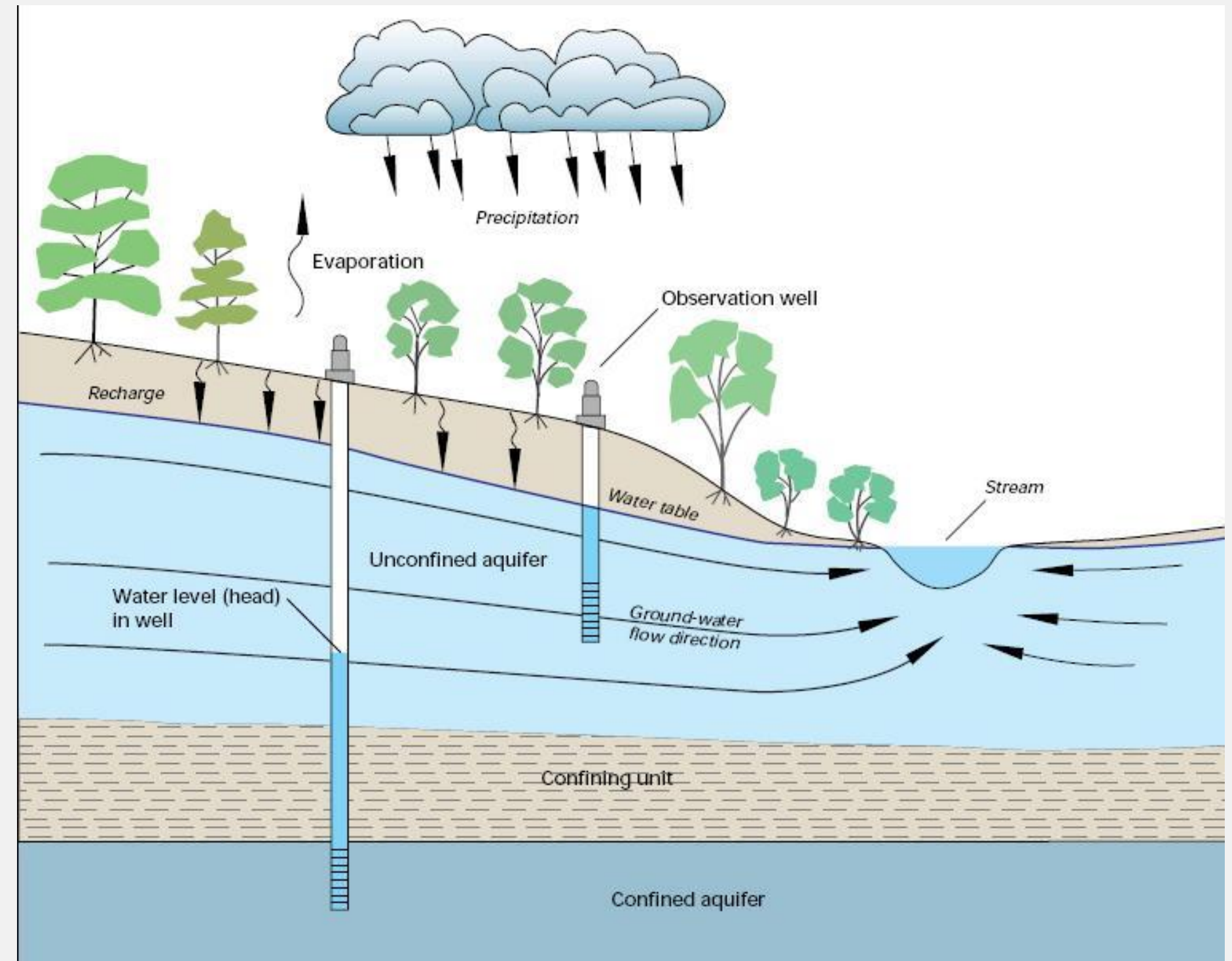
Runoff Components

2. **Interflow** – that which infiltrates into the soil surface and moves laterally through the upper soil horizons towards streams as perched groundwater above major groundwater level

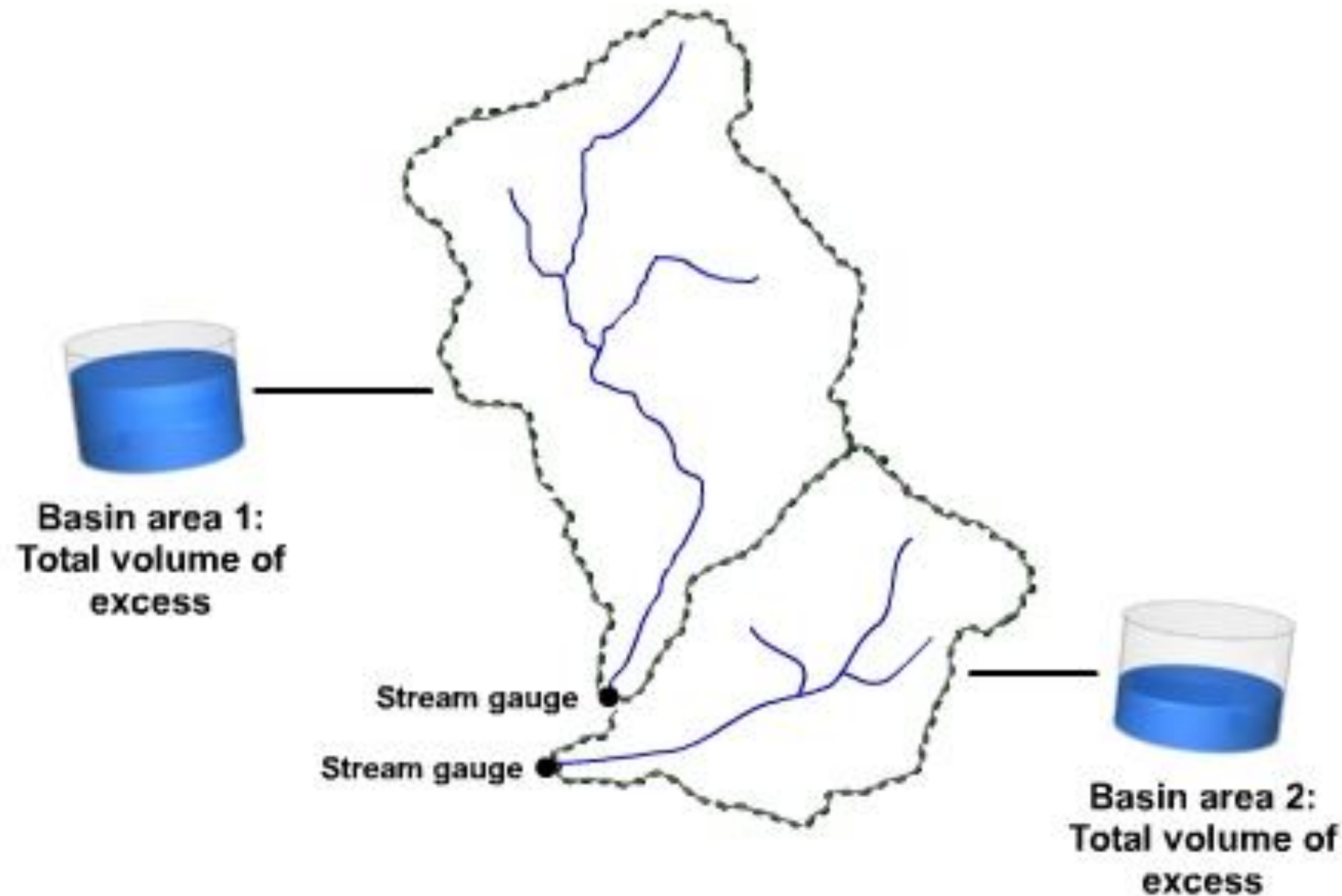


Runoff Components

3. Groundwater Runoff – groundwater discharge into a stream due to deep percolation of the infiltrated water into groundwater aquifers



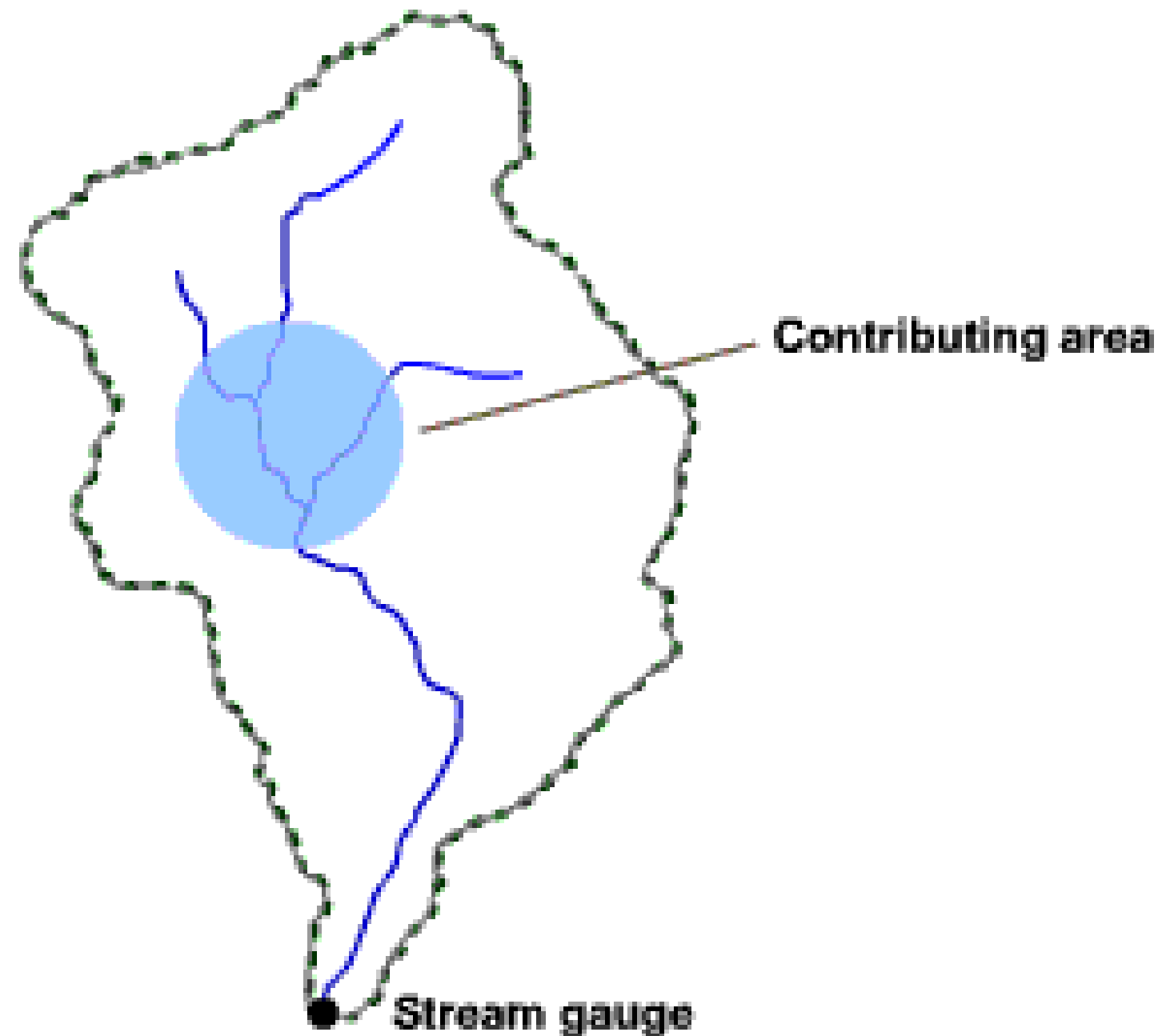
Influence of Basin Area on Runoff Volume



©The COMET Program

- ❖ *when rain falls in a uniform manner over a larger basin and a smaller basin, the **larger basin produces more runoff volume.***

Contributing Area of Storm Determines Runoff

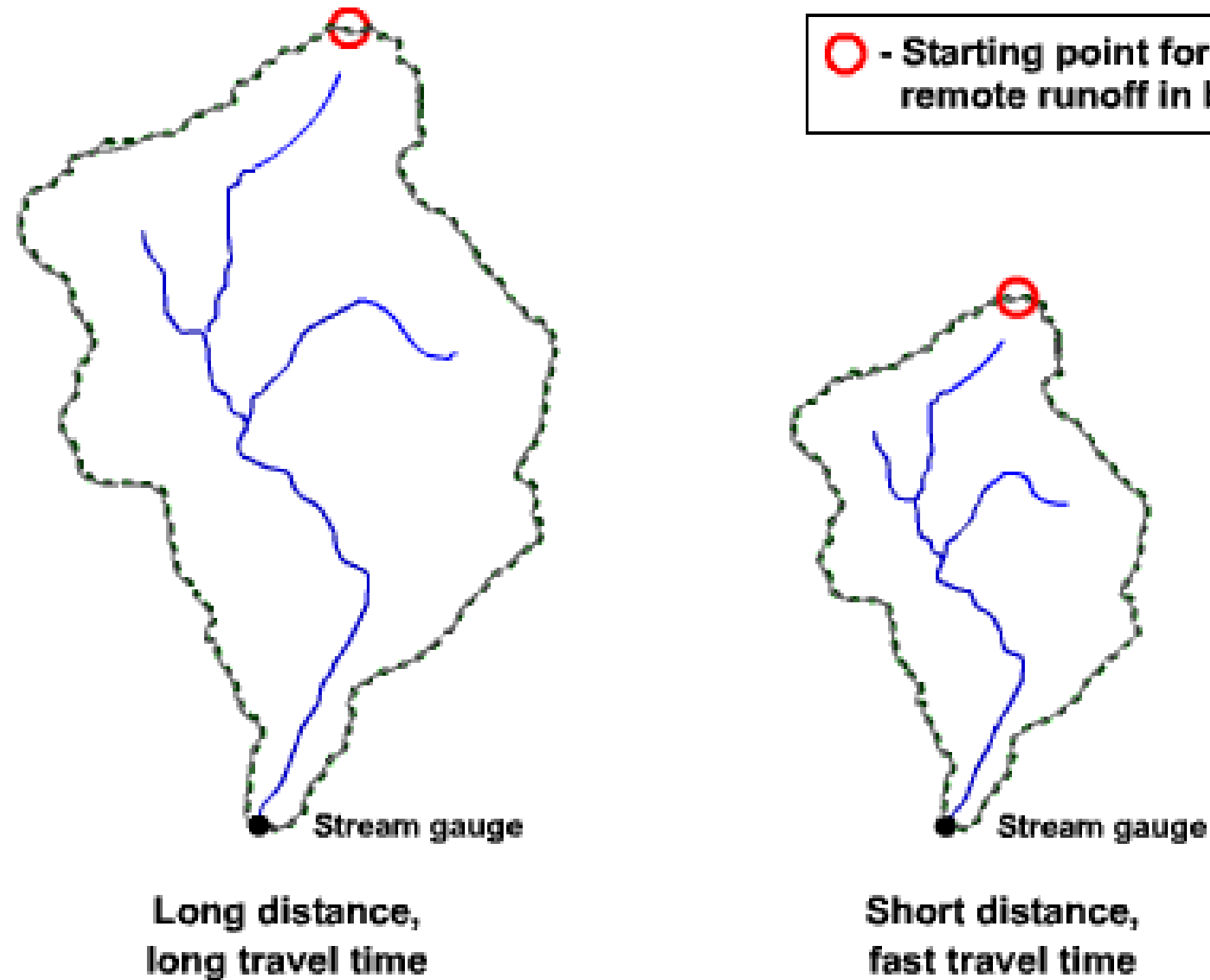


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- ❖ Many storms will cover only part of a basin. For most situations, the **runoff volume will be determined by the contributing area** — that part of the basin covered by a storm — not the total size of a basin.



Influence of Basin Size on Runoff

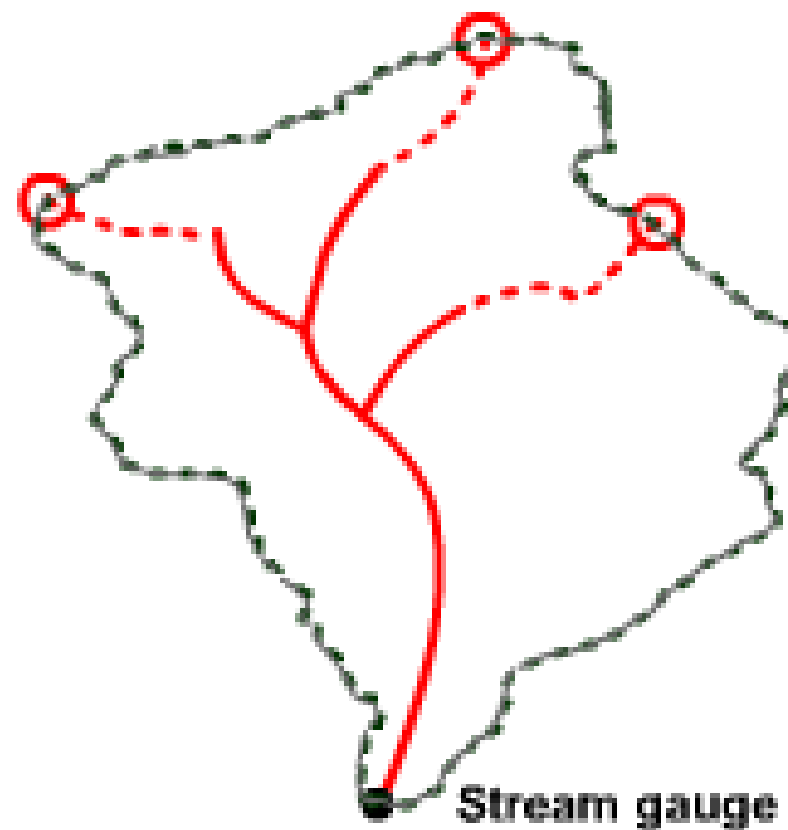


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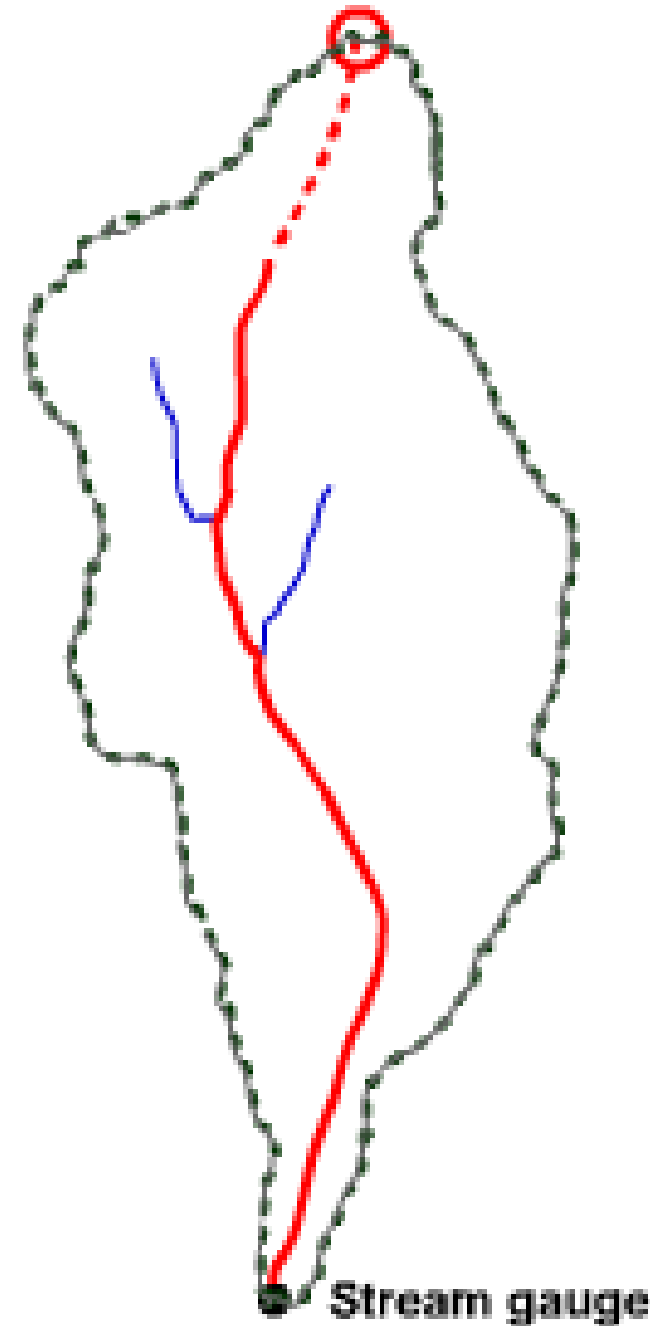
- ❖ Runoff traveling from most upstream point of the **larger basin** will travel a **longer path**, and therefore take longer to reach the basin outlet than runoff traveling from the farthest point in the smaller basin.

Influence of Basin Shape on Runoff

○ - Starting point for most remote runoff in basin



Shorter distance, less
travel time



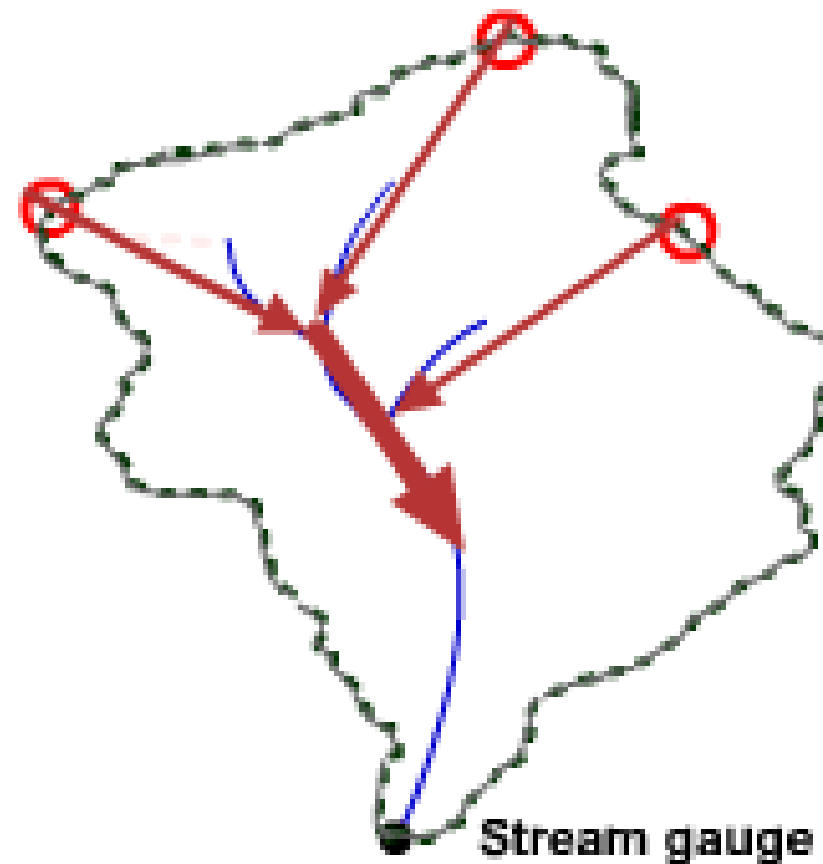
Long distance, long
travel time

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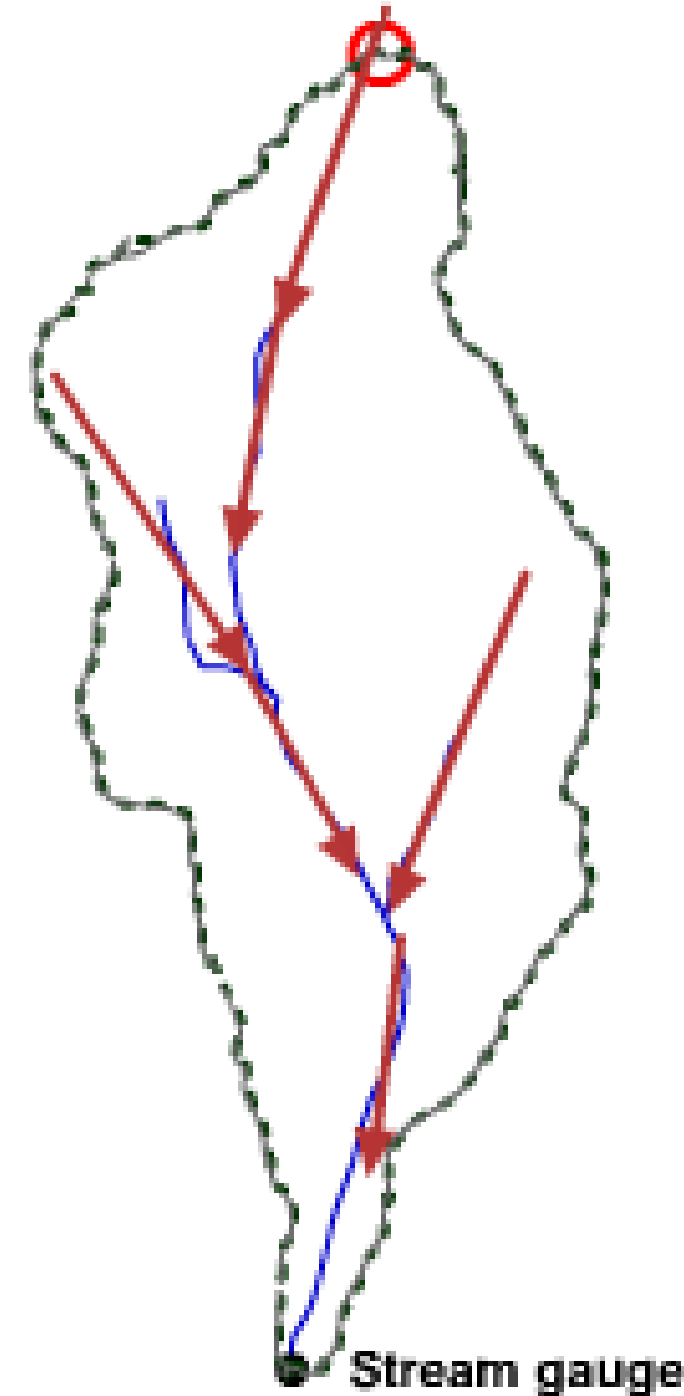
- ❖ The runoff in the more round basin will arrive more quickly at the basin outlet.



Influence of Basin Shape on Peak Flow



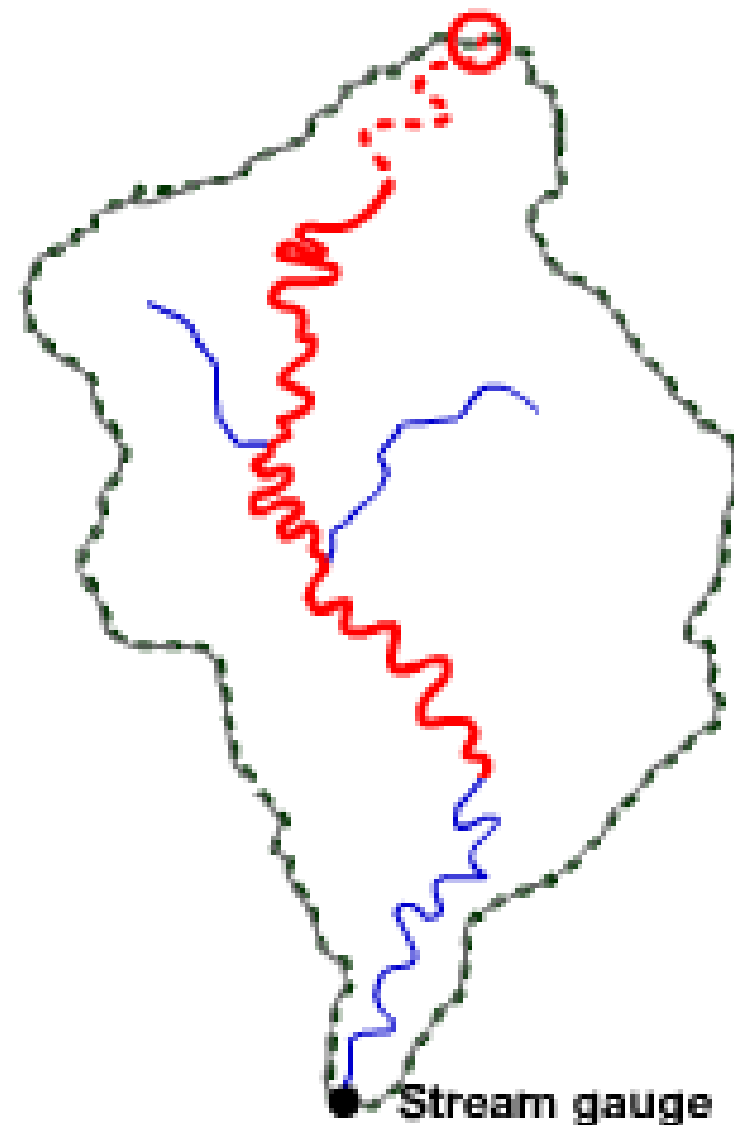
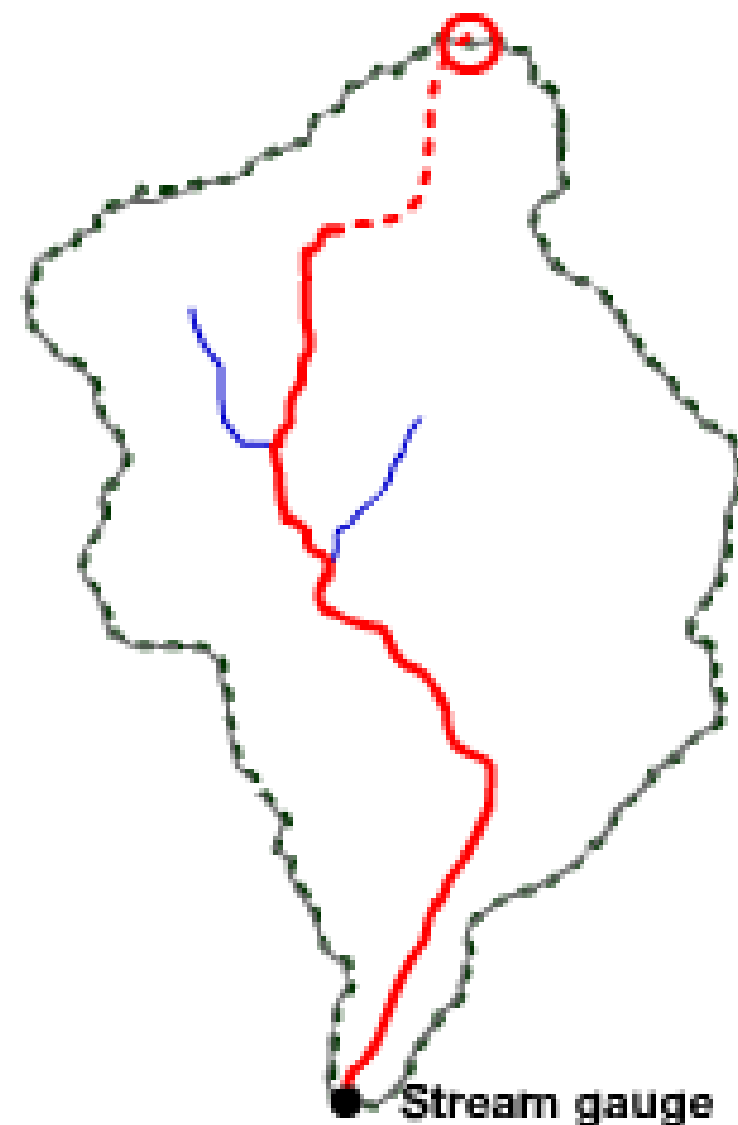
Water from multiple locations more likely to arrive at the same time



Water from multiple locations less likely to arrive at the same time

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Influence of Straight vs. Meandering Flowpath on Runoff

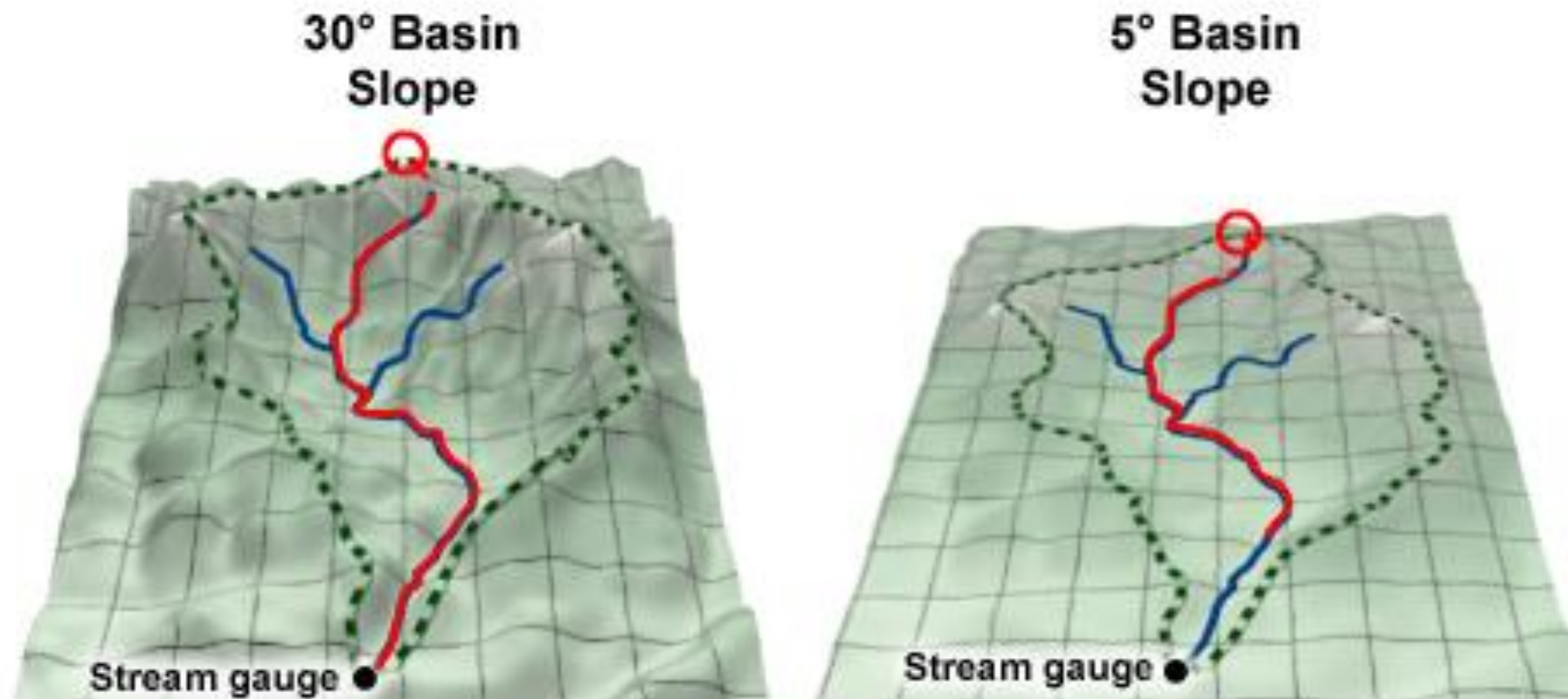


○ - Starting point for most remote runoff in basin

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- ❖ Meanders increase the travel time of runoff through the basin, and may reduce the overall runoff volume.

Influence of Basin Slope on Runoff



○ - Starting point for most remote runoff in basin

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- ❖ As the ground becomes increasingly steep, water will move faster and will have less time in contact with the ground surface, reducing the time during which it could infiltrate.

Factors Affecting Runoff

- Climatic factors
 - Precipitation
 - Form
 - Intensity
 - Duration
 - Time Distribution
 - Areal Distribution
 - Direction of storm movement in relation to the orientation of the watershed
 - Frequency of occurrence
 - Antecedent precipitation
 - Vegetation Interception of Rainfall
 - Evapotranspiration

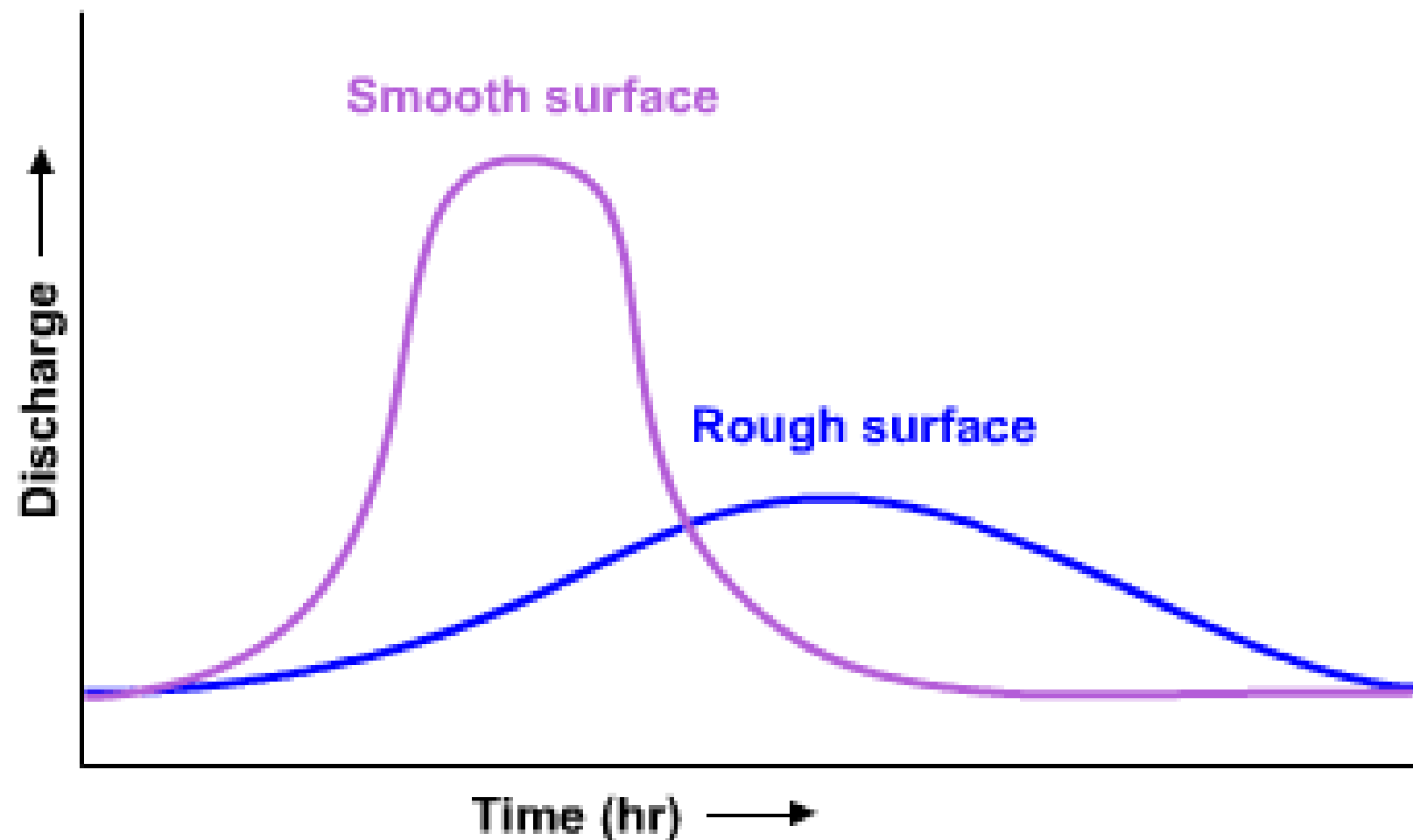
➤ Physiographic factors

▪ Watershed characteristics

- Size
- Shape
- Slope
- Orientation
- Elevation
- Stream Density
- Land use and cover
- Infiltration characteristics
- Geologic Conditions (e.g. permeability, aquifer transmissivity and storage coefficient)
- Topographic conditions (e.g. depression storage, lakes, swamps)
- Artificial Drainage
- Man-made modification (e.g. dams, terraces, etc.)

- Channel characteristics
 - Slope
 - Length
 - Cross-section slope
 - Roughness of channel bed
 - Tributaries
 - Storage Capacity (e.g. backwater effects)

Hydrographs for Rough vs. Smooth Channels

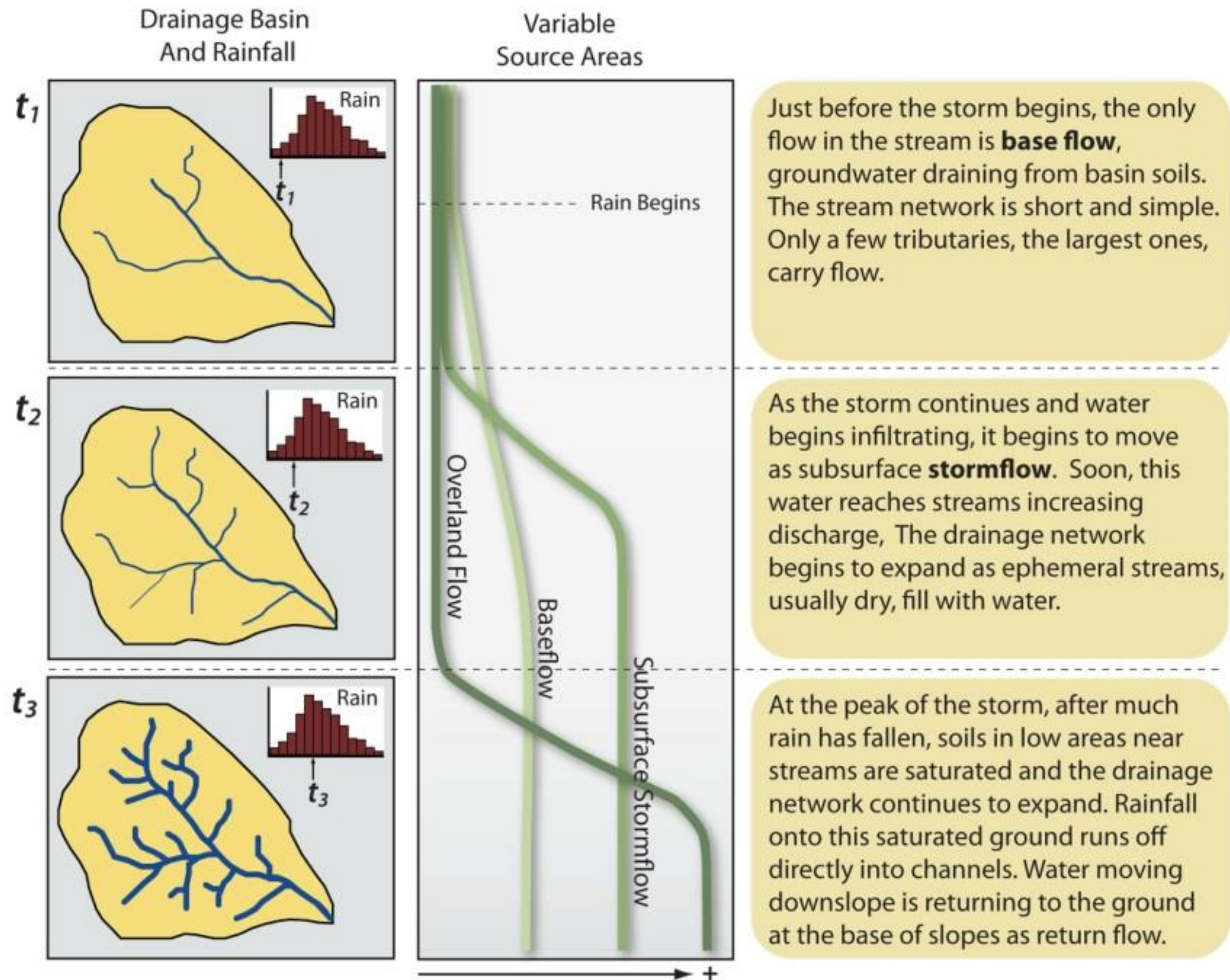


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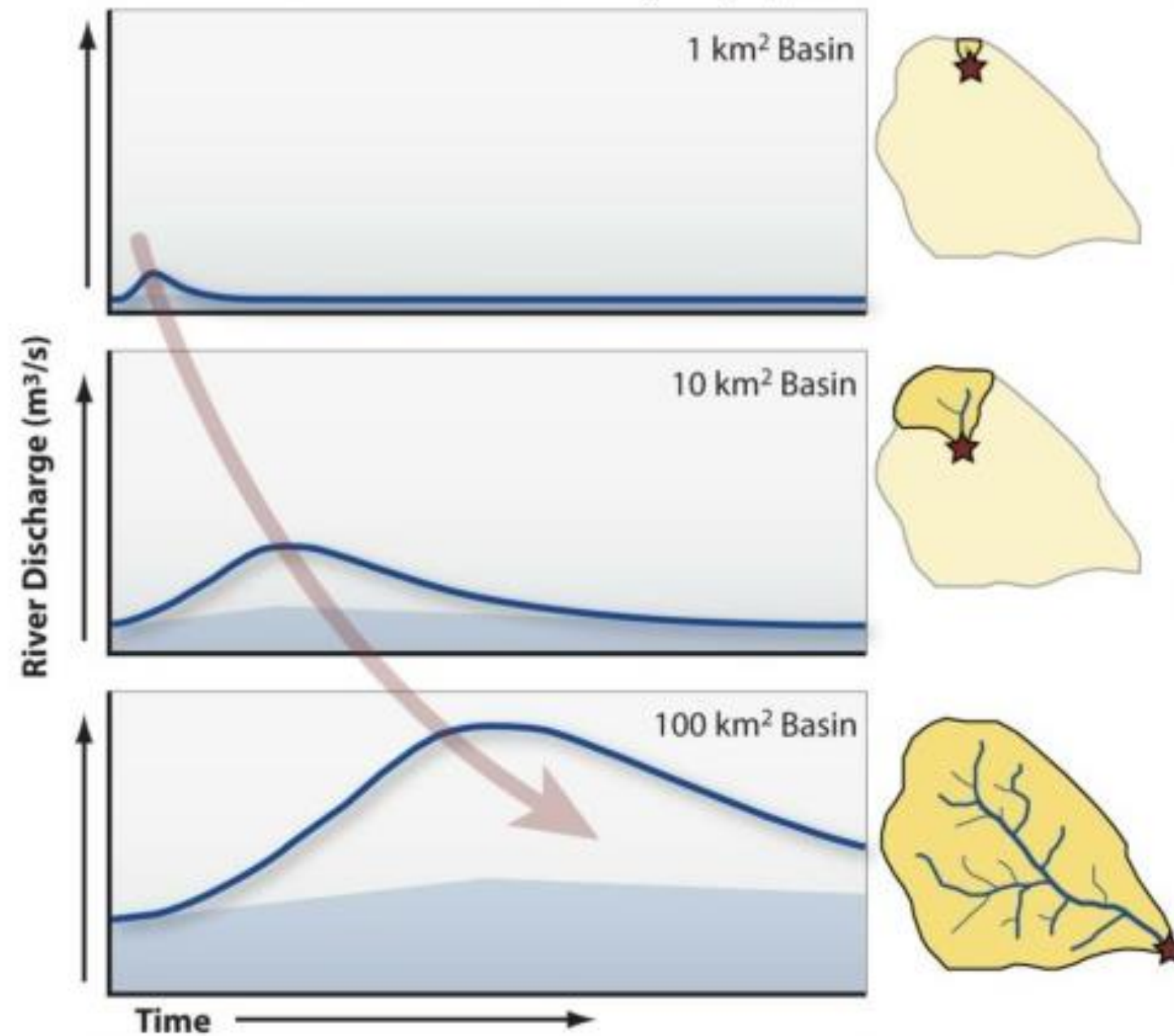
- ❖ The **greater the roughness, the more turbulent the flow.**
- ❖ **More turbulent flow** results in **slower runoff and streamflow velocities.** This allows more time for infiltration, and it also results in a broader flood wave with lower peak discharges than in rapid runoff situations.
- ❖ Reducing channel roughness results in faster streamflow velocities and greater peak flows.

- 
- In very large watersheds, **channel storage** has prominent effects on runoff. In smaller watersheds, **land use influence** is the predominant factor affecting runoff.
 - Many hydrologists are of the consensus that **small watershed** is one which is very sensitive to high intensity rainfall of short duration with land **area of less than 250 sq. km.**



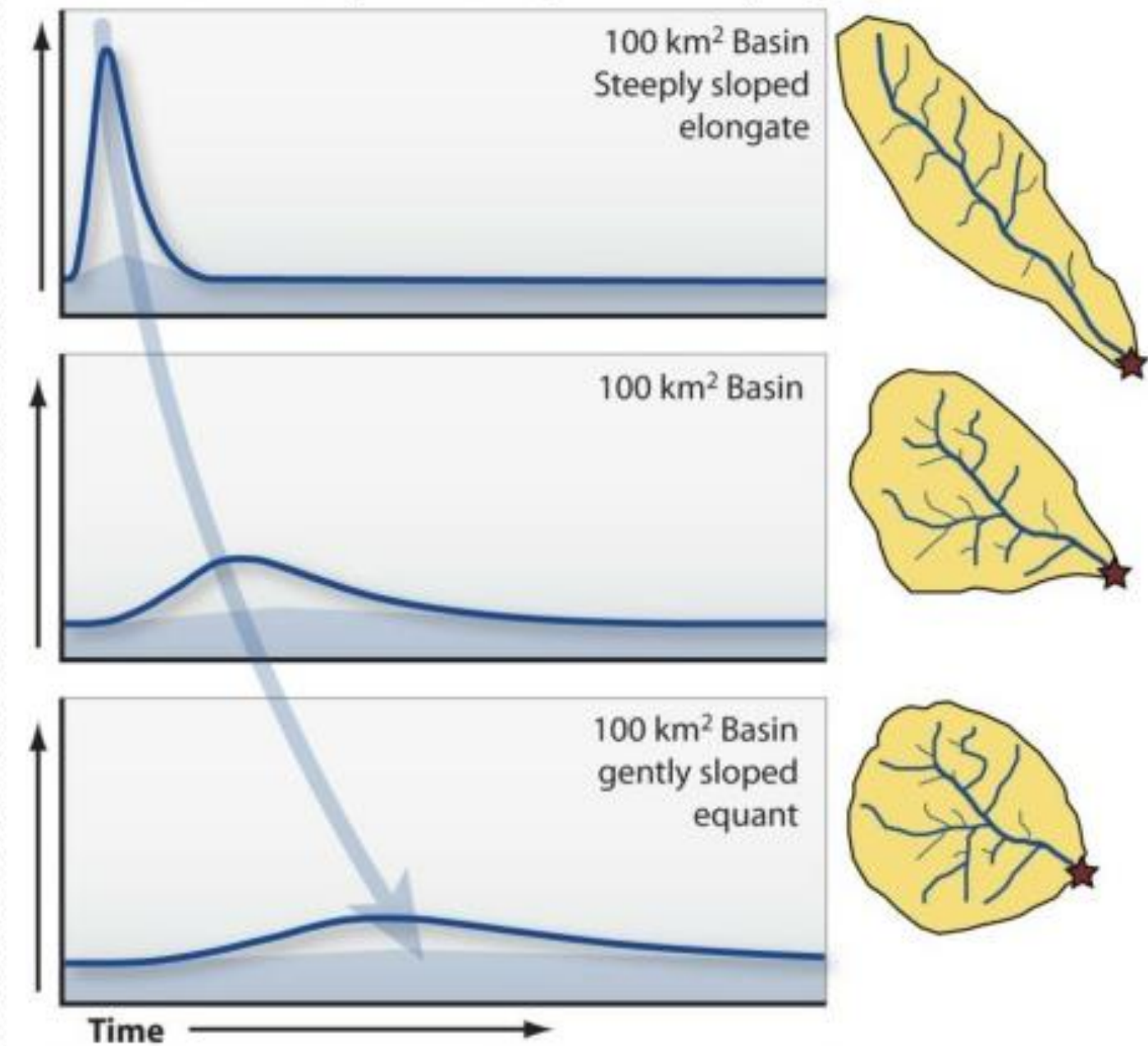


Effects Of Basin **Scale** On The Hydrograph:



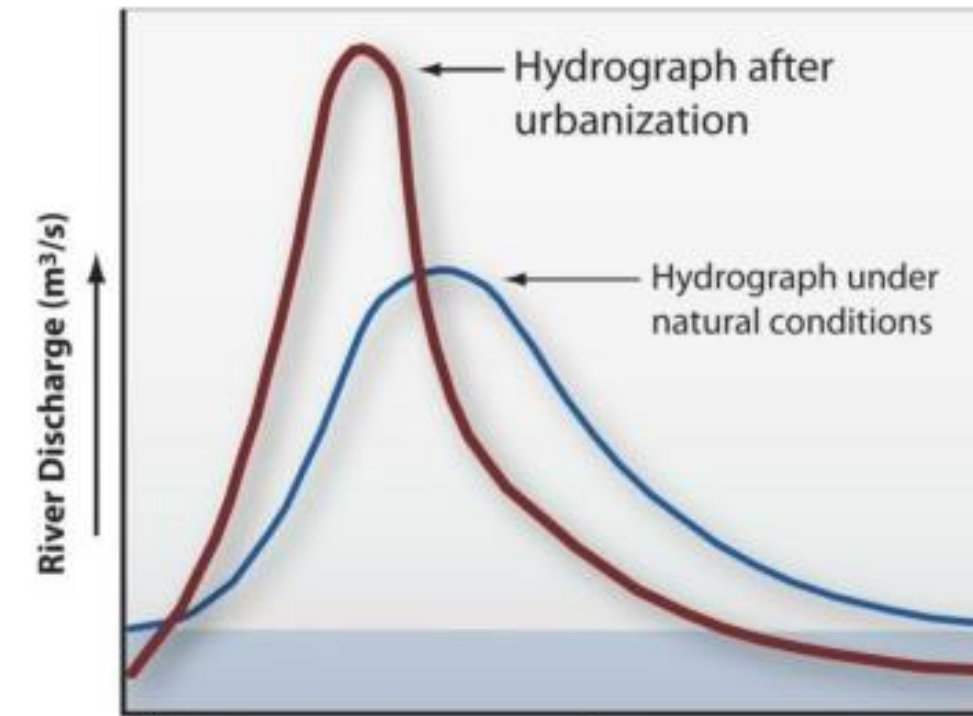
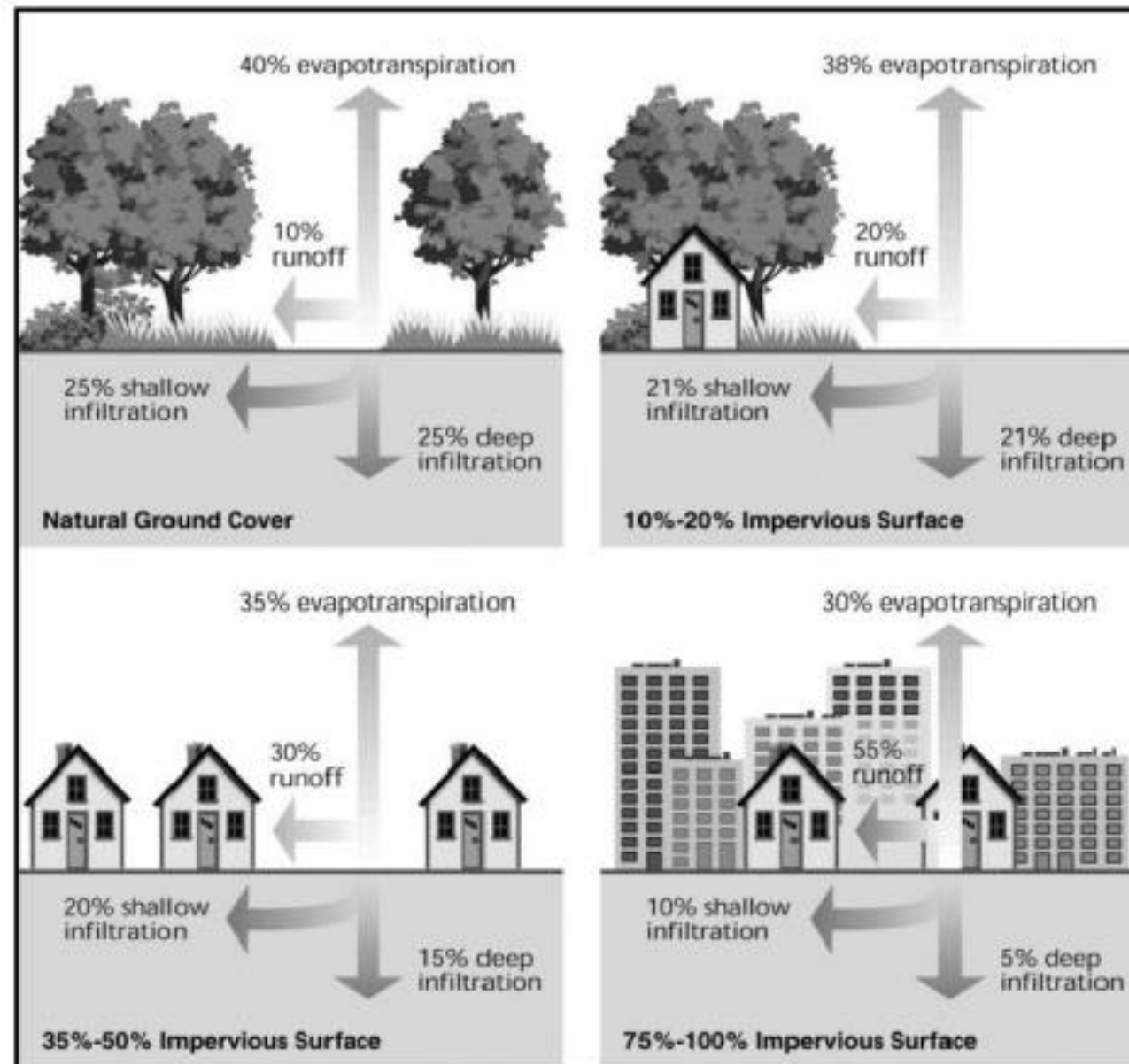
River discharge increases with basin area. Rivers rise and fall more slowly in large basins than in small basins.

Effects Of Basin **Slope** and **Shape** On The Hydrograph:



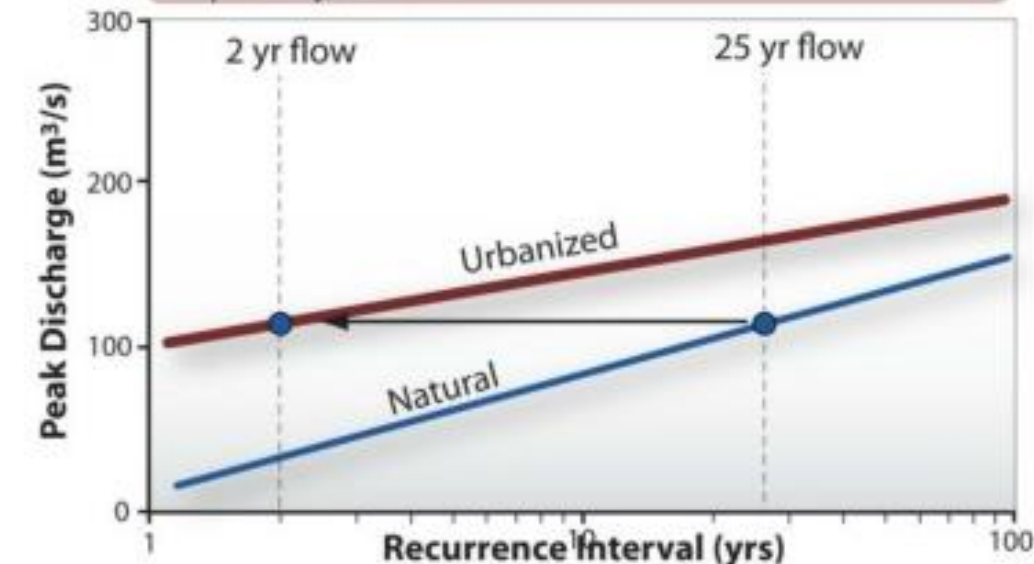
Hydrographs are more peaked and floods more abrupt in narrow, steep basins than in equant, gently sloping basins.

Watershed Urbanization



Before development, rainfall followed a more convoluted path through the landscape - held in detention storage by pit and mound topography, infiltrating into organic-rich forest soil and moving slowly to the channel. The infiltrating water fed baseflow during times when it was not raining. Flood peaks were lower and came later.

After urbanization, rainfall moves rapidly to the channel with little chance to infiltrate during storms, thus baseflow is reduced. Flowing directly off impervious surfaces such as parking lots, runoff enters streams quickly raising their level. Flood peaks now come sooner and are higher, increasing flood hazards and the tempo of geomorphic change. For example, the natural 25 yr flow becomes the much more frequent 2 year flow.



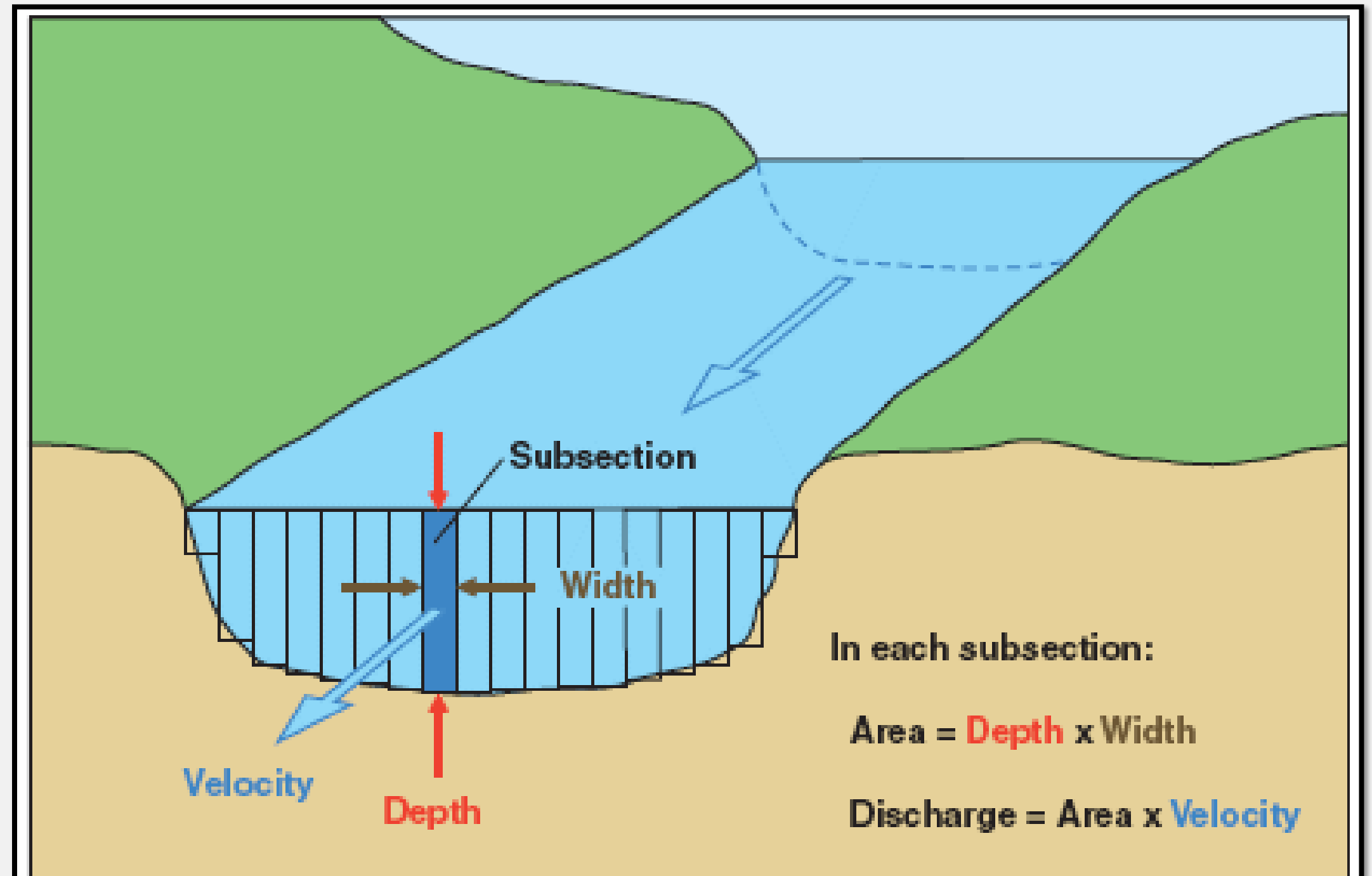
Runoff Measurement and Estimation

Runoff Measurement and Estimation

- A. Direct methods
- B. Indirect measurement
- C. Use of empirical formula



A. Direct Method



Current-meter discharge measurements are made by determining the discharge in each subsection of a channel cross section and summing the subsection discharges to obtain a total discharge.

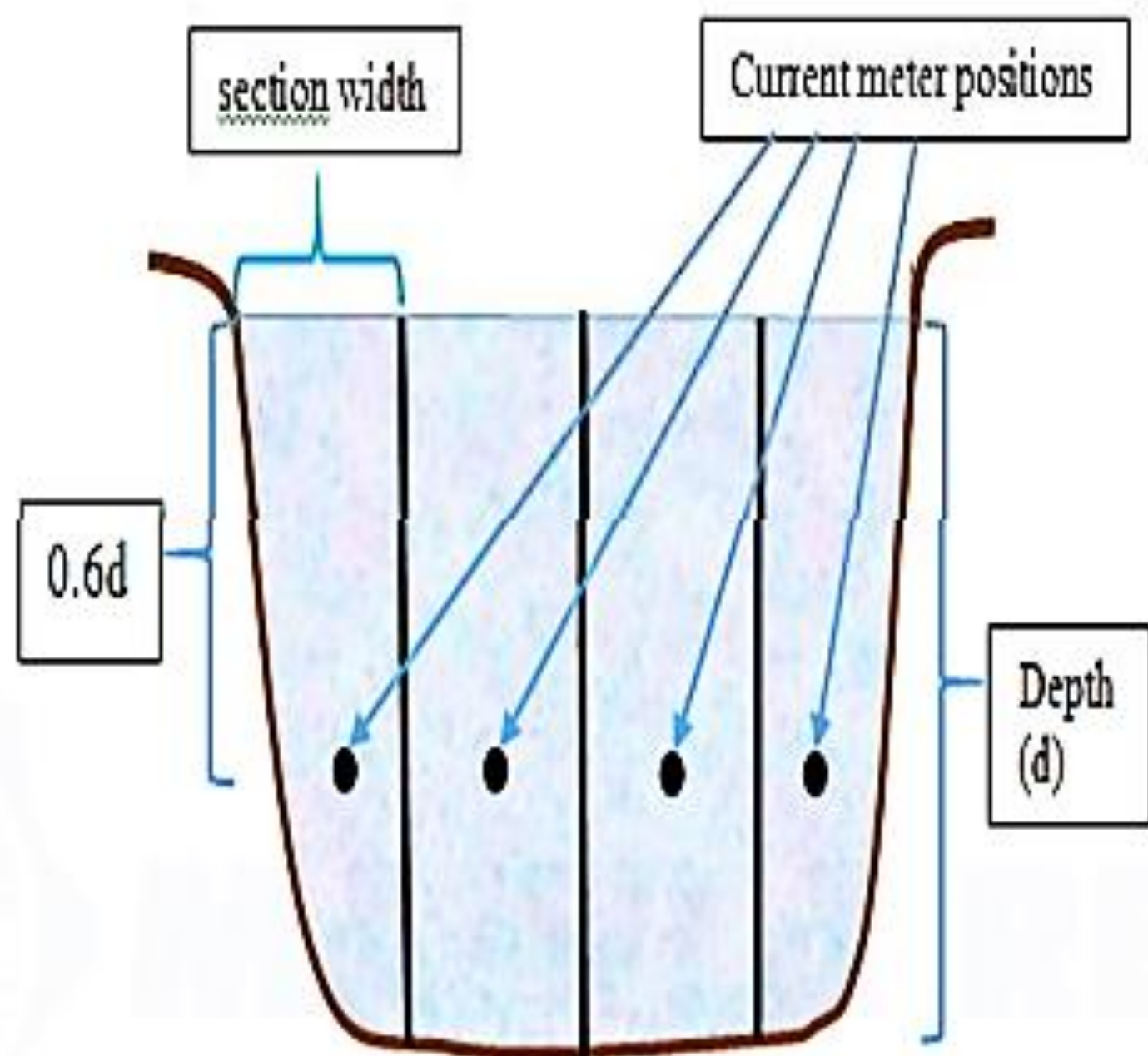


Figure 3-4. A canal cross section with established vertical sections and placements of the current meter for a less than 1 meter depth.

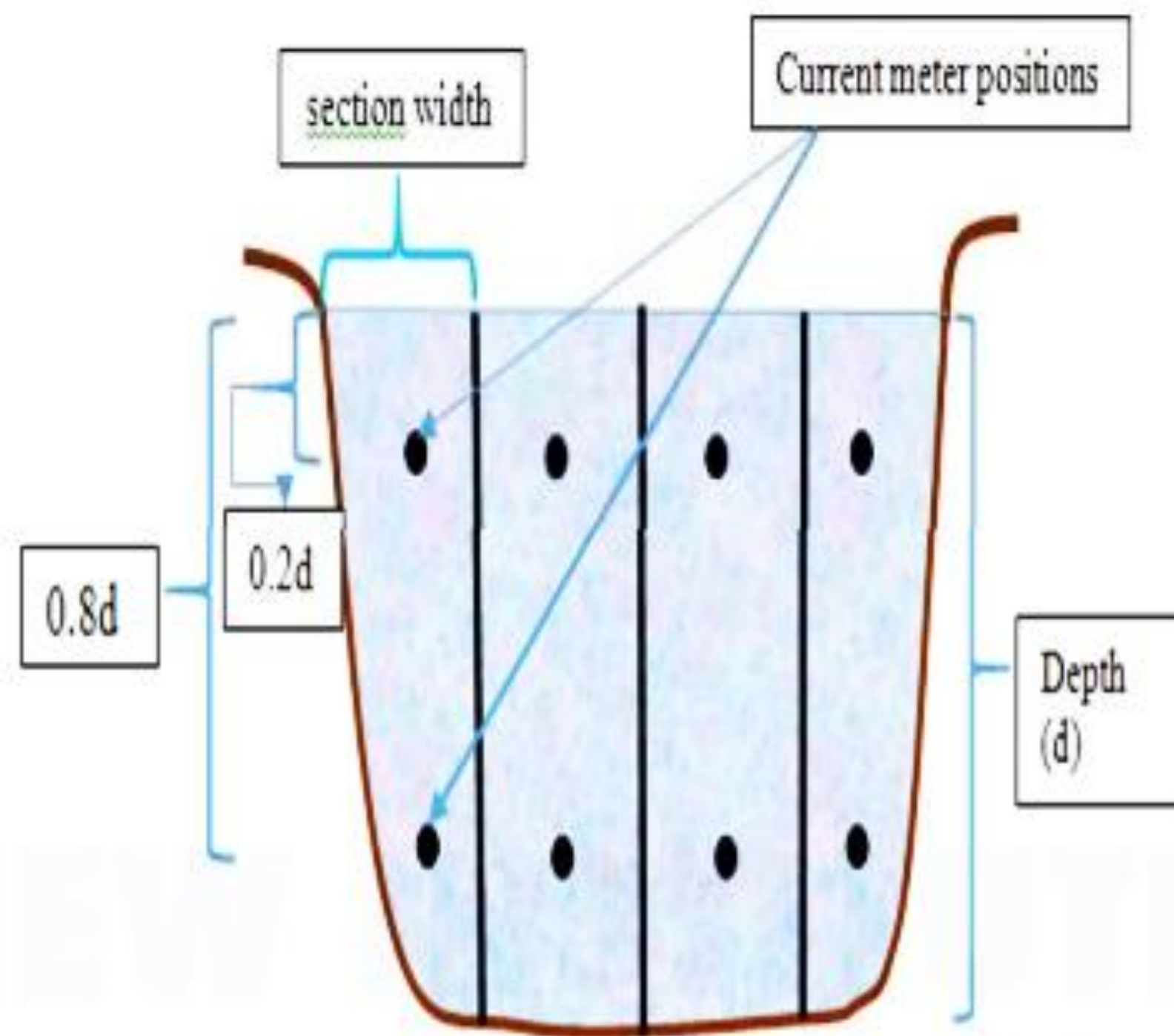


Figure 3-3. A canal cross section with established vertical sections and placements of the current meter for greater than 1 meter depth.

B. Indirect Method

- use of orifice, weir, flume and other control structure
- staff gage, water level recorder



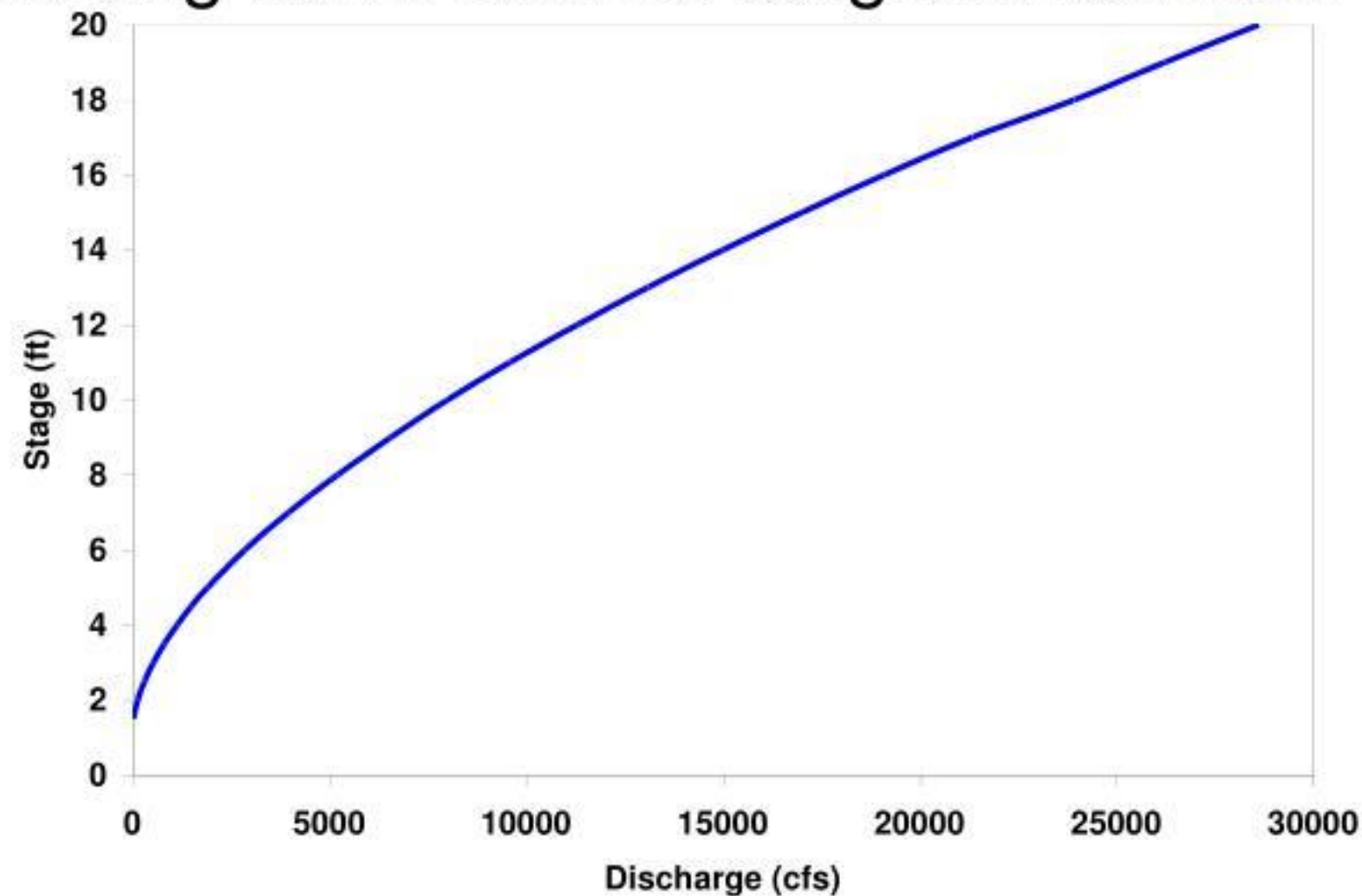
Staff Gauge

used to measure the stage or depth of flow above a datum or reference level.



Rating curve

- It is not feasible to measure flow daily.
- Rating curves are used to estimate flow from stage data
- Rating curve defines stage/streamflow relationship



http://nwis.waterdata.usgs.gov/nwis/measurements/?site_no=08158000



Sample Question:

In discharge measurement, the readings in a staff gage are useless without this:

- a) synthetic hydrograph
- b) unit hydrograph
- c) rating curve
- d) matching curve

C. Use of Empirical Formula

Rational Equation

$$q_p = CiA$$

where:

q_p = **design peak runoff** (m^3/s) for a given recurrence interval (e.g. 10 yrs)

C = **runoff coefficient** or watershed characteristic factor

i = **rainfall intensity** for design recurrence interval and for a duration equal to the time of concentration of a watershed (mm/hr)

A = **watershed area** (ha)



Sample Question:

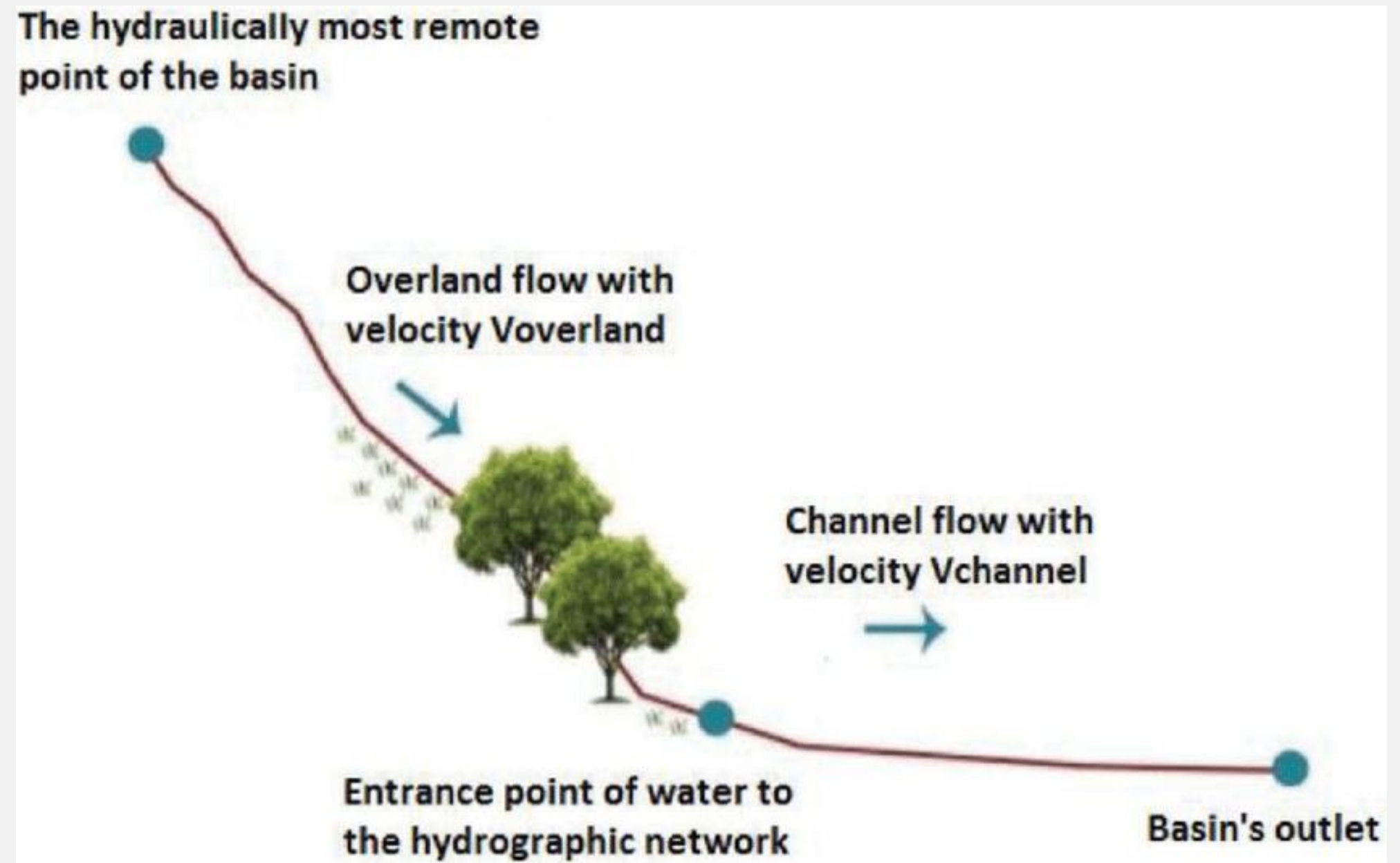
The rational equation is applicable for

- a) flat and small watersheds
- b) flat and large watersheds
- c) sloping and small watersheds
- d) sloping and large watersheds

C. Use of Empirical Formula

Time of Concentration (t_c)

It is the required time for a water particle to flow hydraulically from the most distant point in the watershed to the outlet.



C. Use of Empirical Formula

Time of Concentration (t_c) Kirpich Formula

$$t_c = 0.02L^{0.77}S^{-0.385}$$

where:

t_c = time of concentration (minutes)
interval (e.g. 10 yrs)

L = maximum length of flow (m)

S = watershed gradient (m/m) or the difference in elevation
between the outlet and the most remote point of the watershed

Sample Question:

A 1,000-ha watershed has a main stream length of 1.5 km and a main stream gradient of 2.5° . If a rainfall event start at 5:15PM, when do we expect the peak runoff?

$$L = 1,500 \text{ m}$$
$$S = \frac{\text{rise}}{\text{run}}; \quad \tan\theta = \frac{\text{opp}}{\text{adj}}; \quad S = \tan\theta; \quad S = \tan(2.5^\circ) = 0.0437$$

- a. 4:33 PM
- b. 5:33 PM
- c. 5:51 PM
- d. 6:35 PM

Apply the formula: $t_c = 0.02L^{0.77}S^{-0.385}$

$$t_c = 15 \text{ minutes}$$

$$5:15\text{PM} + 15 \text{ minutes} = 5:33 \text{ PM}$$

B

C. Use of Empirical Formula

Soil Conservation Service (SCS) Method / Curve Number

$$Q = \frac{(P - 0.2S)^2}{(P + 0.8S)}$$

where:

- Q = direct runoff (mm)
- P = storm rainfall (mm)
- S = maximum potential difference between rainfall and runoff and the time the storm began

$$S = \frac{25,400}{CN} - 254$$

**CN = SCS
Curve Number**
function of land use,
antecedent moisture,
soil type, hydrologic
condition

Curve Number	Runoff Potential	Explanation
Low CN (30–60)	Low runoff	Soil absorbs water well (e.g., sandy soil, forest)
High CN (70–100)	High runoff	Soil doesn't absorb much (e.g., clay, paved area)

Sample Question:

What is the CN number of the area with potential maximum retention of 80.

- a. 84.56
- b. 76.04
- c. 102.3
- d. 64.2

$$S = \frac{25,400}{CN} - 254$$

$$80 = \frac{25,400}{CN} - 254$$

$$CN = 76.04$$

Sample Question:

Marikina watershed having a drainage area of 500 sq. km. has been characterized to have an average CN value of 65 in 1970 where it experienced a storm depth of 100mm. After 30 years, the same storm magnitude was felt but this time, due primarily to a change in land use inside the watershed, the CN value now is 84.

What is the percentage change in the depth of runoff?

- a. 133 %
- b. 41%
- c. 80 %
- d. 29 %

Solution to the previous problem:

$$S = \frac{25400}{CN} - 254$$

$$S_{65} = \frac{25400}{65} - 254 = 390.77 - 254 = 136.77 \text{ mm}$$

$$Q_{65} = \frac{(100 - 0.2 \cdot 136.77)^2}{(100 + 0.8 \cdot 136.77)} \approx 25.19 \text{ mm}$$

$$S_{84} = \frac{25400}{84} - 254 = 302.38 - 254 = 48.38 \text{ mm}$$

$$Q_{84} = \frac{(100 - 9.676)^2}{(100 + 38.704)} \approx 58.78 \text{ mm}$$

$$\text{Percentage Change} = \frac{Q_{84} - Q_{65}}{Q_{65}} \times 100$$

$$\text{Percentage Change} = \frac{58.78 - 25.19}{25.19} \times 100 \approx \frac{33.59}{25.19} \times 100 \approx 133.3\%$$

C. Use of Empirical Formula

Time of Concentration (t_c) **SCS Lag Formula**

$$t_c = 0.0134L^{0.8} \left(\frac{1000}{CN} - 10 \right)^{0.7} S^{-0.05}$$

where:

Q = direct runoff (mm)

I = storm rainfall

S = maximum potential difference between rainfall and runoff and the time the storm began

Gumbel Distribution

$$X_T = M + S(0.78 \ln T - 0.45)$$

where:

M = mean of the sample

S = standard deviation

T = return period

X_T = magnitude at a certain T



Sample Question:

Given the annual peak discharges for a 50-ha watershed, compute for the 10- and 50-year peak runoffs.

Year	Streamflow (m^3/s)
1995	1386
1996	1387.4
1997	1364
1998	1285.2
1999	1289
2000	1406
2001	1407
2002	1470
2003	1505
2004	1547

Sample Question:

Given the annual peak discharges for a 50-ha watershed, compute for the 10- and 50-year peak runoffs.

Year	Streamflow (m^3/s)
1995	1386
1996	1387.4
1997	1364
1998	1285.2
1999	1289
2000	1406
2001	1407
2002	1470
2003	1505
2004	1547

$$X_T = M + S(0.78 \ln T - 0.45)$$

$$M = 1,404.6 \text{ m}^3/\text{s}$$

$$SD = 80.344 \text{ m}^3/\text{s}$$

$$Q_{10} = 1404.6 + 80.344(0.78 \ln(10) - 0.45)$$

$$Q_{10} = 1,512.74 \text{ m}^3/\text{s}$$

$$Q_{50} = 1404.6 + 80.344(0.78 \ln(50) - 0.45)$$

$$Q_{50} = 1613.61 \text{ m}^3/\text{s}$$

Sample Question:

The peak runoff in a watershed has a mean of $540 \text{ m}^3/\text{s}$ and a standard deviation of $125 \text{ m}^3/\text{s}$. If the sample data is normally distributed, the magnitude of runoff corresponding to probability of exceedance of the 16% is

- a) $415 \text{ m}^3/\text{s}$
- b) $540 \text{ m}^3/\text{s}$
- c) $665 \text{ m}^3/\text{s}$
- d) $86.4 \text{ m}^3/\text{s}$

Sample Question:

The peak runoff in a watershed has a mean of $540 \text{ m}^3/\text{s}$ and a standard deviation of $125 \text{ m}^3/\text{s}$. If the sample data is normally distributed, the magnitude of runoff corresponding to probability of exceedance of the 16% is

- a) $415 \text{ m}^3/\text{s}$
- b) $540 \text{ m}^3/\text{s}$
- c) $665 \text{ m}^3/\text{s}$
- d) $86.4 \text{ m}^3/\text{s}$

$$X_T = M + S(0.78 \ln T - 0.45)$$

$$M = 540 \text{ m}^3/\text{s}$$

$$SD = 125 \text{ m}^3/\text{s}$$

$$Q_{6.25} = 540 + 125(0.78 \ln(6.25) - 0.45)$$

$$Q_{6.25} = 662.43 \text{ m}^3/\text{s}$$

Sample Question:

Streamflow in river X has a mean of $128 \text{ m}^3/\text{s}$ and a standard deviation of $52 \text{ m}^3/\text{s}$. If the sample data is normally distributed, the magnitude of streamflow corresponding to a return period of 6.25 years is

- a) $128 \text{ m}^3/\text{s}$
- b) $76 \text{ m}^3/\text{s}$
- c) $180 \text{ m}^3/\text{s}$
- d) $52 \text{ m}^3/\text{s}$

Sample Question:

Streamflow in river X has a mean of $128 \text{ m}^3/\text{s}$ and a standard deviation of $52 \text{ m}^3/\text{s}$. If the sample data is normally distributed, the magnitude of streamflow corresponding to a return period of 6.25 years is

- a) $128 \text{ m}^3/\text{s}$
- b) $76 \text{ m}^3/\text{s}$
- c) $180 \text{ m}^3/\text{s}$
- d) $52 \text{ m}^3/\text{s}$

$$X_T = M + S(0.78 \ln T - 0.45)$$

$$M = 128 \text{ m}^3/\text{s}$$

$$SD = 52 \text{ m}^3/\text{s}$$

$$Q_{6.25} = 128 + 52(0.78 \ln(6.25) - 0.45)$$

$$Q_{6.25} = 178.93 \text{ m}^3/\text{s}$$

STREAMFLOW

BASIC EQUATION

$$Q = AV$$

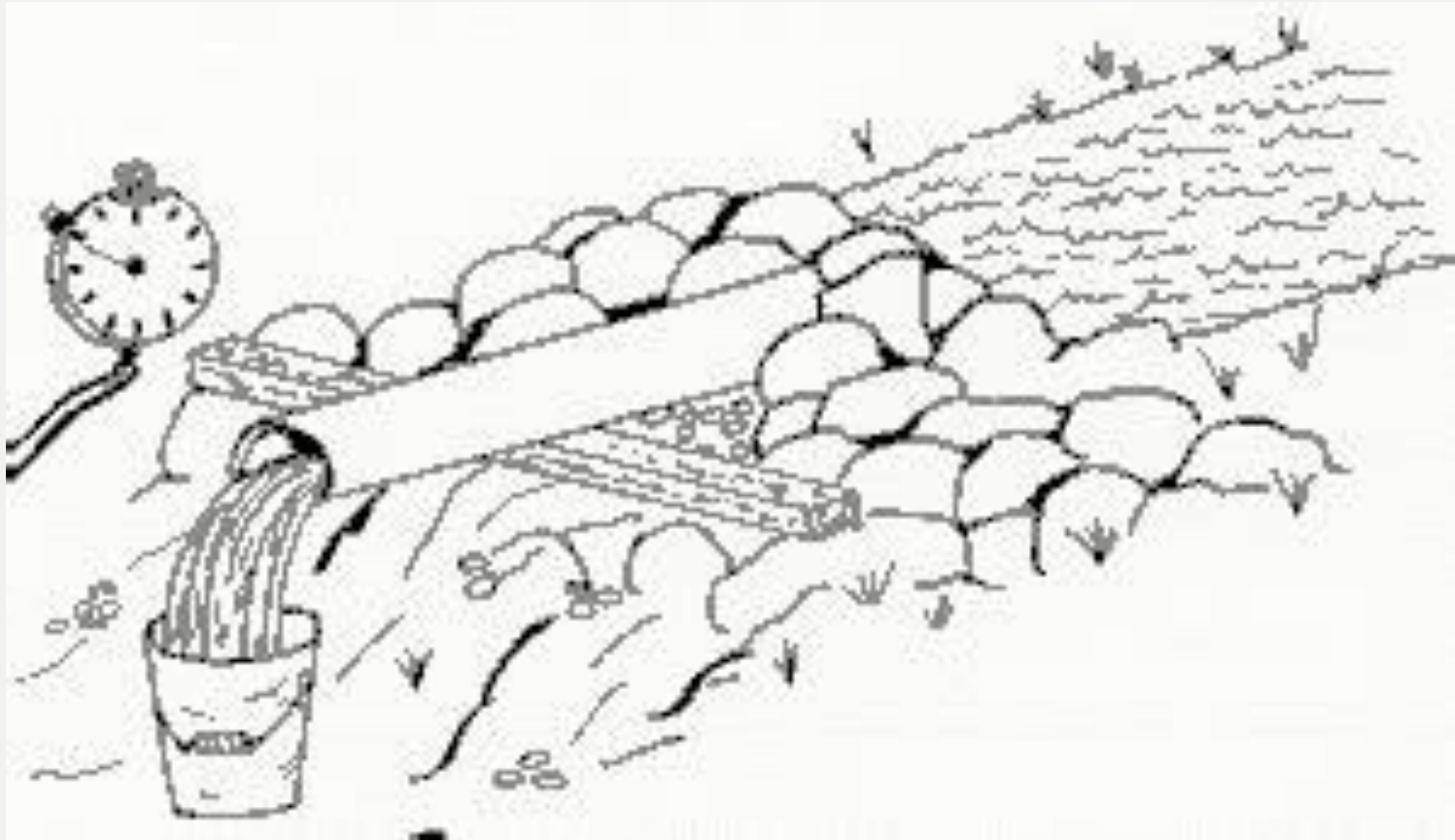


Methods of Flow Measurement

- A. Volumetric Method
- B. Gravimetric Method
- C. Slope-Area Method
- D. Velocity-Area Method
- E. Control-Section Method



Volumetric and Gravimetric Methods



Slope-Area Method

$$V = \frac{k}{n} R^{2/3} S^{1/2}$$

***V** = velocity of flow (m/s or ft/s)*

***R** = hydraulic radius (m or ft)*

***k** = conversion factor (**1**-metric, **1.486**-english)*

***n** = roughness coefficient*

***S** = slope of energy grade line*



Sample Question:

Determine the discharge (lps) for a stream with a total cross-sectional area of 2 m^2 and the wetted perimeter is 4m over a bottom slope of 0.25% if the roughness coefficient of the earthen material forming the stream is 0.035.

Sample Question:

Determine the discharge (lps) for a stream with a total cross-sectional area of 2 m^2 and the wetted perimeter is 4m over a bottom slope of 0.25% if the roughness coefficient of the earthen material forming the stream is 0.035.

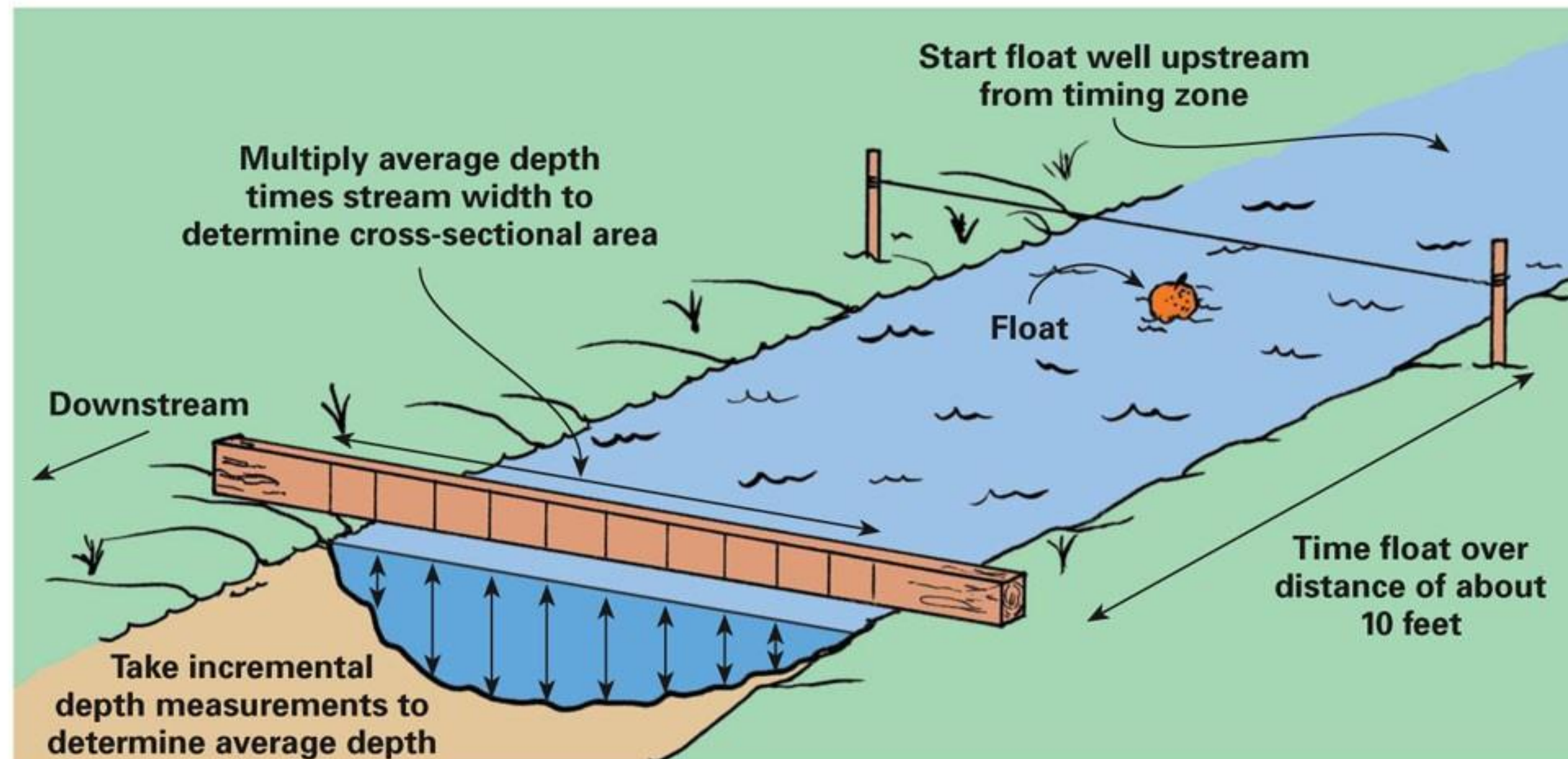
$$Q = AV$$

$$Q = (2) \left(\frac{1}{0.035} \right) \left(\frac{2}{4} \right)^{\frac{2}{3}} (0.0025)^{1/2}$$

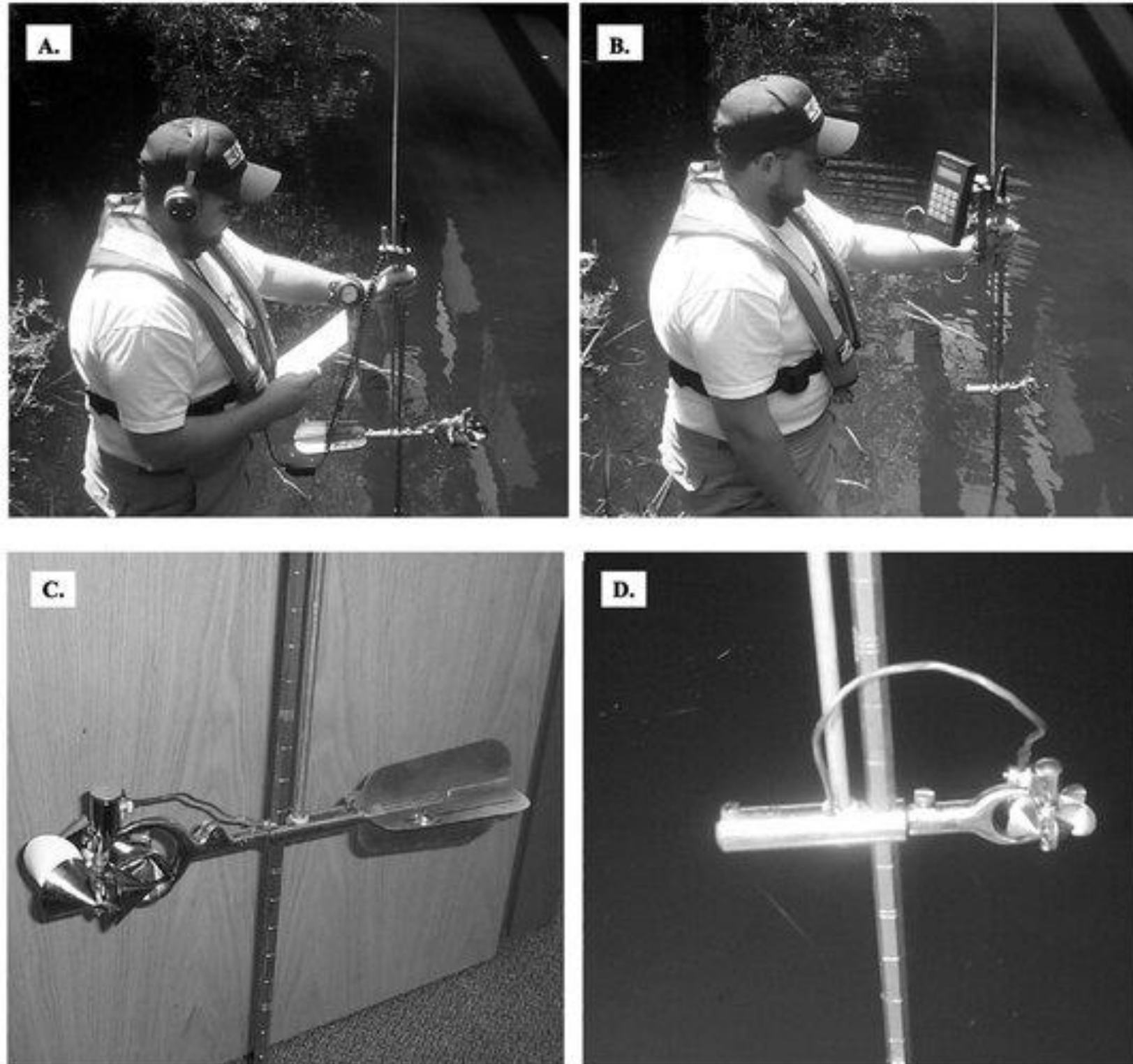
$$Q = 1.8 \text{ cumecs or } 1,800 \text{ lps}$$

Velocity-Area Method

The Float Method of Estimating Flow



Velocity-Area Method



Current Meter Method

$$V = a + bN$$

Where:

v = the velocity

a = constant that accounts for the baseline velocity of the water.

This represents the velocity when the current meter reading n is zero

b = coefficient

n = reading from the meter

Types of Current Meter

Classification	Description
Anemometer and Propeller Type	- uses anemometer cup and propellers for sensing velocity - rated by dragging through tanks of still water at known speeds - does not sense direction of velocity
Electromagnetic	- produces voltage proportional to the velocity - provides direct analog reading of velocity - provides directional measurements
Doppler Type	- measures velocity by the change of source light or sound frequency from the frequency of reflections from moving particles - uses laser light (laser Doppler velocimeters) or sound (acoustic Doppler velocimeter)
Optical Strobe Velocity Meter	- measures velocity using the strobe effect with mirrors mounted around a polygon drum reflecting light coming from the water surface. The rate of rotation of the mirror drum is varied until images become steady. With the rate of rotation and the distance from the mirrors to the water surface known, surface velocity can be determined

Sample Question:

If a current meter calibration shows that $a = 0.5 \text{ m/s}$ and $b = 2.0 \text{ m/s}$ per unit of n , and you obtain a reading of $n=3$, what is the streamflow velocity?

Sample Question:

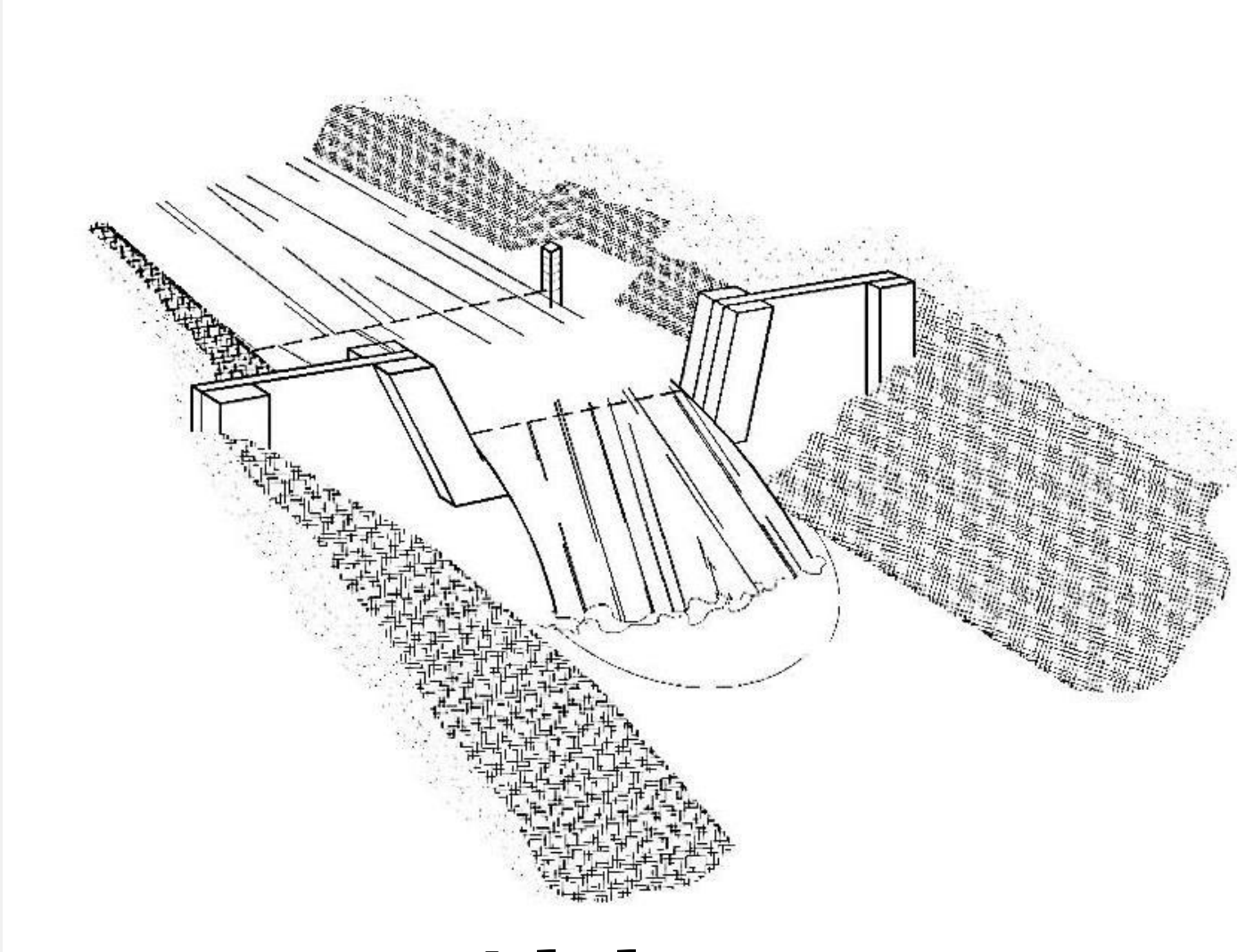
If a current meter calibration shows that $a = 0.5$ m/s and $b = 2.0$ m/s per unit of n , and you obtain a reading of $n=3$, what is the streamflow velocity?

$$v = a + bn$$

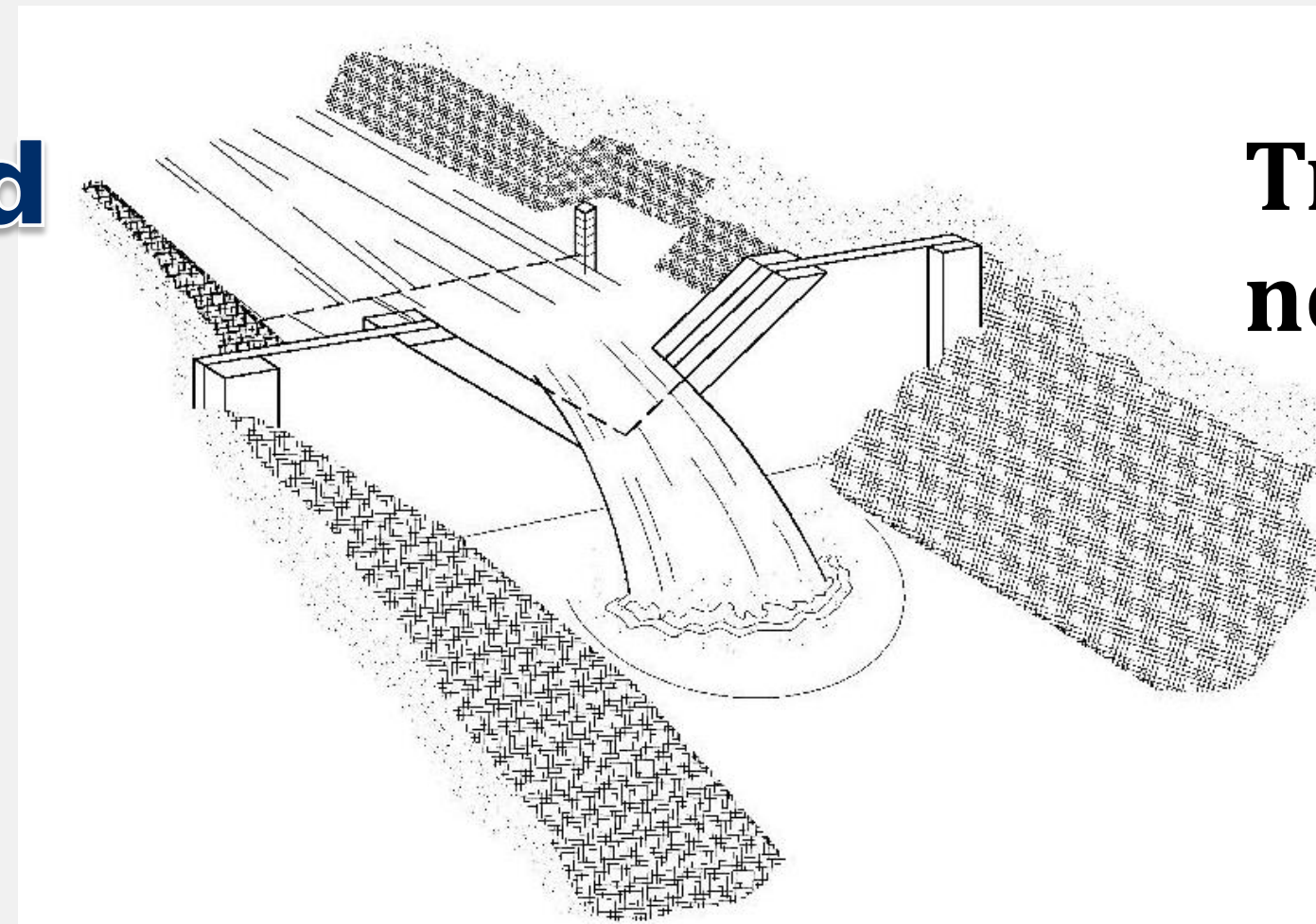
$$v = 0.5 + (2.0 \times 3)$$

$$v = 0.5 + 6.0 = 6.5 \text{ m/s}$$

Control-Section Method



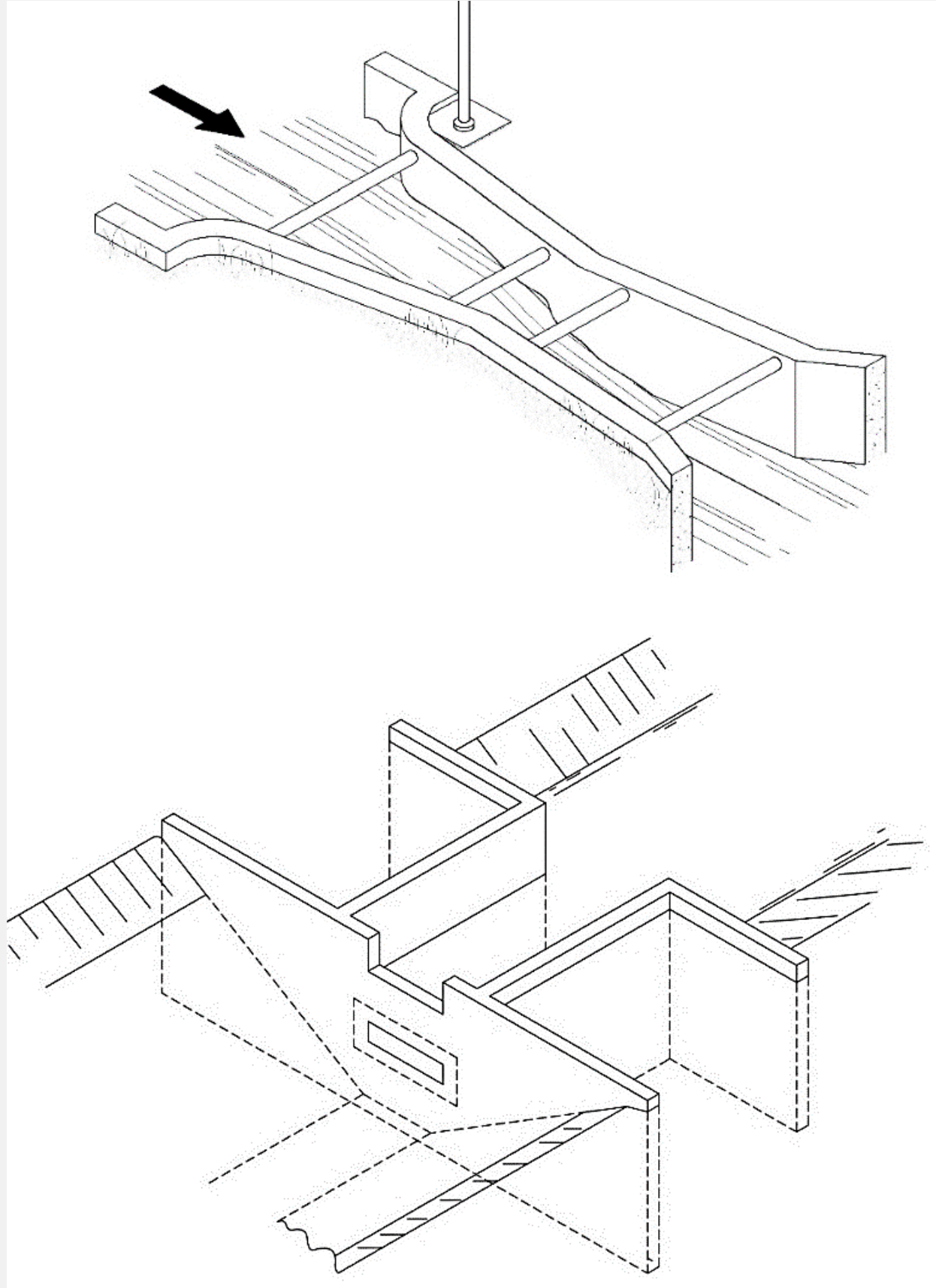
**Trapezoidal-
notch weir**



**Triangular-
notch Weir**



Control-Section Method



Parshall Flume

Submerged Orifice



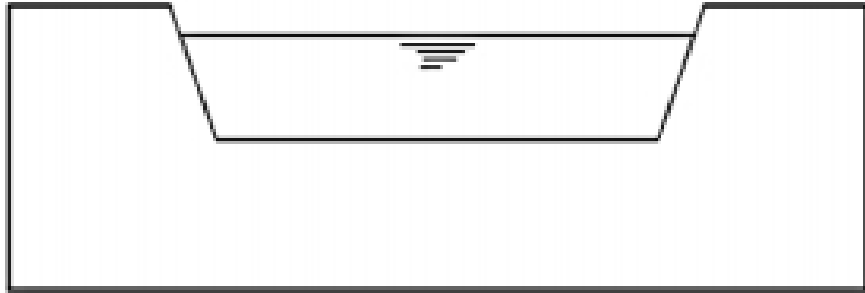
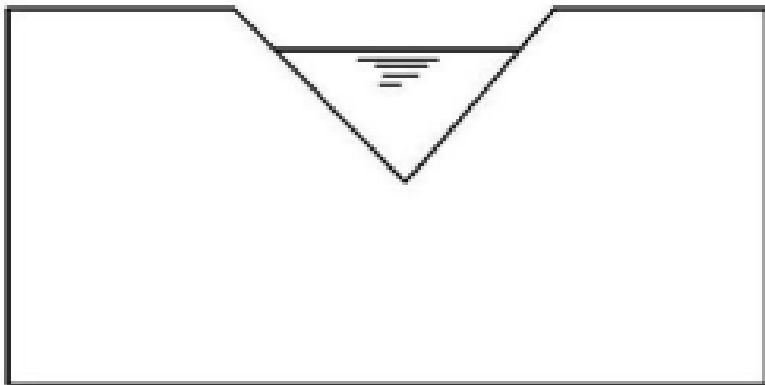
Control Structures

Various Types of Commonly Used Flow Measurement Structures

Table 1. Types of Weir, Description and Discharge Evaluation

Classification	Description	Discharge Evaluation
SHARP-CRESTED/ THIN-PLATE - weir constructed with a crest of vertical thin plate		
Contracted Rectangular		$Q = 1.84 (L - 0.2H)H^{3/2}$ Q = discharge, m³/s L = length of crest, m H = head (difference between the elevation of the weir crest and the elevation of the water surface in the weir pool), m
Supressed Rectangular		$Q = \frac{2}{3} C_d L H \sqrt{2gH}$ $C_d = 0.615 \left(1 + \frac{1}{H + 1.6} \right) \left[1 + 0.5 \left(\frac{H}{H + D} \right)^2 \right]$ Q = discharge, m³/s C_d = discharge coefficient L = length of crest, m g = gravitational acceleration, m²/s H = head, m (mm for computing μ) D = distance from the crest to the bottom of the approach channel, mm

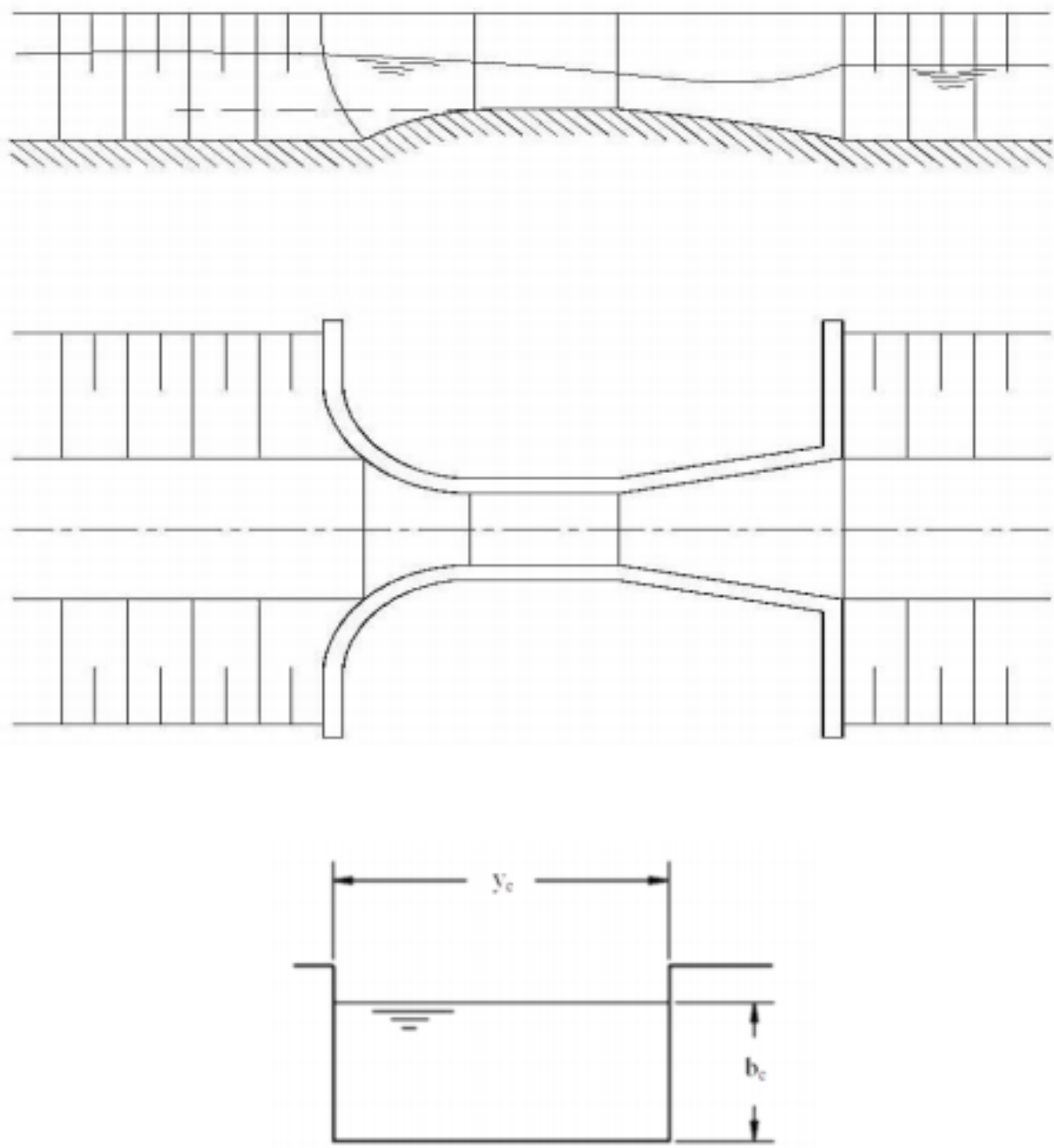
Control Structures

Cipoletti Contracted		$Q = 1.86 LH^{3/2}$ Q = discharge, m^3/s L = length of crest, m H = head (difference between the elevation of the weir crest and the elevation of the water surface in the weir pool), m
V-notch		$Q = \frac{8}{15} \sqrt{2gC_d} H^{5/2}$ Q = discharge, m^3/s g = gravitational acceleration, m^2/s C_d = discharge coefficient H = head (difference between the elevation of the weir crest and the elevation of the water surface in the weir pool), m



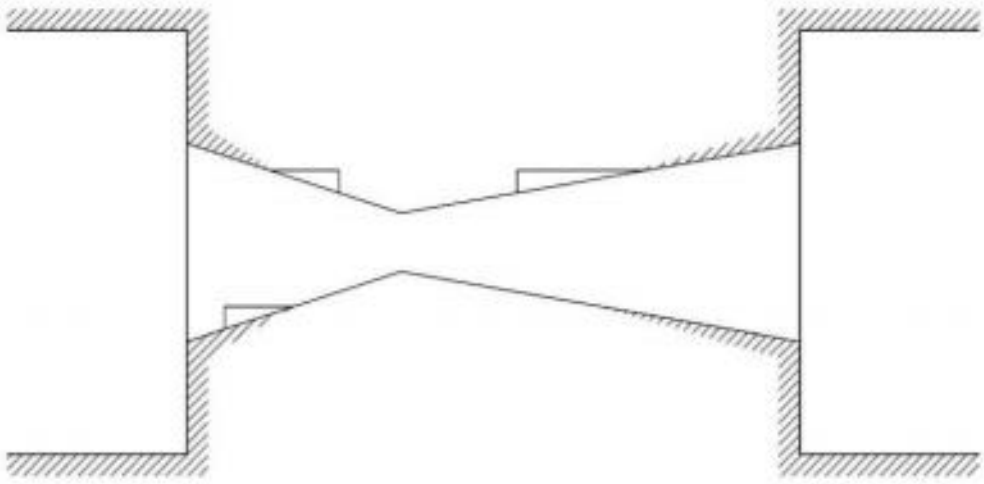
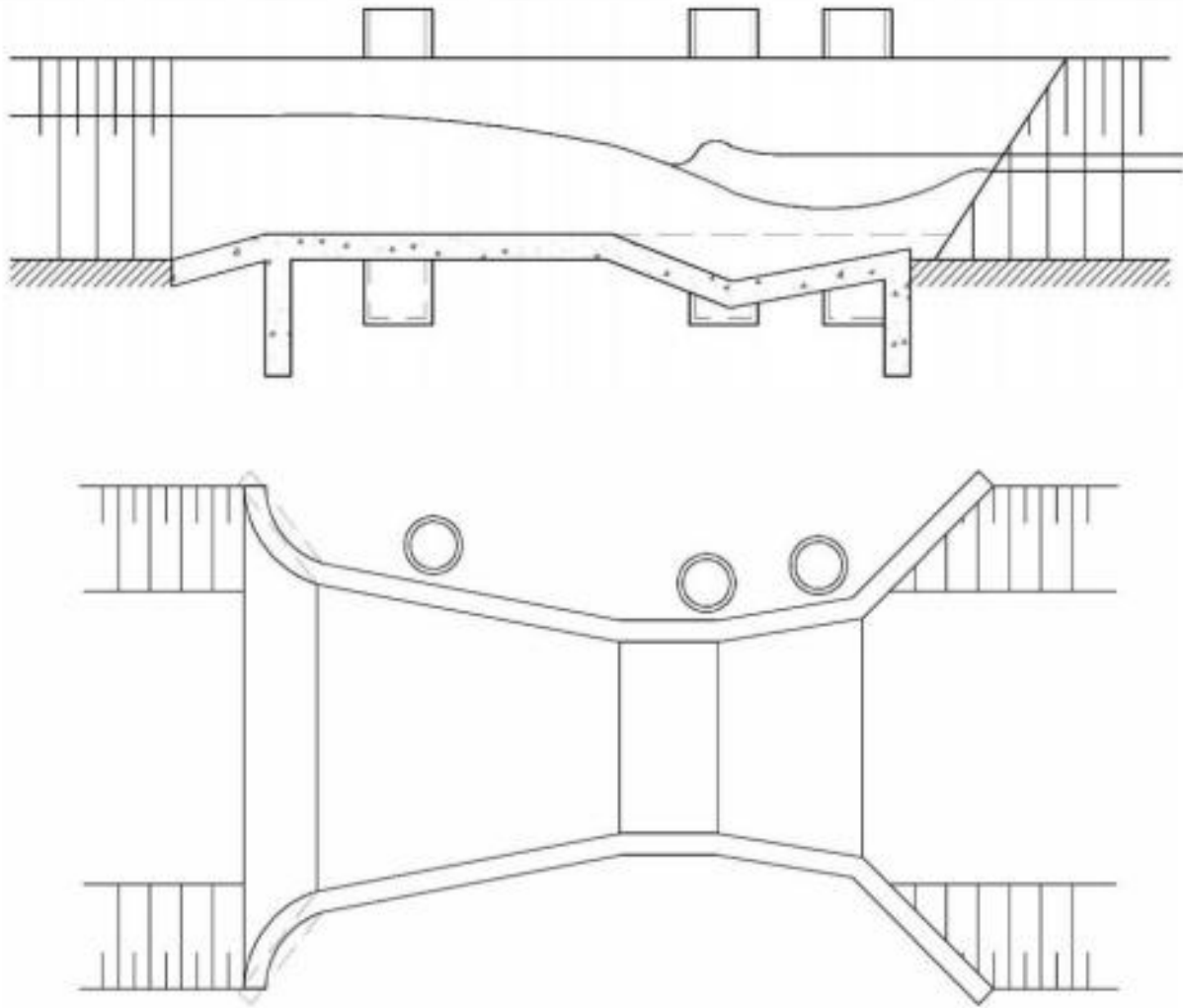
Control Structures

Table 2. Types of Flumes, Description and Discharge Evaluation

Classification	Description	Discharge Evaluation
Long-throated (Rectangular control section)		$Q = C_d C_v \frac{2}{3} \left(\frac{2}{3} g \right)^{0.50} b_c H_1^{1.50}$ Q = discharge, m³/s g = gravitational acceleration, m²/s H₁ = (3/2)y_c = total upstream energy head over the weir crest, m C_d = coefficient of discharge C_v = coefficient due to approach velocity



Control Structures

Cut-throat		$Q = CH_a^n$ $C = KW^{1.025}$ Q = discharge, m³/s C = free flow coefficient H_a = upstream gauge reading (2L/9 from throat) K = flume length coefficient W = throat width, m
Parshall		$Q = kH_a^u$ Q = discharge, m³/s k = dimensional factor which is a function of throat width H_a = upstream gauge reading u = 1.522 to 1.60



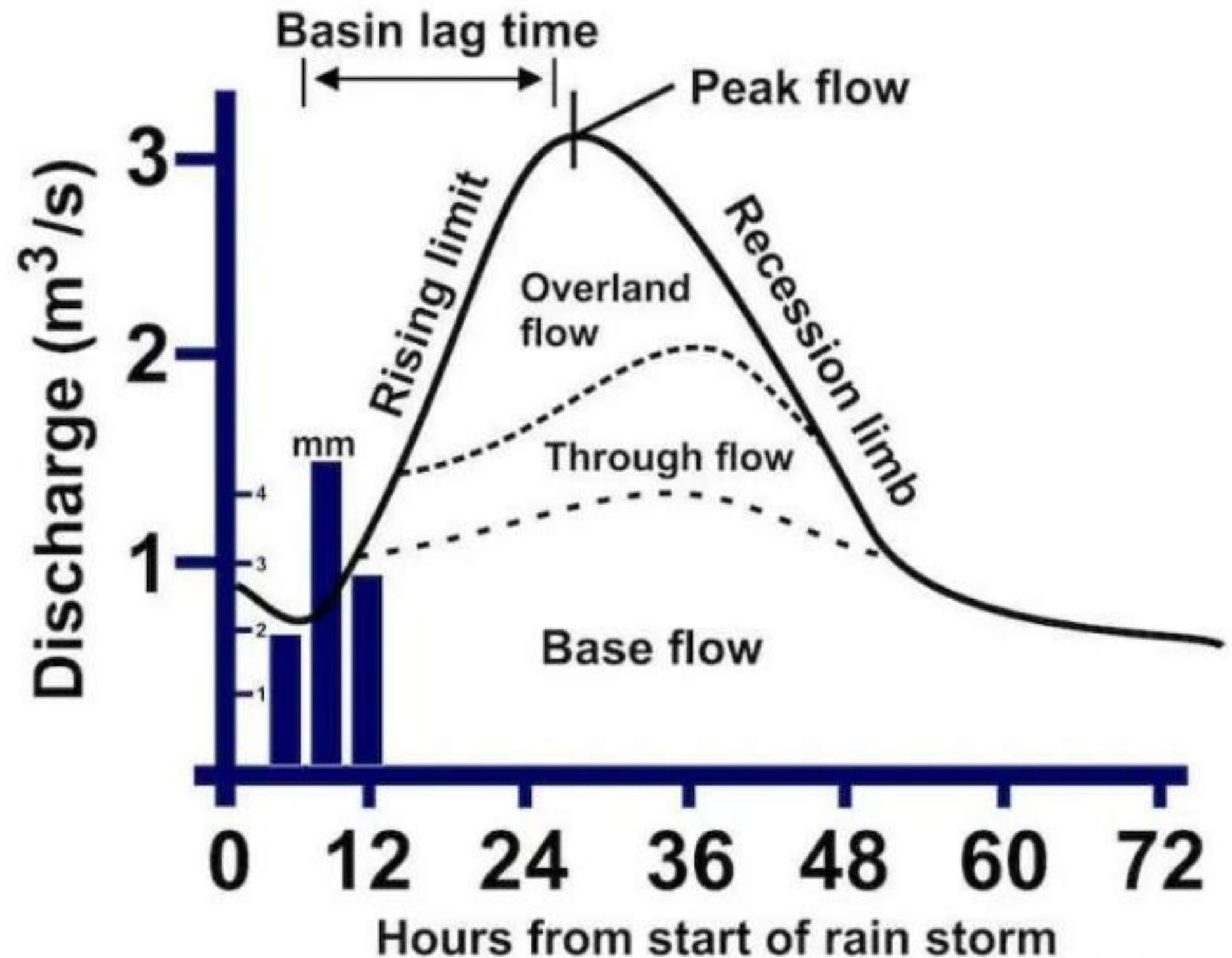
Control Structures

Table 3. Types of Orifices, Description and Discharge Equation

Classification	Description	Discharge Evaluation
Circular Sharp-Edged		$Q = C_d C_v A \sqrt{2g(h_1 - h_2)}$ $Q = C_d C_v A \sqrt{2g\Delta h}$ Q = discharge, m³/s C_d = discharge coefficient C_v = velocity coefficient A = area of the orifice, m² h_1-h_2 = head differential across the orifice Δh = upstream head above the center of the orifice
Rectangular Sharp-Edged		

Hydrograph Analysis and Unit Hydrograph

A **hydrograph** is a graphical representation of the instantaneous runoff rate against time. The area under the hydrograph gives the runoff volume.



Direct Runoff Hydrograph Development

SIX IMPORTANT RUNOFF PROPERTIES

1. Peak runoff rate
2. Dependable water supply
3. Runoff volume
4. Time to peak
5. Recession time
6. Runoff quality

➤ runoff is a highly variable hydrologic process, the estimation of its properties is usually done on a frequency distribution bases



Important Hydrograph Properties

1. **Peak Flow (q_p)** – used to determine the magnitude of floods and a valuable consideration in the design of structures
2. **Time to Peak (t_p)** – used for flood forecasting and water quality studies; watershed response time
3. **Runoff Volume (V)** – total water yield from a storm for a given watershed
4. **Recession Time (t_r)** – time for surface and interflow to recede; duration of flooding
5. **Base time (t_b)** – total of time to peak and recession time;
$$(t_b) = (t_p) + (t_r)$$
6. **Base flow** – low flow, dependable flow, groundwater discharge



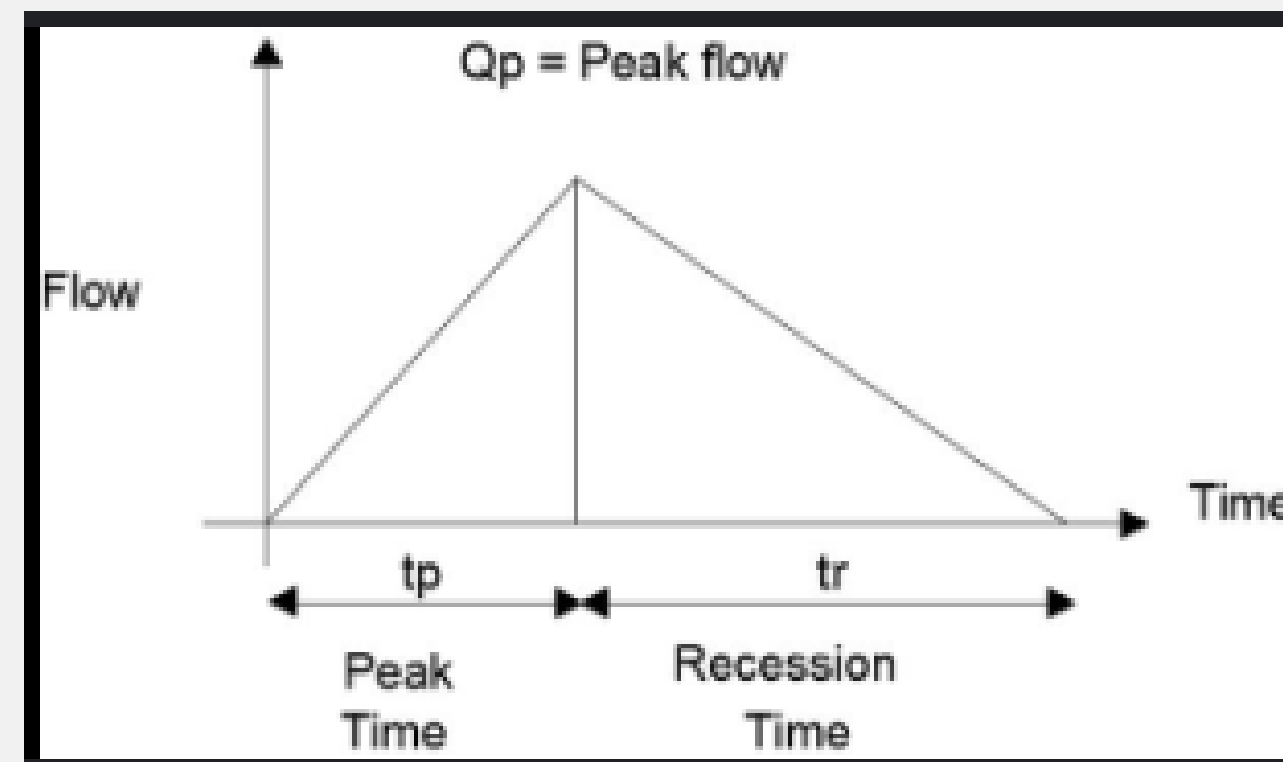
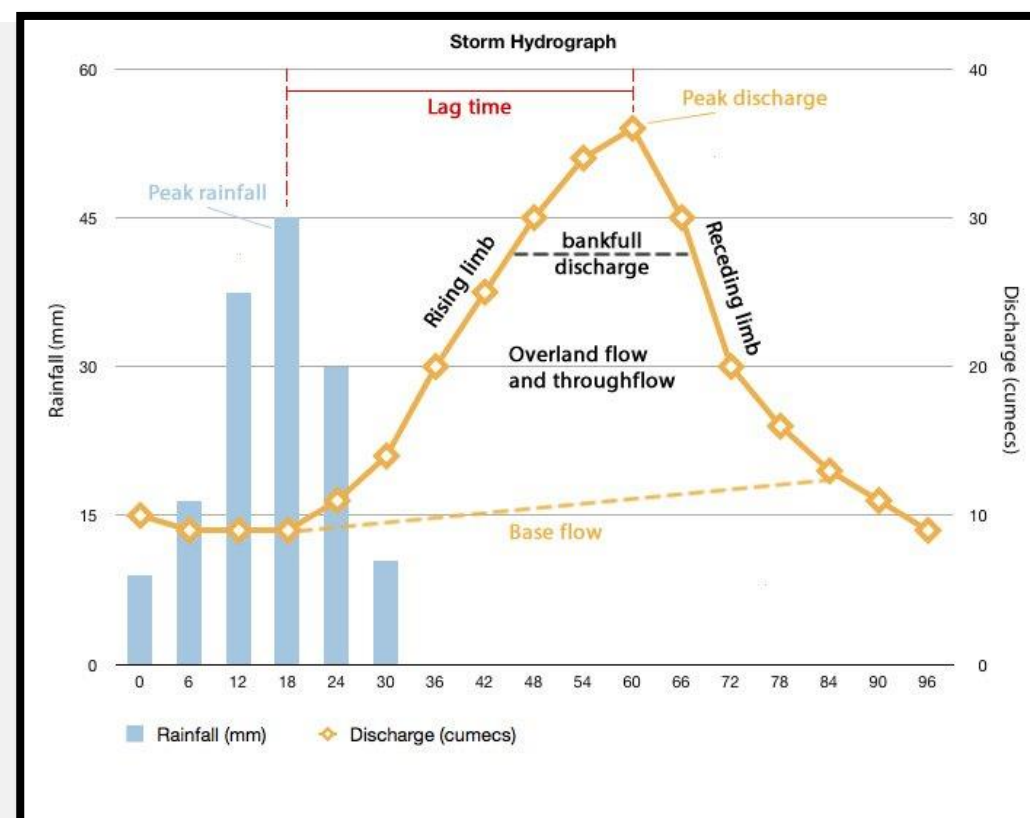
Direct Runoff Hydrograph Development

Basic Hydrograph

- assumes that all hydrographs from all small watersheds (in the US) have similar forms
- it is plotted over 100 arbitrary units of flow and 100 arbitrary units of time

Triangular Hydrograph

- approximation of the basic hydrograph
- developed mainly to simplify flood routing procedures in US

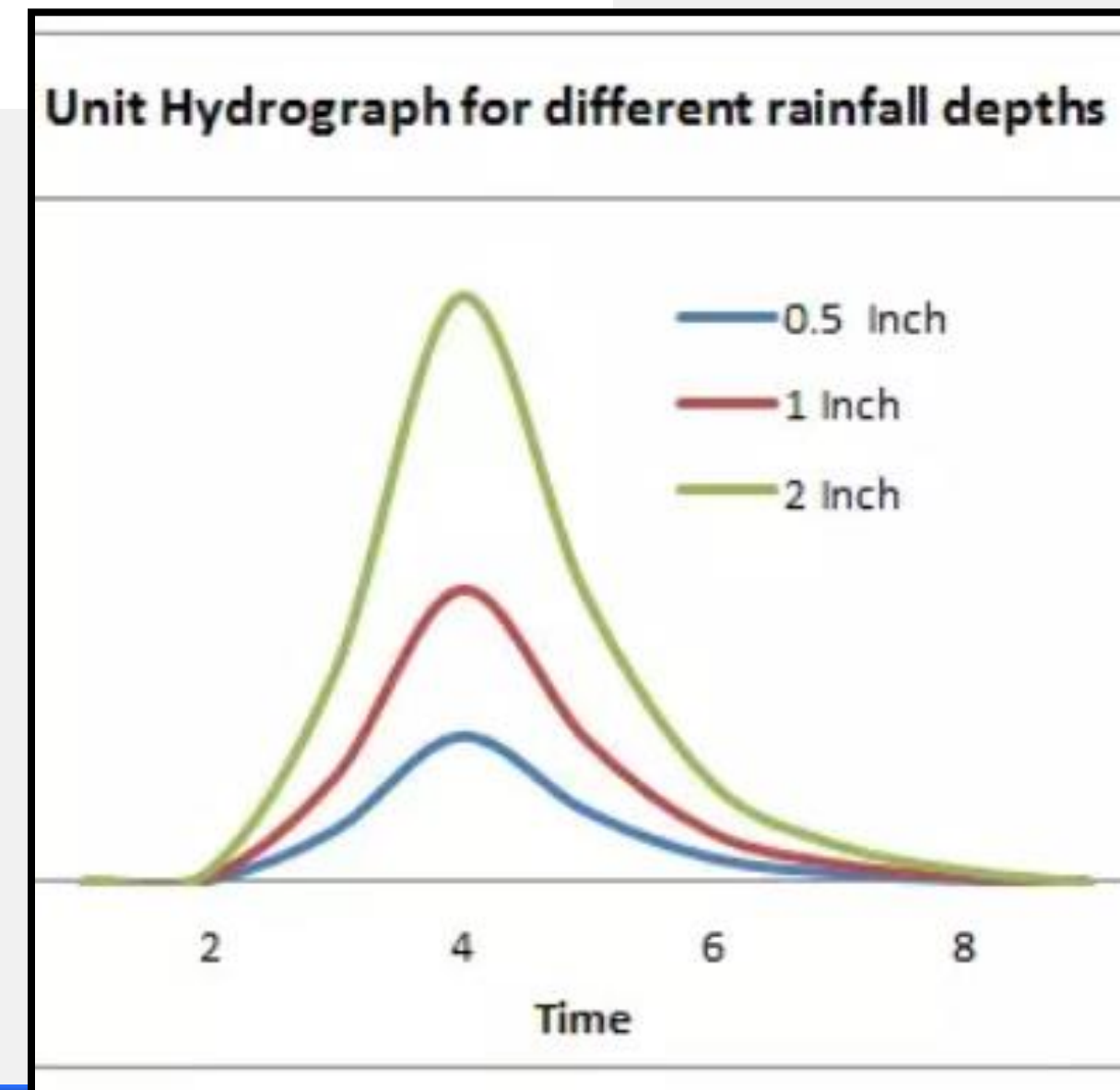


Direct Runoff Hydrograph Development

Unit Hydrograph

- a hydrograph with a unit volume (e.g. 1cm) of direct runoff for a given storm duration
- represent the response of the basin on a given storm duration and characteristics
- enables one to synthesize hydrographs for complex storms by superimposing the hydrographs resulting from the individual components of the storm

Unit hydrograph is a **direct runoff** hydrograph resulting from **one unit (one inch or one cm)** of constant intensity uniform rainfall occurring over the entire watershed



Direct Runoff Hydrograph Development

Synthetic Hydrograph

- prepared using data from a number of watershed to develop dimensionless unit hydrographs which are applicable to ungaged watersheds
- a dimensionless hydrograph is made from natural or unit hydrographs in which the time to peak (t_p) and the peak runoff rate (q_p) are considered and t/t_p is plotted against q/q_p



Sample Question 16:

Portion of precipitation that makes way towards stream channels, lakes, oceans as surface or subsurface flow.

- a. Streamflow
- b. Runoff
- c. Infiltration
- d. Seepage

B

Sample Question 17:

Runoff is mostly affected by this climatic factor

- a. Season of the Year
- b. Rainfall Duration and Intensity
- c. Meteorologic Conditions Before the Storm
- d. Frozen Soil Conditions Before the Storm

B

Sample Question 18:

Determine the 10-year period peak runoff rate (m^3/s) for an 80-ha watershed having a runoff coefficient of 0.4. The maximum length of water flow is 610 meters and the fall along the path is 6.0 meters. The rainfall intensity for that return period is 101.25 mm/hr having duration equal to the time of concentration.

Sample Question 18:

Determine the 10-year period peak runoff rate (m^3/s) for an 80-ha watershed having a runoff coefficient of 0.4. The maximum length of water flow is 610 meters and the fall along the path is 6.0 meters. The rainfall intensity for that return period is 101.25 mm/hr having duration equal to the time of concentration.

Use the formula: $q = CiA$ or $q = 0.0028 CiA$

Where: $C = 0.4$
 $i = 101.25 \text{ mm/hr}$
 $A = 80 \text{ ha}$

$9 \text{ m}^3/\text{s}$

Sample Question 19:

The peak runoff in a 30 ha-watershed corresponding to a rainfall intensity of 60 mm/hr and a runoff coefficient of 0.5 is

- a. $1.2 \text{ m}^3 / \text{s}$
- b. $2.5 \text{ m}^3 / \text{s}$
- c. $4.7 \text{ m}^3 / \text{s}$
- d. $5.5 \text{ m}^3 / \text{s}$

Sample Question 19:

The peak runoff in a 30 ha-watershed corresponding to a rainfall intensity of 60 mm/hr and a runoff coefficient of 0.5 is

- a. $1.2 \text{ m}^3/\text{s}$
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- c. $4.7 \text{ m}^3/\text{s}$
- d. $5.5 \text{ m}^3/\text{s}$

Use the formula: $q = CiA$ or $q = 0.0028 CiA$

Where: $C = 0.5$
 $i = 60 \text{ mm/hr}$
 $A = 30 \text{ ha}$

B

Sample Question 20:

In the Soil Conservation Service method (curve number method), if the Curve number is 100, this indicates that:

- a. rainfall = runoff
- b. rainfall is less than runoff
- c. runoff = 0
- d. rainfall is highly erosive

Sample Question 21:

The lateral movement of infiltrated water when it encounters impervious layer is called:

- a. Percolation
- b. interflow
- c. Base flow
- d. surface runoff

B

Sample Question 22:

In a watershed, the rainfall intensity is 12 mm while the soil infiltration rate is only 10 mm. Determine the surface runoff.

- a. 22 mm
- b. 2 mm
- c. 12 mm
- d. 1.2 mm

B

Sample Question 24:

The velocities at 0.2 and 0.8 of the depth of flow in a **parabolic open channel** are 1.0 m/s and 0.2 m/s, respectively. If the flow depth is 0.5 m and the top width is 1.5 m, the charge is

- a. 500 lps
- b. 450 lps
- c. 300 lps
- d. 250 lps

Use the formula $Q = AV$, where $A = \frac{2}{3}Ty$, and $V = \frac{V_{0.2d} + V_{0.8d}}{2}$

C

Sample Question 23:

The velocities at 0.2 and 0.8 depth of flow in a 1.0-m wide **rectangular open channel** are 1.2 m/s and 0.3 m/s, respectively. If the flow depth is 0.8 m, the discharge is

- a. $0.4 \text{ m}^3/\text{s}$
- b. $0.5 \text{ m}^3/\text{s}$
- c. $0.6 \text{ m}^3/\text{s}$
- d. $0.7 \text{ m}^3/\text{s}$

Use the formula $Q = AV$, where $A = by$, and $V = \frac{V_{0.2d} + V_{0.8d}}{2}$

Sample Question 25:

Hydrograph is made of ____

- a. Direct runoff only
- b. Overland flow, interflow and baseflow
- c. Interflow and base flow
- d. Direct runoff and interflow
- e. None of the above

B

Sample Question 26:

Direct runoff hydrograph from a unit excess precipitation that occurs uniformly over a basin.

- a. Streamflow hydrograph
- b. Total hydrograph
- c. Unit hydrograph
- d. All of the above

C

Sample Question 27:

Unit hydrograph is made of

- a. Direct runoff only
- b. Overland flow, interflow and baseflow
- c. Interflow and baseflow
- d. Direct runoff and interflow
- e. None of the above

A

Sample Question 28:

Prepared using data from a number of watersheds to develop dimensionless unit hydrographs which are applicable to ungagged watersheds.

- a. Unit hydrograph
- b. Basic hydrograph
- c. Synthetic hydrograph
- d. Triangular hydrograph

Sample Question 29:

Given the direct runoff hydrograph in m^3/s taken at 1-hr interval in the mainstream of a 50 sq.km watershed: 0, 2.8, 13.8, 44.8, 77.8, 37.8, 12.8, 4.5, 1.8, 1.3, 0.9, 0.6, 0.2. The equivalent depth of direct runoff is

- a. 10.3 mm
- b. 12.3 mm
- c. 14.3 mm
- d. 163 mm

C

Sample Question 30:

A watershed has an area of 300 ha. Due to a 10 cm rainfall event over the watershed, a streamflow is generated and at the outlet of the watershed it last for 10 hours. Assuming a runoff/rainfall ratio of 0.2 for this event, the average streamflow rate at the outlet in the period of 10 hrs is,

- a. $1.33 \text{ m}^3/\text{sec}$
- b. $1.67 \text{ m}^3/\text{sec}$
- c. $150 \text{ m}^3/\text{min}$
- d. $60,000 \text{ m}^3/\text{hr}$
- e. None of the above

B

Sample Question 31:

Rainfall intensity of 20 mm/hr occurred over a watershed with an area of 100 ha. The rainfall last for 6 hrs. The direct runoff volume in the stream was measured and is equal to 30,000 m³. The precipitation not available to runoff in this case is

- a. 9 cm
- b. 3cm
- c. 17.5 mm
- d. 5 mm
- e. None of the above

A

Sample Question 32:

After a 2-hour storm, a station downstream from a 45 square mile drainage basin measures 9400 cfs as a peak discharge and 3300 acre-feet as total runoff. Find the 2-hour unit hydrograph peak discharge.

- a. 8,800 cfs
- b. 6,800 cfs
- c. 9,400 cfs
- d. 3,300 cfs

Sample Question 32:

After a 2-hour storm, a station downstream from a 45 square mile drainage basin measures 9400 cfs as a peak discharge and 3300 acre-feet as total runoff. Find the 2-hour unit hydrograph peak discharge.

- a. 8,800 cfs
- b. 6,800 cfs
- c. 9,400 cfs
- d. 3,300 cfs

Calculating the runoff from 1-inch rainfall:

$$45 \text{ mi}^2 \left(\frac{5280 \text{ ft}}{1 \text{ mi}} \right)^2 \left(\frac{1 \text{ acre}}{43,560 \text{ ft}^2} \right) \times \left(1 \text{ inch} \times \frac{1 \text{ ft}}{12 \text{ in}} \right) = 2400 \text{ acre} - \text{ft}$$

Finding the total rainfall for 3,300 acre-ft total runoff:

$$\frac{1 \text{ inch}}{2400 \text{ acre} - \text{ft}} = \frac{R}{3,300 \text{ acre} - \text{ft}}$$

R = 1.375 inches

Finally, solving for peak discharge Q for the unit hydrograph:

$$\frac{1.375 \text{ in}}{9400 \text{ cfs}} = \frac{1 \text{ inch}}{Q}$$

Q = 6836 cfs

Sample Question:

Which of the following statements is true?

- a) the average velocity in an open channel is located at 0.6 of the depth measured from the bottom
- b) the average velocity is greater than the surface velocity
- c) the maximum velocity occurs at the surface
- d) the average velocity in an open channel is located at 0.4 of the depth measured from the bottom

c

**TIME FOR
A BREAK**

